

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\121-CNVSMANWAY+JACKETFinal.pvdb

Date of Analysis : Apr 10,2026 3:25pm

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User
 FileName : 121-C_NVS_MANWAY+JACKET_Final Page 2 of 179
 Vessel Design Summary: Step: 35 3:26pm Apr 10,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2250.000 mm ID
 Vessel Design Length, Tangent to Tangent 23.52 m
 Specified Datum Line Distance 0.26 m
 Shell Side Design Temperature 302 °C
 Channel Side Design Temperature 275 °C
 Shell Side Design Pressure 18.001 N/mm²
 Channel Side Design Pressure 18.001 N/mm²
 Shop Shell Side Test Pressure 23.401 N/mm²
 Shop Channel Side Test Pressure 23.401 N/mm²
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 11	2	...	K11789	
Head	SA-387 11	2	...	K11789	
Nozzle	SA-336 F11	3	...	K11572	
Nozzle Flg	SA-182 F11	1	...	K11597	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-182 F11	2	...	K11572	
Tubes	SA-213 T11	...	...	K11597	
Tubesheet	SA-336 F11	3	...	K11572	
Jacket	SA-387 11	2	...	K11789	
Jkt Closure	SA-387 11	2	...	K11789	
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-387 11	2	...	K11789	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (N/mm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
channel sphere-ar	18.001	0.00	19.000	3.0000	N/A	No
Channel - rear	18.002	0.00	19.000	3.0000	N/A	No
shell	18.002	0.00	19.000	3.0000	N/A	No
shell cone -rea	18.001	0.00	19.000	3.0000	N/A	No
Shell-2	18.002	0.00	19.000	3.0000	N/A	No
shell cone- fro	18.001	0.00	19.000	3.0000	N/A	No
shell- front	18.002	0.00	19.000	3.0000	N/A	No
Channel front	18.002	0.00	19.000	3.0000	N/A	No
Channel sphere-ont	18.001	0.00	19.000	3.0000	N/A	No

Liquid Level: 1.90 m Dens.: 51.976 kg/m³ Sp. Gr.: 0.052

Element Types and Properties:



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Element Type	"To" Elev m	Element Length m	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Sphere	0.0	0.3	90.0	75.0	72.4	...	1.00	1.00
Cylinder	2.0	2.0	156.0	156.0	151.0	...	1.00	1.00
Cylinder	4.4	2.0	156.0	156.0	151.0	...	1.00	1.00
Cylinder	5.1	0.7	156.0	156.0	151.0	...	1.00	1.00
Cylinder	17.9	12.9	156.0	156.0	151.0	...	1.00	1.00
Cylinder	18.6	0.7	156.0	156.0	151.0	...	1.00	1.00
Cylinder	20.6	2.0	156.0	156.0	151.0	...	1.00	1.00
Cylinder	23.0	2.0	156.0	156.0	151.0	...	1.00	1.00
Sphere	23.3	0.3	90.0	75.0	72.4	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N4- Tube outle	WRC-297	0.961	Shell	Passed
N1- Shell inle	WRC-297	0.984	Shell	Passed
N2- Shell outl	WRC-297	0.937	Shell	Passed
N3- tube inlet	WRC-297	0.961	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

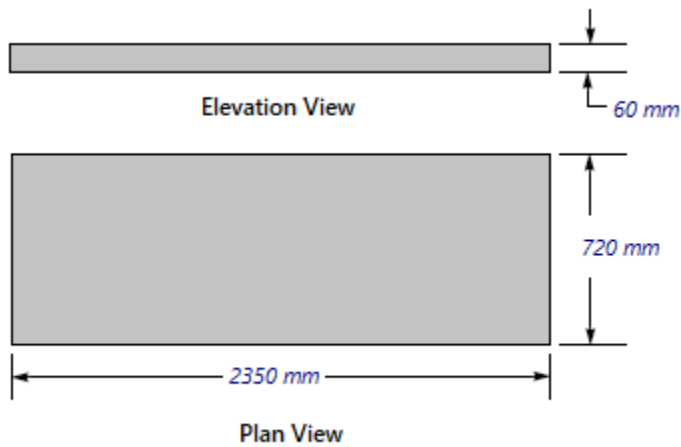
Saddle Width	204.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1800.000 mm
Wear Pad Width	254.000 mm
Wear Pad Thickness	10.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	3760.000 mm

Baseplate Length	2350.000 mm
Baseplate Thickness	60.000 mm
Baseplate Width	720.000 mm
Number of Ribs (including outside ribs)	5
Rib Thickness	20.000 mm
Web Thickness	20.000 mm
Height of Center Web	300.000 mm
Number of Bolts in Baseplate	8

Baseplate Sketch



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**Summary of Maximum Saddle Loads, Operating Case, Un-Factored:**

Maximum Vertical Saddle Load	2765126.25	N
Maximum Transverse Saddle Shear Load	119635.27	N
Maximum Longitudinal Saddle Shear Load	239270.55	N

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	2644010.25	N
Maximum Transverse Saddle Shear Load	8489.55	N
Maximum Longitudinal Saddle Shear Load	3903.10	N

Weights:

Fabricated - Bare w/o removable internals	406945.9	kgm
Shop Test - Fabricated + water (full)	496878.4	kgm
Shipping - Fab. + rem. intls.+ shipping app.	417963.2	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	417963.2	kgm
Empty - Fab. + intls. + details + wghts.	417963.3	kgm
Operating - Empty + operating liquid (no ca)	422363.2	kgm

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 FileName : 121-C_NVS_MANWAY+JACKET_Final Page 5 of 179
 Nozzle Summary: Step: 33 3:25pm Apr 10,2026

Nozzle Calculation Summary (N/mm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
Manway-1	18	...	...	...		OK	Passed
N4- Tube outlet	17	...	...	OK	35.58	Failed	Passed
N1- Shell inlet	14	...	...	OK	35.58	Failed	Failed
N2- Shell outle	18	...	...	OK	35.58	Failed	Passed
N3- tube inlet	17	...	...	OK	35.58	Failed	Passed
Mayway-2	18	...	...	...		OK	Passed

MAWP Summary for Nozzles, Flanges on the Vessel Side (Inner Shell):

Minimum MAWP Nozzles : 17 Nozzle : N3- tube inlet

MAWP Summary for Nozzles and Nozzle Flanges on the Jacket(s) Side:

Minimum MAWP Nozzles : 14 Nozzle : N1- Shell inlet

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
10	Manway-1	0.000	0.000	993.151
20	N4- Tube outlet	1258.000	90.000	993.151
30	N1- Shell inlet	3258.000	90.000	993.151
70	N2- Shell outle	19898.000	270.000	1003.600
80	N3- tube inlet	22258.000	270.000	993.151
90	Mayway-2	0.000	0.000	993.151

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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 Nozzle Schedule: Step: 32 3:25pm Apr 10,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
Manway-1	24.000 in	160	WNF	24.000	59.512	...	...	593.03	1500
N4- Tube outlet	24.000 in	160	None	24.000	59.512	...	...	747.88	...
N1- Shell inlet	24.000 in	160	None	24.000	59.512	...	...	530.34	...
N2- Shell outle	24.000 in	160	None	24.000	59.512	...	...	805.53	...
N3- tube inlet	24.000 in	160	None	24.000	59.512	...	...	747.88	...
Mayway-2	24.000 in	160	WNF	24.000	59.512	...	...	593.03	1500

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
Manway-1	SA-336 F11	54.400	10.000	...	...	...
N4- Tube ou	SA-336 F11	50.000	20.000	...	...	...
N1- Shell i	SA-336 F11	50.000	20.000	...	...	...
N2- Shell o	SA-336 F11	50.000	50.000	...	...	...
N3- tube in	SA-336 F11	50.000	20.000	...	...	...
Mayway-2	SA-336 F11	54.400	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum m	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
Manway-1	...	0.0	450.00	0.00	channel sphere-a
N4- Tube outlet	1.000	90.0	500.00	0.00	Channel - rear
N1- Shell inlet	3.360	90.0	250.00	0.00	shell
N2- Shell outle	19.640	270.0	530.00	0.00	shell- front
N3- tube inlet	22.000	270.0	500.00	0.00	Channel front
Mayway-2	...	0.0	450.00	0.00	Channel sphere-o

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Bill of Material: Step: 31 3:25pm Apr 10,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	SPHERICAL HEAD: 90.0mm THK X 2250.0mm ID X 0.3m	SA-387 11
3	CYLINDER: 156.0mm THK X 2250.0mm ID X 2.0m	SA-387 11
2	CYLINDER: 156.0mm THK X 2250.0mm ID X 0.7m	SA-387 11
2	INSULATION: 1383mm X 150mm THK	Hot insulation
4	INSULATION: 2000mm X 150mm THK	Hot insulation
33	BAFFLES: 1377.0mm DIA X 8.0mm THK	SA-387 11
12	TIE RODS: 19mm DIA X 19000mm	SA-193 B7
2	TUBESHEET: 2562.0mm X 360.0mm THK	SA-336 F11
3439	TUBES: 19000.0mm X 19.0mm DIA X 2.1mm THK	SA-213 T11
1	JACKET CYL: 1600.0mm LONG X 2820.0mm DIA X 190.0mm THK	SA-387 11
2	JACKET SEALING RING: 210.0mm THK	SA-387 11
2	INSULATION: 700mm X 150mm THK	Hot insulation
1	SADDLE: 204.0mm X 120 DEG	SA-387 11
1	SADDLE: 204.0mm X 120 DEG	SA-516 70
1	INSULATION: 12880mm X 150mm THK	Hot insulation
1	JACKET CYL: 1600.0mm LONG X 2820.0mm DIA X 200.0mm THK	SA-387 11
2	CLASS 1500 GR 1.9, 24.0" BLIND FLANGE(S)	SA-182 F11
2	CYLINDRICAL SEGMENT 156.0mm THK X 2807.8mm X 2.0m	SA-387 11
1	CYLINDRICAL SEGMENT 156.0mm THK X 1453.0mm X 2.0m	SA-387 11
1	CYLINDRICAL SEGMENT 156.0mm THK X 6068.9mm X 12.9m	SA-387 11
1	CYLINDRICAL SEGMENT 156.0mm THK X 999.5mm X 12.9m	SA-387 11
1	NAMEPLATE	...



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 FileName : 121-C_NVS_MANWAY+JACKET_Final Page 8 of 179
 MDMT Summary: Step: 34 3:26pm Apr 10,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Left Saddle [24]	A	-8						Yes
Sdl 2 Fr60 [24]	A	-24						No
channel sphere[9]	A	44	38		0.887	75.000	1.00	Yes
Channel sphere[9]	A	44	38		0.887	75.000	1.00	Yes
Manway-1 [1]	A	44	42		0.965	75.000	1.00	Yes
Nozzle Flg [5]	B	-18	-35					
Mayway-2 [1]	A	44	42		0.965	75.000	1.00	Yes
Nozzle Flg [5]	B	-18	-35					
Warmest MDMT:		44	42					

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
 The Basic MDMT includes the (30F) PWHT credit if applicable.

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FileName : 121-C_NVS_MANWAY+JACKET_Final Page 9 of 179
Warnings and Errors: Step: 0 3:25pm Apr 10,2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

Note:

For nozzles that are welded to both the Core and Jacket of jacketed vessels, the 2021 ASME Code requires that the opening in the jacket also be checked. This is accomplished by modelling an additional nozzle in the same location as the nozzle that penetrates through to the vessel core. The flange specification if any can be removed if needed. Other items such as weld sizes, pad specification etc. should also be addressed.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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 Input Echo: Step: 1 3:25pm Apr 10,2026

Units used in this Analysis (SI):

Name	System Unit	Constant	User Unit
Length	Feet	0.3048	m
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	0.0006	m ²
Moment	ft. lbs.	1.3563	N-m
Stress	lbs./sq.in.	6.8946	kPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	N/mm ²
Elast. Modulus	lbs./sq.in.	0.0069	N/mm ²
Pipe Density	lbs./cu.in.	27680.0000	kg/m ³
Ins. Density	lbs./cu.ft.	16.0183	kg/m ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	0.4470	m/sec
Tray Weight	lbs./sq.ft.	47.8790	N/m ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	0.3048	m
Volume	in.3	0.1600E-04	m ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	18.001	N/mm ²
Channel Side Design Pressure	18.001	N/mm ²
Shell Side Design Temperature	302.0	°C
Channel Side Design Temperature	275.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Sour	
Service Type, Channel Side	Sour	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	N/mm ²
User defined MAWP, Channel Side	0	N/mm ²
User defined MAPnc, Shell Side	0	N/mm ²
User defined MAPnc, Channel Side	0	N/mm ²
User defined Test Pres., Shell Side	0	N/mm ²
User defined Test Pres., Channel Side	0	N/mm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No



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Input Echo:

Step: 1 3:25pm Apr 10,2026

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	27 m/sec
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0.3 m
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 m
Distance Upwind of Crest Lh	0 m
Distance from Crest to the Vessel x	0 m
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 m

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year



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 FileName : 121-C_NVS_MANWAY+JACKET_Final Page 12 of 179
 Input Echo: Step: 1 3:25pm Apr 10,2026

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	Yes
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/9
Element From Node	10
Element To Node	20
Element Type	Spherical
Description	channel sphere-ar
Distance "FROM" to "TO"	0.258 m
Inside Diameter	2250 mm
Element Thickness	75 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	90 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	18.001 N/mm ²
Design Temperature Internal Pressure	275 °C
Design External Pressure	0 N/mm ²
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Allowable Stress, Ambient	147989 kPa
Allowable Stress, Operating	147989 kPa
Allowable Stress, Hydrotest	278979 kPa
Material Density	7750.4 kg/m ³
P Number Thickness	16.002 mm
Yield Stress, Operating	248985 kPa
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-3
UNS Number	K11789
Class / Thickness / Grade	2 ::
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	-1.125 m
Height/Length of Insulation	1.383 m
Thickness of Insulation	150 mm
Density	352.4 kg/m ³
Element From Node	10
Detail Type	Nozzle
Detail ID	Manway-1
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	24 in.
Nozzle Schedule	160



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Nozzle Class	1500
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	38702 N
Grade of Attached Flange	GR 1.9
Nozzle Matl	SA-336 F11

Element#	2/9
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Channel - rear
Distance "FROM" to "TO"	2 m
Inside Diameter	2250 mm
Element Thickness	156 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	156 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	18.001 N/mm ²
Design Temperature Internal Pressure	275 °C
Design External Pressure	0 N/mm ²
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Liquid	1.6 m
Liquid Density	51.976 kg/m ³

Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Insulation	2 m
Thickness of Insulation	150 mm
Density	352.4 kg/m ³

Element From Node	20
Detail Type	Nozzle
Detail ID	N4- Tube outlet
Dist. from "FROM" Node / Offset dist	1 m
Nozzle Diameter	24 in.
Nozzle Schedule	160
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	17071 N
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11

Element#	3/9
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	shell



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Distance "FROM" to "TO"	2	m
Inside Diameter	2250	mm
Element Thickness	156	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	156	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	18.001	N/mm ²
Design Temperature Internal Pressure	302	°C
Design External Pressure	0	N/mm ²
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Liquid	1.6 m
Liquid Density	62.971 kg/m ³

Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 30
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Insulation	2 m
Thickness of Insulation	150 mm
Density	352.4 kg/m ³

Element From Node	30
Detail Type	Nozzle
Detail ID	N1- Shell inlet
Dist. from "FROM" Node / Offset dist	1 m
Nozzle Diameter	24 in.
Nozzle Schedule	160
Nozzle Class	None
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	12669 N
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11

Element#	4/9
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	shell cone -rea
Distance "FROM" to "TO"	0.7 m
Inside Diameter	2250 mm
Element Thickness	156 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	156 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	18.001 N/mm ²
Design Temperature Internal Pressure	302 °C
Design External Pressure	0 N/mm ²
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0



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Weld is pre-Heated	No
Element From Node	40
Detail Type	Insulation
Detail ID	Ins: 40
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Insulation	0.7 m
Thickness of Insulation	150 mm
Density	352.4 kg/m³

Element#	5/9
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Shell-2
Distance "FROM" to "TO"	12.88 m
Inside Diameter	2250 mm
Element Thickness	156 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	156 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	18.001 N/mm²
Design Temperature Internal Pressure	302 °C
Design External Pressure	0 N/mm²
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	1.06 m
Width of Saddle	204 mm
Height of Saddle at Bottom	1800 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	254 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 m

Element From Node	50
Detail Type	Saddle
Detail ID	Sdl 2 Fr60
Dist. from "FROM" Node / Offset dist	12.34 m
Width of Saddle	204 mm
Height of Saddle at Bottom	1800 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	254 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.40000001
Moment Factor	3.0
Dimension E at base (optional)	0 mm



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Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 m
Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Liquid	1.6 m
Liquid Density	62.971 kg/m ³
Element From Node	50
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Insulation	12.88 m
Thickness of Insulation	150 mm
Density	352.4 kg/m ³

Element#	6/9
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	shell cone- fro
Distance "FROM" to "TO"	0.7 m
Inside Diameter	2250 mm
Element Thickness	156 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	156 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	18.001 N/mm ²
Design Temperature Internal Pressure	302 °C
Design External Pressure	0 N/mm ²
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 60
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Insulation	0.7 m
Thickness of Insulation	150 mm
Density	352.4 kg/m ³

Element#	7/9
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	shell- front
Distance "FROM" to "TO"	2 m
Inside Diameter	2250 mm
Element Thickness	156 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	156 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	18.001 N/mm ²
Design Temperature Internal Pressure	302 °C



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Design External Pressure	0	N/mm ²
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid: 70	
Dist. from "FROM" Node / Offset dist	0	m
Height/Length of Liquid	1.9	m
Liquid Density	62.971	kg/m ³

Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	m
Height/Length of Insulation	2	m
Thickness of Insulation	150	mm
Density	352.4	kg/m ³

Element From Node	70	
Detail Type	Nozzle	
Detail ID	N2- Shell outle	
Dist. from "FROM" Node / Offset dist	1	m
Nozzle Diameter	24	in.
Nozzle Schedule	160	
Nozzle Class	None	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	17517	N
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	

Element#	8/9	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	Channel front	
Distance "FROM" to "TO"	2	m
Inside Diameter	2250	mm
Element Thickness	156	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	156	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	18.001	N/mm ²
Design Temperature Internal Pressure	275	°C
Design External Pressure	0	N/mm ²
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid: 80	
Dist. from "FROM" Node / Offset dist	0	m
Height/Length of Liquid	1.6	m
Liquid Density	51.976	kg/m ³



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Element From Node	80
Detail Type	Insulation
Detail ID	Ins: 80
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Insulation	2 m
Thickness of Insulation	150 mm
Density	352.4 kg/m³

Element From Node	80
Detail Type	Nozzle
Detail ID	N3- tube inlet
Dist. from "FROM" Node / Offset dist	1 m
Nozzle Diameter	24 in.
Nozzle Schedule	160
Nozzle Class	150
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	17071 N
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11

Element#	9/9
Element From Node	90
Element To Node	100
Element Type	Spherical
Description	Channel sphere-ont
Distance "FROM" to "TO"	0.258 m
Inside Diameter	2250 mm
Element Thickness	75 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	90 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	18.001 N/mm²
Design Temperature Internal Pressure	275 °C
Design External Pressure	0 N/mm²
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	90
Detail Type	Insulation
Detail ID	Ins: 90
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Insulation	1.383 m
Thickness of Insulation	150 mm
Density	352.4 kg/m³

Element From Node	90
Detail Type	Nozzle
Detail ID	Mayway-2
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	24 in.
Nozzle Schedule	160
Nozzle Class	1500
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	38702 N
Grade of Attached Flange	GR 1.9
Nozzle Matl	SA-336 F11



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XY Coordinate Calculations: Step: 2 3:25pm Apr 10,2026

XY Coordinate Calculations:

From	To	X (Horiz.) m	Y (Vert.) m	DX (Horiz.) m	DY (Vert.) m
channel sphere-		0.258	...	0.258	...
Channel - rear		2.258	...	2	...
shell		4.618	...	2	...
shell cone -rea		5.318	...	0.7	...
Shell-2		18.198	...	12.88	...
shell cone- fro		18.898	...	0.7	...
shell- front		20.898	...	2	...
Channel front		23.258	...	2	...
Channel sphere-		23.516	...	0.258	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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 Jacket/VB Calcs: Jacket Step: 3 3:25pm Apr 10,2026

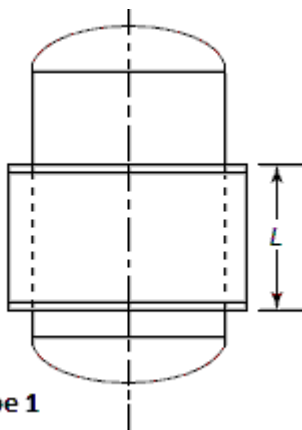
Jacket Design per ASME VIII, Division 2, 4.11.2: Jacket

Design per: Figure 4.11.1 Type 1
 Attachment per: Table 4.11.1, Detail 6, Figure: (a)

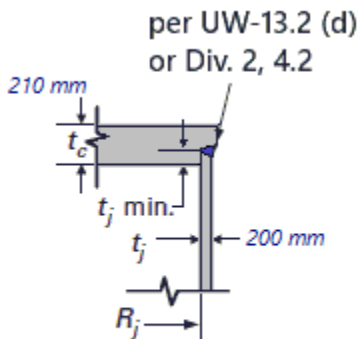
Input Values:

Jacket Design Internal Pressure	Pj	18.00	N/mm ²
Jacket Design Temperature (Internal)		302	C
Jacket Design External Pressure	Pje	0.00	N/mm ²
Jacket Design Temperature (External)		150	C
Static fluid Pressure in this Jacket	pStatic	0.00	N/mm ²
Closure Bar Material		SA-387 11	
Closure Bar Design Stress	[S or f]	147988.73	kPa
Closure Bar Thickness New	tc	210.0000	mm
Closure Bar Corrosion Allowance	cc	3.0000	mm
Inner Shell Outside Diameter	Dso	2562.000	mm
Inner Shell Thickness New	ts	156.0000	mm
Corroded Local Shell Thickness at Closure	ts,c	153.0000	mm
Inner Shell Corrosion Allow. Inside	csi	3.0000	mm
Inner Shell Corrosion Allow. Outside	cso	0.0000	mm
Inner Shell Vacuum Pressure		0.0000	N/mm ²
Jacket Material		SA-387 11	
Jacket Design Stress	Sj	147988.73	kPa
Jacket Inside Diameter New	Dji	2820.000	mm
Jacket Thickness New	tj	190.0000	mm
Jacket Corrosion Allowance Inside	cji	3.0000	mm
Jacket Length	L	1600.00	mm
Jacket Design Length for External Pressure	SL	1600.00	mm
Specific Gravity of contents in Jacket	sg	0.000	
Jacket Long. Weld Joint Efficiency	E	1.00	

This has been specified as a vapor/distribution belt.



Type 1



Div. 2 Detail 6 Fig. (a)

Results:

Jacket Inside Radius [Rj]:

$$= (D_{ji} + 2 \cdot c_i) / 2$$

$$= (2820.0 + 2 \cdot 3.0) / 2$$

$$= 1413.000 \text{ mm}$$

Jacket Corroded Thickness [tjc]:

$$= t_j - c_{ji}$$

$$= 190.0 - 3.0$$

$$= 187.000 \text{ mm}$$

Shell Outside Radius [Rs]:

$$= (D_{so} - 2 \cdot c_{so}) / 2$$

$$= (2562.0 - 2 \cdot 0.0) / 2$$

$$= 1281.000 \text{ mm}$$

Shell Corroded Thickness [tsc]:

$$= t_s - c_{si} - c_{so}$$

$$= 156.0 - 3.0 - 0.0$$

$$= 153.000 \text{ mm}$$

Jacket Space Width [j]:

$$= (D_{ji} + 2 \cdot c_{ji} - D_{so} + 2 \cdot c_{so}) / 2$$

$$= (2820.0 + 2 \cdot 3.0 - 2562.0 + 2 \cdot 0.0) / 2$$

$$= 132.000 \text{ mm}$$

Jacket Thickness Calculations for Cylinder:

Longitudinal Joint: User Defined

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (18.002 \cdot 1413.0) / (147988.73 \cdot 1.0 - 0.6 \cdot 18.002)$$

$$= 185.4052 + 3.0000 = 188.4052 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 N/mm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (147988.73 \cdot 1.0 \cdot 187.0) / (1413.0 + 0.6 \cdot 187.0)$$

$$= 18 - 0.001 = 18 \text{ N/mm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (147988.73 \cdot 1.0 \cdot 190.0) / (1410.0 + 0.6 \cdot 190.0)$$

$$= 18 \text{ N/mm}^2$$



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 Jacket/VB Calcs: Jacket Step: 3 3:25pm Apr 10,2026

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (18.002 \cdot (1413.0 + 0.6 \cdot 187.0)) / (1.0 \cdot 187.0)$$

$$= 146818.781 \text{ kPa}$$

Jacket Minimum Design Metal Temperature Calculations:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Required Closure Thickness [trc]:

$$= 1.414 \cdot \sqrt{(P_j \cdot R_s \cdot j / S)} + c$$

$$= 1.414 \cdot \sqrt{(18.002 \cdot 1281.0 \cdot 132.0 / 147988.734)} + 3.0$$

$$= 205.789 \text{ mm}$$

Maximum Allowable Working Pressure, Closure Bar [CBMawp]:

$$= S / (j \cdot R) \cdot ((t_r - c) / 1.414)^2 - p_{\text{Static}}$$

$$= 147988.734 / (132.0 \cdot 1281.0) \cdot ((210.0 - 3.0) / 1.414)^2 - 0.001$$

$$= 18.756 \text{ N/mm}^2$$

Forming Strain/Elongation Results:

Jacket Forming Strain, Cylinders formed from Plate ($50 \cdot t_{\text{nom}} / R_f(1 - R_f / R_o)$): 6.312 %

Note: Please check Code requirements as the computed elongation is > 5%.

External Pressure results for the Outer Jacket Cylinder:

Maximum External Pressure per Code Case 2286:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 1600.0 / \sqrt{(1600.0 \cdot 187.0)}$$

$$= 2.925$$

Compute the Factor [Ch]:

Since $M_x > 1.5$ and $M_x < 13$, $Ch = 0.92 / (M_x - 0.579)$

$$= 0.92 / (2.925 - 0.579)$$

$$= 0.3921$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.392 \cdot 196439(187.0 / 3200.0)$$

$$= 7202085.000 \text{ kPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 7202085.0 / 245665.4$$

$$= 29.317$$

Predicted Buckling Stress [Fic]:

Since $Fr \geq 2.439$, $Fic = F_y$

$$= 245665.391 \text{ kPa}$$

Factor of Safety [FS]:

Since $Fic [245665.39] \geq F_y [245665.39]$, $FS = 1.667$

Allowable Circumferential Compressive Stress [Fha]:

Since $Fr > 2.439$, $Fha = F_y / FS$

$$= 245665.391 / 1.667$$

$$= 147369.750 \text{ kPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha(t / D_o)$$



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Jacket/VB Calcs: Jacket

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$$= 2 \cdot 147.378(187.0/3200.0)$$
$$= 17.225 \text{ N/mm}^2$$

Summary of Results (mm):

	Required	Actual	Result
Jacket Thk Internal	188.405	190.000	Passed
Closure Bar Thickness	205.789	210.000	Passed

MAWP Results for the Jacket/Vapor/Distribution Belt and Closure Bar:

Jacket Maximum Allowable Working Pressure	18.15	N/mm ²
Closure Bar Maximum Allowable Working Pressure	18.76	N/mm ²
Overall MAWP of Jacket Components	18.15	N/mm ²

Jacket Cylinder External Maximum Allowable Working Pressure:

Jacket Ext. Maximum Allowable Working Pressure	17.22	N/mm ²
--	-------	-------------------

Jacket Hydrotest Pressure Calculations:

Jacket Hydrotest Pressure per UG-99(b) Note 35:

$$= 1.3 \cdot P \cdot \text{Stress Ratio}$$
$$= 1.3 \cdot 18.001 \cdot 1.0$$
$$= 23.401 \text{ N/mm}^2$$

Jacket Hydrotest Pressure per UG-99(b):

$$= 1.3 \cdot \min(\text{Core EMAWP, Jacket MAWP}) \cdot \text{Stress Ratio}$$
$$= 1.3 \cdot \min(689.493, 18.145) \cdot 1.0$$
$$= 23.589 \text{ N/mm}^2$$

Note:

For vessels/jackets constructed of different materials, be sure to check the stresses due to differential thermal expansion, especially for vessels at moderate to high temperatures.

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 Jacket/VB Calcs: Jacket Step: 4 3:25pm Apr 10,2026

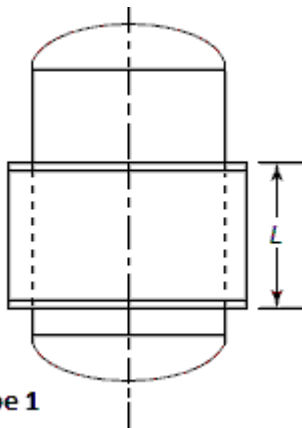
Jacket Design per ASME VIII, Division 2, 4.11.2: Jacket

Design per: Figure 4.11.1 Type 1
 Attachment per: Table 4.11.1, Detail 6, Figure: (a)

Input Values:

Jacket Design Internal Pressure	Pj	18.00	N/mm ²
Jacket Design Temperature (Internal)		302	C
Jacket Design External Pressure	Pje	0.00	N/mm ²
Jacket Design Temperature (External)		150	C
Static fluid Pressure in this Jacket	pStatic	0.00	N/mm ²
Closure Bar Material		SA-387 11	
Closure Bar Design Stress	[S or f]	147988.73	kPa
Closure Bar Thickness New	tc	210.0000	mm
Closure Bar Corrosion Allowance	cc	3.0000	mm
Inner Shell Outside Diameter	Dso	2562.000	mm
Inner Shell Thickness New	ts	156.0000	mm
Corroded Local Shell Thickness at Closure	ts,c	153.0000	mm
Inner Shell Corrosion Allow. Inside	csi	3.0000	mm
Inner Shell Corrosion Allow. Outside	cso	0.0000	mm
Inner Shell Vacuum Pressure		0.0000	N/mm ²
Jacket Material		SA-387 11	
Jacket Design Stress	Sj	147988.73	kPa
Jacket Inside Diameter New	Dji	2820.000	mm
Jacket Thickness New	tj	200.0000	mm
Jacket Corrosion Allowance Inside	cji	3.0000	mm
Jacket Length	L	1600.00	mm
Jacket Design Length for External Pressure	SL	1600.00	mm
Specific Gravity of contents in Jacket	sg	0.000	
Jacket Long. Weld Joint Efficiency	E	1.00	

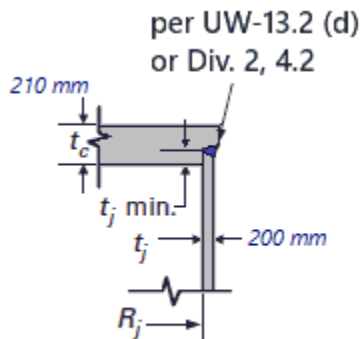
This has been specified as a vapor/distribution belt.



Type 1



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 Jacket/VB Calcs: Jacket Step: 4 3:25pm Apr 10, 2026



Div. 2 Detail 6 Fig. (a)

Results:

Jacket Inside Radius [Rj]:

$$= (D_{ji} + 2 \cdot c_i) / 2$$

$$= (2820.0 + 2 \cdot 3.0) / 2$$

$$= 1413.000 \text{ mm}$$

Jacket Corroded Thickness [tjc]:

$$= t_j - c_{ji}$$

$$= 200.0 - 3.0$$

$$= 197.000 \text{ mm}$$

Shell Outside Radius [Rs]:

$$= (D_{so} - 2 \cdot c_{so}) / 2$$

$$= (2562.0 - 2 \cdot 0.0) / 2$$

$$= 1281.000 \text{ mm}$$

Shell Corroded Thickness [tsc]:

$$= t_s - c_{si} - c_{so}$$

$$= 156.0 - 3.0 - 0.0$$

$$= 153.000 \text{ mm}$$

Jacket Space Width [j]:

$$= (D_{ji} + 2 \cdot c_{ji} - D_{so} + 2 \cdot c_{so}) / 2$$

$$= (2820.0 + 2 \cdot 3.0 - 2562.0 + 2 \cdot 0.0) / 2$$

$$= 132.000 \text{ mm}$$

Jacket Thickness Calculations for Cylinder:

Longitudinal Joint: User Defined

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (18.002 \cdot 1413.0) / (147988.73 \cdot 1.0 - 0.6 \cdot 18.002)$$

$$= 185.4073 + 3.0000 = 188.4073 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 N/mm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (147988.73 \cdot 1.0 \cdot 197.0) / (1413.0 + 0.6 \cdot 197.0)$$

$$= 19 - 0 = 19 \text{ N/mm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (147988.73 \cdot 1.0 \cdot 200.0) / (1410.0 + 0.6 \cdot 200.0)$$

$$= 19 \text{ N/mm}^2$$



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 Jacket/VB Calcs: Jacket Step: 4 3:25pm Apr 10,2026

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (18.002 \cdot (1413.0 + 0.6 \cdot 197.0)) / (1.0 \cdot 197.0)$$

$$= 139915.734 \text{ kPa}$$

Jacket Minimum Design Metal Temperature Calculations:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Required Closure Thickness [trc]:

$$= 1.414 \cdot \sqrt{(P_j \cdot R_s \cdot j / S)} + c$$

$$= 1.414 \cdot \sqrt{(18.002 \cdot 1281.0 \cdot 132.0 / 147988.734)} + 3.0$$

$$= 205.790 \text{ mm}$$

Maximum Allowable Working Pressure, Closure Bar [CBMawp]:

$$= S / (j \cdot R) \cdot ((t_r - c) / 1.414)^2 - p_{\text{Static}}$$

$$= 147988.734 / (132.0 \cdot 1281.0) \cdot ((210.0 - 3.0) / 1.414)^2 - 0.001$$

$$= 18.756 \text{ N/mm}^2$$

Forming Strain/Elongation Results:

Jacket Forming Strain, Cylinders formed from Plate ($50 \cdot t_{\text{nom}} / R_f(1 - R_f / R_o)$): 6.623 %

Note: Please check Code requirements as the computed elongation is > 5%.

External Pressure results for the Outer Jacket Cylinder:

Maximum External Pressure per Code Case 2286:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 1600.0 / \sqrt{(1610.0 \cdot 197.0)}$$

$$= 2.841$$

Compute the Factor [Ch]:

Since $M_x > 1.5$ and $M_x < 13$, $Ch = 0.92 / (M_x - 0.579)$

$$= 0.92 / (2.841 - 0.579)$$

$$= 0.4067$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.407 \cdot 196439(197.0 / 3220.0)$$

$$= 7820338.000 \text{ kPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 7820338.0 / 245665.4$$

$$= 31.833$$

Predicted Buckling Stress [Fic]:

Since $Fr \geq 2.439$, $Fic = F_y$

$$= 245665.391 \text{ kPa}$$

Factor of Safety [FS]:

Since $Fic [245665.39] \geq F_y [245665.39]$, $FS = 1.667$

Allowable Circumferential Compressive Stress [Fha]:

Since $Fr > 2.439$, $Fha = F_y / FS$

$$= 245665.391 / 1.667$$

$$= 147369.750 \text{ kPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha(t / D_o)$$



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Jacket/VB Calcs: Jacket

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$$= 2 \cdot 147.378(197.0/3220.0)$$
$$= 18.033 \text{ N/mm}^2$$

Summary of Results (mm):

	Required	Actual	Result
Jacket Thk Internal	188.407	200.000	Passed
Closure Bar Thickness	205.790	210.000	Passed

MAWP Results for the Jacket/Vapor/Distribution Belt and Closure Bar:

Jacket Maximum Allowable Working Pressure	19.04	N/mm ²
Closure Bar Maximum Allowable Working Pressure	18.76	N/mm ²
Overall MAWP of Jacket Components	18.76	N/mm ²

Jacket Cylinder External Maximum Allowable Working Pressure:

Jacket Ext. Maximum Allowable Working Pressure	18.03	N/mm ²
--	-------	-------------------

Jacket Hydrotest Pressure Calculations:

Jacket Hydrotest Pressure per UG-99(b) Note 35:

$$= 1.3 \cdot P \cdot \text{Stress Ratio}$$
$$= 1.3 \cdot 18.001 \cdot 1.0$$
$$= 23.401 \text{ N/mm}^2$$

Jacket Hydrotest Pressure per UG-99(b):

$$= 1.3 \cdot \min(\text{Core EMAWP, Jacket MAWP}) \cdot \text{Stress Ratio}$$
$$= 1.3 \cdot \min(689.493, 18.756) \cdot 1.0$$
$$= 24.383 \text{ N/mm}^2$$

Note:

For vessels/jackets constructed of different materials, be sure to check the stresses due to differential thermal expansion, especially for vessels at moderate to high temperatures.

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 Internal Pressure Calculations: Step: 5 3:25pm Apr 10,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd N/mm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kPa	Element#
channel sphere-a		18.001	90	3	2250	147989	1
Channel - rear		18.002	156	3	2250	147989	2
shell		18.002	156	3	2250	147989	3
shell cone -rea		18.001	156	3	2250	147989	4
Shell-2		18.002	156	3	2250	147989	5
shell cone- fro		18.001	156	3	2250	147989	6
shell- front		18.002	156	3	2250	147989	7
Channel front		18.002	156	3	2250	147989	8
Channel sphere-o		18.001	90	3	2250	147989	9

Element Required Thickness and MAWP :

From	To	Design Pressure N/mm ²	M.A.W.P. Corroded N/mm ²	M.A.P. New & Cold N/mm ²	Minimum Thickness mm	Required Thickness mm	Element#
channel sphere-a		18.001	18.6551	19.4733	75	72.4445	1
Channel - rear		18.001	18.5625	18.946	156	151.008	2
shell		18.001	18.5624	18.946	156	151.009	3
shell cone -rea		18.001	18.5634	18.946	156	151.001	4
Shell-2		18.001	18.5624	18.946	156	151.009	5
shell cone- fro		18.001	18.5634	18.946	156	151.001	6
shell- front		18.001	18.5622	18.946	156	151.011	7
Channel front		18.001	18.5625	18.946	156	151.008	8
Channel sphere-o		18.001	18.6551	19.4733	75	72.4445	9

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:
 For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 14 N/mm², < design pressure: 18.001 N/mm²
 Shell Side MAPnc = 18 N/mm²
 Channel Side MAWP = 17 N/mm², < design pressure: 18.001 N/mm²
 Channel Side MAPnc = 19 N/mm²

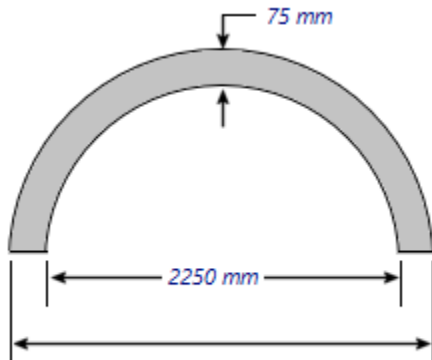
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: channel sphere-ar

Spherical Head Design Information:			
Design Pressure P	18.001 N/mm ²	Int. Design Temperature	275 °C
Material	SA-387 11	Ext. Design Temperature	82 °C
Length	0.258 m	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	147988.7 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	147988.7 kPa
Inside Diameter Di	2250.000 mm	Actual Stress	142800.0 kPa
Spec. Min. Thk t,e, or sa	75.0000 mm	Spec. Nominal Thk.	90.000 mm
Surface Area	11.2 m ²	SG of Contents	0.000
Element Volume	3.9 m ³	Weight of Contents	0.0 N
Empty Weight	71834.6 N	Operating Weight	71834.6 N
		Comp. All. Ope. Stress	115075.3 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.125 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.062 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (18.001 \cdot 1128.0) / (2 \cdot 147988.73 \cdot 1.0 - 0.2 \cdot 18.001)$$

$$= 69.4445 + 3.0000 = 72.4445 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 147988.73 \cdot 1.0 \cdot 72.0) / (1128.0 + 0.2 \cdot 72.0)$$

$$= 19 \text{ N/mm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 147988.73 \cdot 1.0 \cdot 75.0) / (1125.0 + 0.2 \cdot 75.0)$$



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 Element: channel sphere- Step: 6 3:25pm Apr 10,2026

= 19 N/mm²

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (18.001 \cdot (1128.0 + 0.2 \cdot 72.0)) / (2 \cdot 1.0 \cdot 72.0)$$

$$= 142800.031 \text{ kPa}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.769 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Spherical Portion:

Govrn. thk, tg = 75.0, tr = 63.895, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.887$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	44 °C
Min Metal Temp. at Required thickness (UCS 66.1)	38 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	16.579 N/mm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	75.000 mm	Basic MDMT	44 °C
Stress/Thk. Ratio	0.887	Temperature Difference	6 °C
UCS-66(c) red.	No	Computed Min. Temp.	38 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

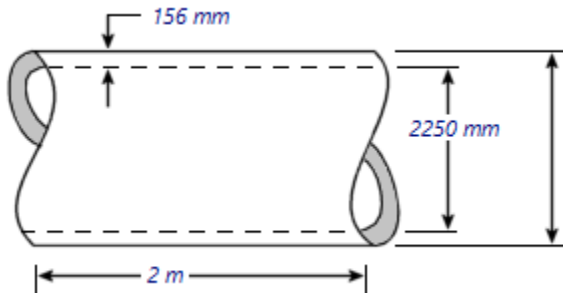
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.125 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.062 mm

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Channel - rear

Cylindrical Shell Design Information:			
Design Pressure P	18.002 N/mm ²	Int. Design Temperature	275 °C
Material	SA-387 11	Ext. Design Temperature	82 °C
Length	2.000 m	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	147988.7 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	147988.7 kPa
Inside Diameter Di	2250.000 mm	Actual Stress	143512.4 kPa
Spec. Min. Thk t,e, or sa	156.0000 mm	Spec. Nominal Thk.	156.000 mm
Surface Area	16.1 m ²	SG of Contents	0.052
Element Volume	7.8 m ³	Weight of Contents	3082.7 N
Empty Weight	179234.5 N	Operating Weight	182317.2 N
		Comp. All. Ope. Stress	126519.3 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (18.002 \cdot 1128.0) / (147988.73 \cdot 1.0 - 0.6 \cdot 18.002) \\
 &= 148.0078 + 3.0000 = 151.0078 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 N/mm²

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147988.73 \cdot 1.0 \cdot 153.0) / (1128.0 + 0.6 \cdot 153.0) \\
 &= 19 - 0.001 = 19 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147988.73 \cdot 1.0 \cdot 156.0) / (1125.0 + 0.6 \cdot 156.0) \\
 &= 19 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (18.002 \cdot (1128.0 + 0.6 \cdot 153.0)) / (1.0 \cdot 153.0) \\
 &= 143512.438 \text{ kPa}
 \end{aligned}$$



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Element: Channel - rear Step: 7 3:25pm Apr 10,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 6.484 %
Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Warning:
The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

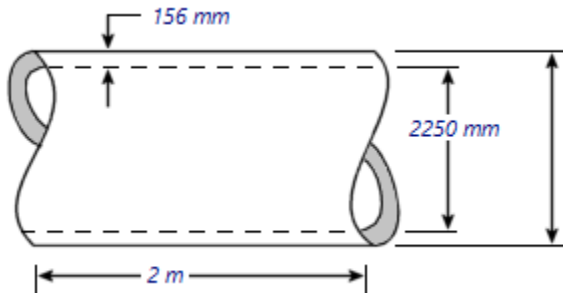
MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	16.579 N/mm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	156.000 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: shell

Cylindrical Shell Design Information:			
Design Pressure P	18.002 N/mm ²	Int. Design Temperature	302 °C
Material	SA-387 11	Ext. Design Temperature	150 °C
Length	2.000 m	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	147988.7 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	147988.7 kPa
Inside Diameter Di	2250.000 mm	Actual Stress	143513.8 kPa
Spec. Min. Thk t,e, or sa	156.0000 mm	Spec. Nominal Thk.	156.000 mm
Surface Area	16.1 m ²	SG of Contents	0.063
Element Volume	5.9 m ³	Weight of Contents	3700.0 N
Empty Weight	179234.5 N	Operating Weight	182934.5 N
		Comp. All. Ope. Stress	121128.6 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (18.002 \cdot 1128.0) / (147988.73 \cdot 1.0 - 0.6 \cdot 18.002) \\
 &= 148.0093 + 3.0000 = 151.0093 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 N/mm²

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147988.73 \cdot 1.0 \cdot 153.0) / (1128.0 + 0.6 \cdot 153.0) \\
 &= 19 - 0.001 = 19 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147988.73 \cdot 1.0 \cdot 156.0) / (1125.0 + 0.6 \cdot 156.0) \\
 &= 19 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (18.002 \cdot (1128.0 + 0.6 \cdot 153.0)) / (1.0 \cdot 153.0) \\
 &= 143513.797 \text{ kPa}
 \end{aligned}$$



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Element: shell Step: 8 3:25pm Apr 10,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 6.484 %
Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Warning:
The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

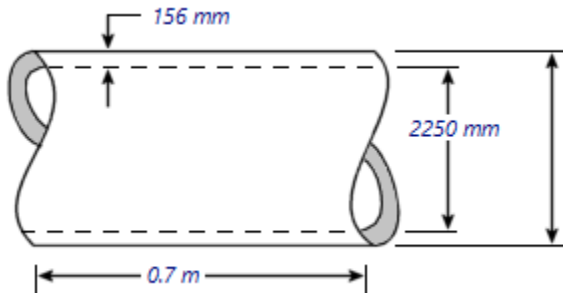
MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	14.494 N/mm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	156.000 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: shell cone -rea

Cylindrical Shell Design Information:			
Design Pressure P	18.001 N/mm ²	Int. Design Temperature	302 °C
Material	SA-387 11	Ext. Design Temperature	150 °C
Length	0.700 m	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	147988.7 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	147988.7 kPa
Inside Diameter Di	2250.000 mm	Actual Stress	143505.9 kPa
Spec. Min. Thk t,e, or sa	156.0000 mm	Spec. Nominal Thk.	156.000 mm
Surface Area	5.6 m ²	SG of Contents	0.000
Element Volume	2.0 m ³	Weight of Contents	0.0 N
Empty Weight	62732.1 N	Operating Weight	62732.1 N
		Comp. All. Ope. Stress	121128.6 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (18.001 \cdot 1128.0) / (147988.73 \cdot 1.0 - 0.6 \cdot 18.001) \\
 &= 148.0005 + 3.0000 = 151.0005 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147988.73 \cdot 1.0 \cdot 153.0) / (1128.0 + 0.6 \cdot 153.0) \\
 &= 19 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147988.73 \cdot 1.0 \cdot 156.0) / (1125.0 + 0.6 \cdot 156.0) \\
 &= 19 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (18.001 \cdot (1128.0 + 0.6 \cdot 153.0)) / (1.0 \cdot 153.0) \\
 &= 143505.906 \text{ kPa}
 \end{aligned}$$

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 6.484 %



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Element: shell cone -rea Step: 9 3:25pm Apr 10,2026

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	14.494 N/mm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	156.000 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

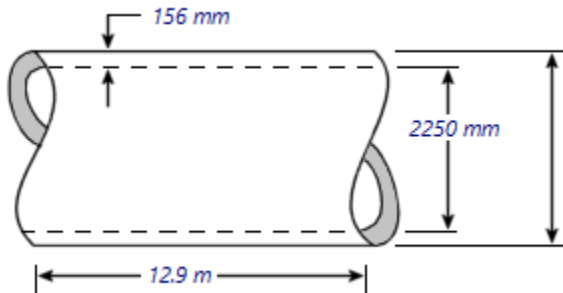
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 FileName : 121-C_NVS_MANWAY+JACKET_Final Page 38 of 179
 Element: Shell-2 Step: 10 3:25pm Apr 10,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:			
Design Pressure P	18.002 N/mm ²	Int. Design Temperature	302 °C
Material	SA-387 11	Ext. Design Temperature	150 °C
Length	12.880 m	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	147988.7 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	147988.7 kPa
Inside Diameter Di	2250.000 mm	Actual Stress	143513.8 kPa
Spec. Min. Thk t,e, or sa	156.0000 mm	Spec. Nominal Thk.	156.000 mm
Surface Area	103.6 m ²	SG of Contents	0.063
Element Volume	37.7 m ³	Weight of Contents	23827.9 N
Empty Weight	1154270.4 N	Operating Weight	1178098.2 N
		Comp. All. Ope. Stress	121128.6 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (18.002 \cdot 1128.0) / (147988.73 \cdot 1.0 - 0.6 \cdot 18.002) \\
 &= 148.0093 + 3.0000 = 151.0093 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 N/mm²

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147988.73 \cdot 1.0 \cdot 153.0) / (1128.0 + 0.6 \cdot 153.0) \\
 &= 19 - 0.001 = 19 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147988.73 \cdot 1.0 \cdot 156.0) / (1125.0 + 0.6 \cdot 156.0) \\
 &= 19 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (18.002 \cdot (1128.0 + 0.6 \cdot 153.0)) / (1.0 \cdot 153.0) \\
 &= 143513.797 \text{ kPa}
 \end{aligned}$$



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FileName : 121-C_NVS_MANWAY+JACKET_Final Page 39 of 179
Element: Shell-2 Step: 10 3:25pm Apr 10,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 6.484 %
Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Warning:
The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	14.494 N/mm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	156.000 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

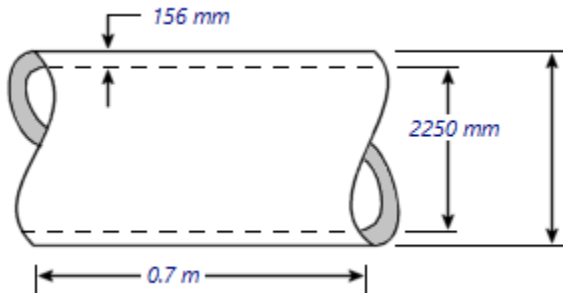
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 FileName : 121-C_NVS_MANWAY+JACKET_Final Page 40 of 179
 Element: shell cone- fro Step: 11 3:25pm Apr 10,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: shell cone- fro

Cylindrical Shell Design Information:			
Design Pressure P	18.001 N/mm ²	Int. Design Temperature	302 °C
Material	SA-387 11	Ext. Design Temperature	150 °C
Length	0.700 m	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	147988.7 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	147988.7 kPa
Inside Diameter Di	2250.000 mm	Actual Stress	143505.9 kPa
Spec. Min. Thk t,e, or sa	156.0000 mm	Spec. Nominal Thk.	156.000 mm
Surface Area	5.6 m ²	SG of Contents	0.000
Element Volume	2.0 m ³	Weight of Contents	0.0 N
Empty Weight	62732.1 N	Operating Weight	62732.1 N
		Comp. All. Ope. Stress	121128.6 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= (18.001 \cdot 1128.0) / (147988.73 \cdot 1.0 - 0.6 \cdot 18.001)$
 $= 148.0005 + 3.0000 = 151.0005 \text{ mm}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (147988.73 \cdot 1.0 \cdot 153.0) / (1128.0 + 0.6 \cdot 153.0)$
 $= 19 \text{ N/mm}^2$

Maximum Allowable Pressure, New and Cold [MAPNC]:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (147988.73 \cdot 1.0 \cdot 156.0) / (1125.0 + 0.6 \cdot 156.0)$
 $= 19 \text{ N/mm}^2$

Actual stress at given pressure and thickness, corroded [Sact]:
 $= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$
 $= (18.001 \cdot (1128.0 + 0.6 \cdot 153.0)) / (1.0 \cdot 153.0)$
 $= 143505.906 \text{ kPa}$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 6.484 %



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FileName : 121-C_NVS_MANWAY+JACKET_Final Page 41 of 179
Element: shell cone- fro Step: 11 3:25pm Apr 10,2026

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

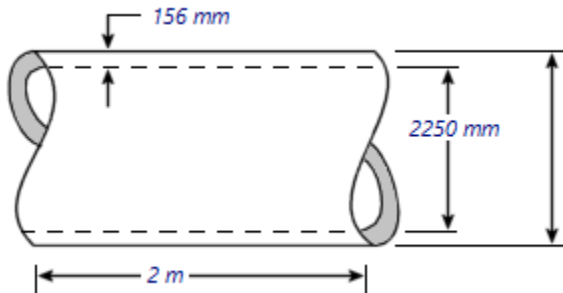
MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	14.494 N/mm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	156.000 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: shell- front

Cylindrical Shell Design Information:			
Design Pressure P	18.002 N/mm ²	Int. Design Temperature	302 °C
Material	SA-387 11	Ext. Design Temperature	150 °C
Length	2.000 m	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	147988.7 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	147988.7 kPa
Inside Diameter Di	2250.000 mm	Actual Stress	143515.3 kPa
Spec. Min. Thk t,e, or sa	156.0000 mm	Spec. Nominal Thk.	156.000 mm
Surface Area	16.1 m ²	SG of Contents	0.063
Element Volume	5.9 m ³	Weight of Contents	3700.0 N
Empty Weight	179234.5 N	Operating Weight	182934.5 N
		Comp. All. Ope. Stress	121128.6 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (18.002 \cdot 1128.0) / (147988.73 \cdot 1.0 - 0.6 \cdot 18.002) \\
 &= 148.0110 + 3.0000 = 151.0109 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 N/mm²

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147988.73 \cdot 1.0 \cdot 153.0) / (1128.0 + 0.6 \cdot 153.0) \\
 &= 19 - 0 = 19 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147988.73 \cdot 1.0 \cdot 156.0) / (1125.0 + 0.6 \cdot 156.0) \\
 &= 19 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (18.002 \cdot (1128.0 + 0.6 \cdot 153.0)) / (1.0 \cdot 153.0) \\
 &= 143515.266 \text{ kPa}
 \end{aligned}$$



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FileName : 121-C_NVS_MANWAY+JACKET_Final Page 43 of 179
Element: shell- front Step: 12 3:25pm Apr 10,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 6.484 %
Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Warning:
The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

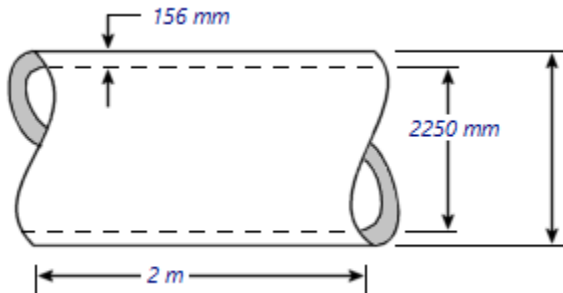
MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	14.494 N/mm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	156.000 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Channel front

Cylindrical Shell Design Information:			
Design Pressure P	18.002 N/mm ²	Int. Design Temperature	275 °C
Material	SA-387 11	Ext. Design Temperature	82 °C
Length	2.000 m	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	147988.7 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	147988.7 kPa
Inside Diameter Di	2250.000 mm	Actual Stress	143512.4 kPa
Spec. Min. Thk t,e, or sa	156.0000 mm	Spec. Nominal Thk.	156.000 mm
Surface Area	16.1 m ²	SG of Contents	0.052
Element Volume	7.8 m ³	Weight of Contents	3082.7 N
Empty Weight	179234.5 N	Operating Weight	182317.2 N
		Comp. All. Ope. Stress	121128.6 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (18.002 \cdot 1128.0) / (147988.73 \cdot 1.0 - 0.6 \cdot 18.002) \\
 &= 148.0078 + 3.0000 = 151.0078 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 N/mm²

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147988.73 \cdot 1.0 \cdot 153.0) / (1128.0 + 0.6 \cdot 153.0) \\
 &= 19 - 0.001 = 19 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147988.73 \cdot 1.0 \cdot 156.0) / (1125.0 + 0.6 \cdot 156.0) \\
 &= 19 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (18.002 \cdot (1128.0 + 0.6 \cdot 153.0)) / (1.0 \cdot 153.0) \\
 &= 143512.438 \text{ kPa}
 \end{aligned}$$



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Element: Channel front Step: 13 3:25pm Apr 10,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 6.484 %
Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Warning:
The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

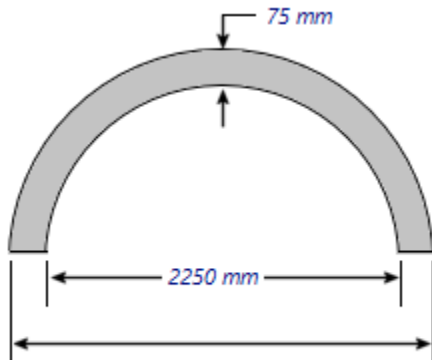
MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	16.579 N/mm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	156.000 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Channel sphere-ont

Spherical Head Design Information:			
Design Pressure P	18.001 N/mm ²	Int. Design Temperature	275 °C
Material	SA-387 11	Ext. Design Temperature	82 °C
Length	0.258 m	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	147988.7 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	147988.7 kPa
Inside Diameter Di	2250.000 mm	Actual Stress	142800.0 kPa
Spec. Min. Thk t,e, or sa	75.0000 mm	Spec. Nominal Thk.	90.000 mm
Surface Area	11.2 m ²	SG of Contents	0.000
Element Volume	3.9 m ³	Weight of Contents	0.0 N
Empty Weight	71834.6 N	Operating Weight	71834.6 N
		Comp. All. Ope. Stress	126519.3 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.125 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.062 mm

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\
 &= (18.001 \cdot 1128.0) / (2 \cdot 147988.73 \cdot 1.0 - 0.2 \cdot 18.001) \\
 &= 69.4445 + 3.0000 = 72.4445 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)} \\
 &= (2 \cdot 147988.73 \cdot 1.0 \cdot 72.0) / (1128.0 + 0.2 \cdot 72.0) \\
 &= 19 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)} \\
 &= (2 \cdot 147988.73 \cdot 1.0 \cdot 75.0) / (1125.0 + 0.2 \cdot 75.0)
 \end{aligned}$$



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FileName : 121-C_NVS_MANWAY+JACKET_Final Page 47 of 179
Element: Channel sphere- Step: 14 3:25pm Apr 10,2026

= 19 N/mm²

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned} &= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t) \\ &= (18.001 \cdot (1128.0 + 0.2 \cdot 72.0)) / (2 \cdot 1.0 \cdot 72.0) \\ &= 142800.031 \text{ kPa} \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.769 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Spherical Portion:

Govrn. thk, tg = 75.0, tr = 63.895, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.887$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	44 °C
Min Metal Temp. at Required thickness (UCS 66.1)	38 °C

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	16.579 N/mm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	75.000 mm	Basic MDMT	44 °C
Stress/Thk. Ratio	0.887	Temperature Difference	6 °C
UCS-66(c) red.	No	Computed Min. Temp.	38 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.125 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.062 mm

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	18.843 N/mm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	23.401 N/mm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	23.964 N/mm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	15.944 N/mm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	25.741 N/mm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	25.741 N/mm ²
App 27-4	= M.A.W.P.	14.494 N/mm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	21.552 N/mm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	23.401 N/mm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	24.608 N/mm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	18.236 N/mm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	25.741 N/mm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	25.741 N/mm ²
App 27-4	= M.A.W.P.	16.579 N/mm ²

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

$$= \text{Test Factor} \cdot \text{Design Pressure} \cdot \text{Stress Ratio}$$

$$= 1.3 \cdot 18.001 \cdot 1.0$$

$$= 23.401 \text{ N/mm}^2$$

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

$$= \text{Test Factor} \cdot \text{Design Pressure} \cdot \text{Stress Ratio}$$

$$= 1.3 \cdot 18.001 \cdot 1.0$$

$$= 23.401 \text{ N/mm}^2$$

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kPa & N/mm²):

From To	Stress	Allowable	Ratio	Pressure
channel sphere-ar	178007.6	278978.8	0.638	23.42
Channel - rear	182962.3	278978.8	0.656	23.42
shell	182962.3	278978.8	0.656	23.42
shell cone -rea	182962.3	278978.8	0.656	23.42
Shell-2	182962.3	278978.8	0.656	23.42
shell cone- fro	182962.3	278978.8	0.656	23.42
shell- front	182962.3	278978.8	0.656	23.42
Channel front	182962.3	278978.8	0.656	23.42
Channel sphere-ont	178007.6	278978.8	0.638	23.42

Stress ratios for Nozzle and Pad Materials (kPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Manway-1	Nozzle	147988.73	147988.73	1.000
N4- Tube outlet	Nozzle	147988.73	147988.73	1.000
N1- Shell inlet	Nozzle	147988.73	147988.73	1.000
N2- Shell outle	Nozzle	147988.73	147988.73	1.000
N3- tube inlet	Nozzle	147988.73	147988.73	1.000
Mayway-2	Nozzle	147988.73	147988.73	1.000



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Shop Pressure Test Results: Step: 15 3:25pm Apr 10,2026

Minimum 1.000

Stress ratios for Jacket/Vapor Belt Materials:

Description	Cylinder/Closure	Ambient	Operating	Ratio
Jacket	Cylinder	147988.73	147988.73	1.000
Jacket	Closure	147988.73	147988.73	1.000
Jacket	Cylinder	147988.73	147988.73	1.000
Jacket	Closure	147988.73	147988.73	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kPa):

Description	Ambient	Operating	Ratio
channel sphere-ar	147988.73	147988.73	1.000
Channel - rear	147988.73	147988.73	1.000
shell	147988.73	147988.73	1.000
shell cone -rea	147988.73	147988.73	1.000
Shell-2	147988.73	147988.73	1.000
shell cone- fro	147988.73	147988.73	1.000
shell- front	147988.73	147988.73	1.000
Channel front	147988.73	147988.73	1.000
Channel sphere-ont	147988.73	147988.73	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (kPa):

Description	Ambient	Operating	Ratio
Tube Material	117991.02	110493.01	1.068
Tubesheet Material	147988.73	147988.73	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kPa):

Description	Ambient	Operating	Ratio
Manway-1	110590.50	278978.75	0.396
N4- Tube outlet	110590.50	278978.75	0.396
N1- Shell inlet	115121.46	278978.75	0.413
N2- Shell outle	115121.46	278978.75	0.413
N3- tube inlet	110590.50	278978.75	0.396
Mayway-2	110590.50	278978.75	0.396

Elements Suitable for Test Pressure.

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Ring MDMT Calcs: Step: 16 3:25pm Apr 10, 2026

MDMT Calculations for Stiffening Rings:

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: shell

Shell Test Pressure		23.40 N/mm ²
Shell Thickness	ts	156.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2250.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K11789
Shell Allowable Stress at Temperature	Ss	147988.73 kPa
Shell Allowable Stress at Ambient		147988.73 kPa

Channel Data:

Channel Description: Channel - rear

Channel Type:		Cylinder
Channel Test Pressure		23.40 N/mm ²
Channel Thickness	tc	156.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2250.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	147988.73 kPa
Channel Allowable Stress at Ambient		147988.73 kPa

Tube Data:

Number of Tube Holes	Nt	3439
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	19000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		18280.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 T11
Tube Material UNS Number		K11597
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		117991.02 kPa
Tube Allowable Stress At Ambient		117991.02 kPa
Tube Yield Stress At design Temperature	Syt	204984.41 kPa
Tube Pitch (Center to Center Spacing)	P	35.0000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	3.0000 mm
Groove Weld Leg	ag	3.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	1078.200 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	100.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	3.6 m ²



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Perimeter of Tube Layout Cp 6982.6 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-336 F11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K11572
Tubesheet Allowable Stress at Temperature	S	147988.73 kPa
Tubesheet Allowable Stress at Ambient	Tt	147988.73 kPa
Thickness of Tubesheet	h	360.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	2562.000 mm

Area of the Untubed Lanes AL 0.000 m²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 795.000 mm
Tube End condition corresponding to Span (l) k 0.80

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
Perform Differential Pressure Design Yes
Differential Design Pressure Pdif 2.4501 N/mm²
Run Multiple Load Cases Yes

Mean Shell Metal Temp. along Shell len. Tsm 21.11 °C
Mean Tube Metal Temp. along Tube length Ttm 21.11 °C
Junction Stress Reduction option Perform Plastic Calculation

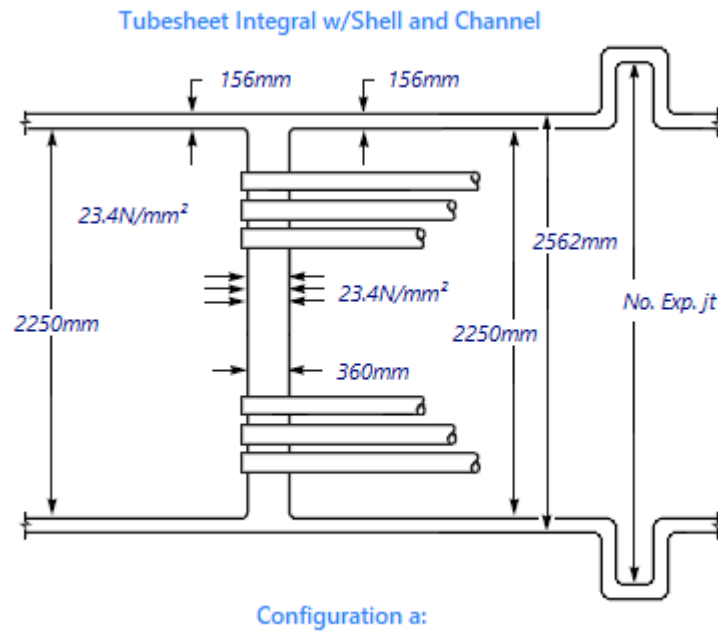
Tubesheet Integral with Both
Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



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ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20409E+06 N/mm ²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20409E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20409E+06 N/mm ²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20409E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20409E+06 N/mm ²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20409E+06 N/mm ²
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20409E+06 N/mm ²
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20409E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20409E+06 N/mm ²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20409E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20409E+06 N/mm ²

Note:



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The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Differential pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (2.45 \cdot (19.05/2 - 0)) / (117991.02 \cdot 1.0 + 0.4 \cdot 2.45)$
 $= 0.1962 + 0.0000 = 0.1962 \text{ mm}$

Tube Required Thickness under External Pressure (Differential pressure) :

Determine Parameter [Mx]:
 $= L / \sqrt{(Ro \cdot t)}$
 $= 18280.0 / \sqrt{(9.525 \cdot 2.11)}$
 $= 4077.580$

Compute the Factor [Ch]:
 Since $Mx > 2 \cdot (Do/t)^{0.94}$, $Ch = 0.55 / (Do/t)$
 $= 0.55 \cdot (19.05/2.11)$
 $= 0.0609$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:
 $= 1.6 \cdot Ch \cdot Ey(t / Do)$
 $= 1.6 \cdot 0.061 \cdot 204092 (2.11/19.05)$
 $= 2203223.000 \text{ kPa}$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:
 $= Fhe / Fy$
 $= 2203223.0/204984.4$
 $= 10.748$

Predicted Buckling Stress [Fic]:
 Since $Fr \geq 2.439$, $Fic = Fy$
 $= 204984.406 \text{ kPa}$

Factor of Safety [FS]:
 Since $Fic [204984.41] \geq Fy [204984.41]$, $FS = 1.667$

Allowable Circumferential Compressive Stress [Fha]:
 Since $Fr > 2.439$, $Fha = Fy/FS$
 $= 204984.406/1.667$
 $= 122966.039 \text{ kPa}$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot Fha(t / Do)$
 $= 2 \cdot 122.973 (2.11/19.05)$
 $= 27.241 \text{ N/mm}^2$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.4552 mm
Allowable Internal Pressure at Corroded thickness	28.68 N/mm ²
Required Internal Design Pressure	2.45 N/mm ²
Allowable External Pressure at Corroded thickness	27.24 N/mm ²
Required External Design Pressure	2.45 N/mm ²
Required Thickness due to Shell Side pressure	0.4552 mm

Note: The tubes were evaluated using the specified differential pressure.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Note: Shell/Channel Design Pressures modified by Input differential pressure:
 Pdif = 2.450 Ps = 23.401 Pt = 20.951 N/mm²



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Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 100.0/360.0$$

$$= 0.2778 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole diameter $[d^*]$:

$$= \max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(19.05 - 2 \cdot 2.11 (204092/204092) \cdot (0.278); 19.05 - 2 \cdot 2.11)$$

$$= 18.1154 \text{ mm}$$

Equivalent outer tube limit circle diameter $[Do]$:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 1078.2 + 19.05$$

$$= 2175.450 \text{ mm}$$

Determine the basic ligament efficiency for shear $[\mu]$:

$$= (p - dt) / p$$

$$= (35.0 - 19.05) / 35.0$$

$$= 0.456$$

Equivalent outer tube limit radius $[ao]$:

$$= Do / 2$$

$$= 2175.45/2$$

$$= 1087.7250 \text{ mm}$$

Effective tube pitch $[p^*]$:

$$= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 35.0 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 1000248 ; 4 \cdot 2175.45 \cdot 35.0) / (\pi \cdot 2175.45^2)) }$$

$$= 35.0000 \text{ mm}$$

Effective ligament efficiency for bending $[\mu^*]$:

$$= (p^* - d^*) / p^*$$

$$= (35.0 - 18.1154) / 35.0$$

$$= 0.4824$$

Ratio $[ps]$:

$$= a_s / a_o$$

$$= 1125.0/1087.725$$

$$= 1.0343$$

Ratio $[pc]$:

$$= a_c / a_o$$

$$= 1125.0/1087.725$$

$$= 1.0343$$

Compute parameter $[xt]$:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2$$

$$= 1 - 3439 ((19.05 - 2 \cdot 2.11) / (2 \cdot 1087.725))^2$$

$$= 0.8402$$

Determine parameter $[xs]$:

$$= 1 - Nt(dt / (2 \cdot a_o))^2$$

$$= 1 - 3439 (19.05 / (2 \cdot 1087.725))^2$$

$$= 0.7363$$



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Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts(D_s + ts)E_s/L$$

$$= \pi \cdot 156.0 (2250.0 + 156.0)204092/18280.0$$

$$= 13163851.0000 \text{ N/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt(D_t - tt)E_t/L$$

$$= \pi \cdot 2.11 (19.05 - 2.11)204092/18280.0$$

$$= 1253.5989 \text{ N/mm}$$

Stiffness Factor [Ks,t]:

$$= K_s / (N_t \cdot K_t)$$

$$= 13163851 / (3439 \cdot 1253.599)$$

$$= 3.05346$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s/K_t)$$

$$= 1 / (1 + 13163851/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + ts) ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((2250.0 + 156.0) 156.0)$$

$$= 0.0030 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.003 \cdot 204092 \cdot 156.0^3 / (6(1 - 0.3^2))$$

$$= 421068448.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + ts) \cdot ks/h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2/2)$$

$$= 6(2250.0 + 156.0) \cdot 0.42107E+09/360.0^3 \cdot (1 + 360.0 \cdot 0.003 + (360.0 \cdot 0.003)^2/2)$$

$$= 343781.5000 \text{ N/mm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot ts) \cdot (1 - D_s \cdot \nu_s / (2(D_s + ts)))$$

$$= 2250.0^2 / (4 \cdot 204092 \cdot 156.0) \cdot (1 - 2250.0 \cdot 0.3 / (2(2250.0 + 156.0)))$$

$$= 0.34177E-04 \text{ mm/kPa}$$

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / ((D_c + tc) tc)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((2250.0 + 156.0) 156.0)$$

$$= 0.0030 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot tc^3 / (6(1 - \nu_c^2))$$

$$= 0.003 \cdot 204092 \cdot 156.0^3 / (6(1 - 0.3^2))$$

$$= 421068448.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + tc/\cos(\theta)) \cdot kc/h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta)/2 + (h \cdot \beta_c)^2/2 \cdot CqDelta)$$

$$= 6(2250.0 + 156.0/\cos(0)) \cdot 0.42107E+09/360.0^3 \cdot (1.0 + 360.0 \cdot 0.003 \cdot (1.0 + 1.0)/2 + (360.0 \cdot 0.003)^2/2 \cdot 1.0)$$



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$$= 343781.5 \text{ N/mm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot T_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 2250.0^2 / (4 \cdot 204092 \cdot 156.0) \cdot (1 - 2250.0 \cdot 0.3 / (2(2250.0 + 156.0)))$$

$$= 0.34177\text{E-}04 \text{ mm/kPa}$$

Channel coefficient [ζ_{etac}]:

$$\zeta_{etac} = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 10.285714 ; \mu^* = 0.482417 ; E^* = 106360 \text{ N/mm}^2$$

$$E^*/E = 0.521142 ; v^* = 0.300817$$

Note: As h/p (10.286) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$= ((24(1 - v^*)N_t \cdot E_t \cdot t_t(dt - t_t)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.301^2)3439 \cdot 204092 \cdot 2.11$$

$$(19.05 - 2.11)1087.725^2) / (106360 \cdot 18280.0 \cdot 360.0^3))^{1/4}$$

$$= 1.6348$$

Values from Table 4.18.3

$$Z_d = 0.334027 ; Z_v = 0.178792 ; Z_m = 0.728293$$

$$Z_a = 0.582124\text{E}+00 ; Z_w = 0.178792$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2562.0 / 2175.45$$

$$= 1.1777$$

Compute coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.3) / 106360 \cdot (343781.5 + 343781.5 +$$

$$204092 \cdot \ln(1.18))$$

$$= 4.7392$$

Compute parameter [ϕ]:

$$= (1 + v^*)^F$$

$$= (1 + 0.3008)^{4.7392}$$

$$= 6.1649$$

Compute parameter [Q_1]:

$$= (\psi_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.0343 - 1 - 6.1649 \cdot 0.1788) / (1 + 6.1649 \cdot 0.7283)$$

$$= -0.194535$$

Compute Parameter [Q_{z1}]:

$$= (Z_d + Q_1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.33403 + -0.19453 \cdot 0.17879) / 2 \cdot 1.63479^4$$

$$= 1.0687$$

Compute parameter [Q_{z2}]:

$$= (Z_v + Q_1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.17879 + -0.19453 \cdot 0.72829) / 2 \cdot 1.63479^4$$

$$= 0.1325$$

Compute parameter [U]:

$$= (Z_w + (\psi_s - 1)Z_m)X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.1788 + (1.0343 - 1)0.7283)1.63479^4 / (1 + 6.1649 \cdot 0.7283)$$

$$= 0.2651$$



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ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:
 = 0 (config a)

Compute Parameter [γ]:
 = 0.000 mm (For Pressure only cases)

Calculate parameter [ωs]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.0343 \cdot 0.42107E+09 \cdot 0.003 \cdot 0.342E-04 (1 + 360.0 \cdot 0.003)$$

$$= 91334.8125 \text{ mm}^2$$

Calculate parameter [ω*s]:

$$= a o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s$$

$$= 1087.725^2 (1.034^2 - 1) \cdot (1.034 - 1) / 4 - 91334.812$$

$$= -90628.1953 \text{ mm}^2$$

Parameter [ωc]:

$$= \rho_c \cdot k_c [\delta C (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \Theta + h \cdot C_q \Theta + h \cdot \beta C / 2)]$$

$$= 1.034 \cdot 0.42107E+09 [0.342E-04 (1.0 + 0.003 \cdot 1.0 \cdot 360.0) - 0 (1.0/360.0 + 1.0 \cdot 0.003/2)]$$

$$= 91334.8125 \text{ mm}^2$$

Parameter [ω*c]:

$$= a o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega_c$$

$$= 1087.72498^2 [(1.03427^2 + 1) \cdot (1.03427 - 1) / 4 - (1.03427 - 1) / 2] - 91334.81250$$

$$= -90628.1953 \text{ mm}^2$$

Pressure [P*S]:
 = 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:
 = 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/Kst \cdot (D_s/D_o)^2] v_s - [(\rho_s^2 - 1) / (J \cdot Kst)] - [(1 - J) / (2J \cdot Kst)] [(D_j^2 - (D_s)^2) / D_o^2] \}$$

$$= 23.401 \{ 0.736 + 2(1 - 0.736) 0.3 + [2/3.053 \cdot (2250.0/2175.45)^2] 0.3 - [(1.034^2 - 1) / (1.0 \cdot 3.053)] - [(1 - 1.0) / (2 \cdot 1.0 \cdot 3.053)] [(0^2 - (2250.0)^2) / 2175.45^2] \}$$

$$= 25.3175 \text{ N/mm}^2$$

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot Kst)] P_t$$

$$= [0.84 + 2(1 - 0.84) 0.3 + 1/(1 \cdot 3.053)] 20.951$$

$$= 26.4734 \text{ N/mm}^2$$

Pressure [Pγ]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a o^2)$$

$$= 3439 \cdot 1253.599 \cdot 0 / (3.142 \cdot 1087.725^2)$$

$$= 0.0000 \text{ N/mm}^2$$

Pressure [Pw]:

$$= -\gamma_b \cdot U \cdot W^* / (2 \cdot \pi \cdot a o^2)$$

$$= -0 \cdot 0.265 \cdot 0 / (2 \cdot \pi \cdot 1087.725^2)$$

$$= -0.0000 \text{ N/mm}^2$$

Calculate the pressure [Prim]:

$$= -(U/a o^2) \cdot (\omega^* s \cdot P_s - \omega^* c \cdot P_t)$$

$$= -(0.265/1087.725^2) \cdot (-90628.195 \cdot 23.401 - -90628.195 \cdot 20.951)$$

$$= 0.0498 \text{ N/mm}^2$$



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Calculate the pressure [Pw]:

$$= (U/ao^2) \cdot (\omega_s \cdot P^*s - \omega_c \cdot P^*c)$$

$$= (0.265/1087.725^2) \cdot (91334.8125 \cdot 0 - 91334.8125 \cdot 0)$$

$$= 0.0000 \text{ N/mm}^2$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (ps - 1) Qz2)) \cdot (P^*s - P^*t + P_y + P_\omega + P_w + P_{rim})$$

$$= 0.1000E+01 \cdot 3.053 / (1 + 1.0 \cdot 3.053 (1.069 + (1.034 - 1) 0.133)) \cdot (25.318 - 26.473 + 0 + 0 + 0 + 0.05)$$

$$= -0.7897 \text{ N/mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega^*s \cdot P^*s - \omega^*c \cdot P^*t) - (\omega_s \cdot P^*s - \omega_c \cdot P^*c)) + W^* \cdot y_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-90628.195 \cdot 23.401 - -90628.195 \cdot 20.951) - (91334.812 \cdot 0 - 91334.812 \cdot 0)) \cdot 1 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 6.1649 \cdot 0.72829)$$

$$= -40444.2969 \text{ N}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.195 + 2 \cdot -40444.297 / (-0.79 \cdot 1087.725^2)$$

$$= -0.10795$$

Fm Value from Table 4.18.3 = 0.138359

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu^*) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.1384 / 0.4824) \cdot (2 \cdot 1087.725 / (360.0 - 0))^2 \cdot -0.79$$

$$= -12404.6221 \text{ kPa}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot Sy = 1.5 \cdot 0.95 \cdot 309976.41 = 441716.34 \text{ kPa}, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu^*) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.0086 / 0.4824) \cdot (2 \cdot 1087.725 / (4.655 - 0))^2 \cdot -0.08$$

$$= -441706.8125 \text{ kPa}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + Cats + Catc$$

$$= 4.6552 + 0 + 0$$

$$= 4.6552 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:

$$= (2 \cdot \mu \cdot h / ao) \cdot (0.8 \cdot \beta T \cdot Sy)$$

$$= (2 \cdot 0.456 \cdot 360.0 / 1087.725) \cdot 235582.047$$

$$= 71063.7734 \text{ kPa}$$

Shear Stress is not required to be computed; [Pe] <= Taulimit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\text{Tubesheet thickness (Incl. Corr.)} = 3.8100 \text{ mm}$$

$$\text{Tubesheet Shear Stress} = 19409.5234 \text{ kPa}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(HreqB; HreqS)$$

$$= \max(4.6552; 3.81)$$



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= 4.6552 mm

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

Ftmin = 0.8425, Ftmax = 1.124

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((23.4 \cdot 0.7363 - 20.95 \cdot 0.8402) - (-0.79) \cdot 0.843) /$$

$$(0.8402 - 0.7363)$$

$$= 2816.5400 \text{ kPa}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((23.4 \cdot 0.7363 - 20.95 \cdot 0.8402) - (-0.79) \cdot 1.124) /$$

$$(0.8402 - 0.7363)$$

$$= 4955.2979 \text{ kPa}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 4955.298 \text{ kPa}$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= \beta_T \cdot S_{y,t} = 194735.1719 \text{ kPa}$$

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 4955.3 \cdot 0.0001$$

$$= 556.42 \text{ N}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (19.05 - 2.11) 2.11$$

$$= 0.000 \text{ m}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (19.05 - 2.11) 194735.172$$

$$= 21866.457 \text{ N}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 3.0 (19.05 + 0.67 \cdot 3.0) 194735.17$$

$$= 21866.4570 \text{ N}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 194735.172 / 194735.172$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 3.0 (19.05 + 0.67 \cdot 3.0) 194735.17$$

$$= 21258.1211 \text{ N}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 19.05)^2 + 1.07 \cdot 2.11 \cdot (19.05 - 2.11) 1.0 \cdot 1.0} - 0.75 \cdot 19.05 \right)$$

$$= 2.562 \text{ mm}$$



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Minimum Required Fillet Weld Leg [afr]: 1.2810 mm
 Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 21866.46 is >= tube strength 21866.46 N

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= \beta T \cdot S_{y,s} = 0.95 \cdot 309976.41 = 441716.34 \text{ kPa}$

Axial Membrane Stress in Shell $[\sigma_{s,m}]$:
 $= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 1087.725^2 / ((2250.0 + 156.0) 156.0) \cdot [-0.79 + (1.034^2 - 1)$
 $(23.4 - 20.95)] + 1125.0^2 \cdot 20.95 / ((2250.0 + 156.0) 156.0)$
 $= 68692.3047 \text{ kPa}$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 156.0 mm, the minimum Shell length adjacent to the tubesheet should be 1066.415 mm

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	12404.6	<= 441716.3 kPa	Pass
Tubesheet Shear Stress	2472.4	<= 235582.0 kPa	Pass
Maximum Tube Stress	4955.3	<= 194735.2 kPa	Pass
Maximum Force on any one Tube	556.4	<= 21866.5 N	Pass
Axial Membrane Stress in Shell	68692.3	<= 441716.3 kPa	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	4.655	360.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	3.000	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D3uc	4.655	...	12405	441716	2472	235582	Ps+Pt-Th D3	Ok
Max:	4.6552	...	mm	0.028		0.010	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+/-)Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx



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[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:		Fail
D3uc	68692	441716	...	...	:	...	...	...	...	:		Ok
Max RATIO	0.156			...	:		...		...	:		

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D3uc	4955	194735	...	...	556	21866	...	441716	...	441716	Ok
Max RATIO	0.025			...		0.025		...		...	

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	4.655	360.000	Ok
Tube thickness :	0.455	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	3.000	Ok
Tube-tubesheet groove weld leg :	1.281	3.000	Ok

Min Shell length of thk, (156.000) adj. to tubesheet: 1066.415 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -2.482 mm
 Due to Pressure + Thermal Shell Compresses by : -2.482 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.7000	...	...	...	...	...
2	-2.4816	...	...	...	...	...
3	-1.8549	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...



External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length m	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kPa
10	20	No Calc	2400	72	No Calc	No Calc
20	Tsht	2.63297	2562	153	No Calc	No Calc
Tsht	30	No Calc	...	153	No Calc	No Calc
30	Jckt	0.20003	2562	153	No Calc	No Calc
Jckt	Jckt	1.6	2562	153	No Calc	No Calc
Jckt	40	14.68	2562	153	No Calc	No Calc
40	50	14.68	2562	153	No Calc	No Calc
50	60	14.68	2562	153	No Calc	No Calc
60	70	14.68	2562	153	No Calc	No Calc
70	Jckt	14.68	2562	153	No Calc	No Calc
Jckt	Jckt	1.6	2562	153	No Calc	No Calc
Jckt	Tsht	0.19997	2562	153	No Calc	No Calc
Tsht	80	No Calc	...	153	No Calc	No Calc
80	90	2.63303	2562	153	No Calc	No Calc
90	100	No Calc	2400	72	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure N/mm ²	External M.A.W.P. N/mm ²
10	20	75	No Calc	...	17.7202
20	Tsht	156	No Calc	...	20.7543
Tsht	30	156	No Calc	...	No Calc
30	Jckt	156	No Calc	...	19.4892
Jckt	Jckt	156	No Calc	...	19.4892
Jckt	40	156	No Calc	...	18.6953
40	50	156	No Calc	...	18.6953
50	60	156	No Calc	...	18.6953
60	70	156	No Calc	...	18.6953
70	Jckt	156	No Calc	...	18.6953
Jckt	Jckt	156	No Calc	...	19.4892
Jckt	Tsht	156	No Calc	...	19.4892
Tsht	80	156	No Calc	...	No Calc
80	90	156	No Calc	...	20.7543
90	100	75	No Calc	...	17.7202

Minimum

18

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners m	Allowable Length Bet. Stiffeners m	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	Tsht	2.63297	No Calc	No Calc	No Calc
Tsht	30	No Calc	No Calc	No Calc	No Calc
30	Jckt	0.20003	No Calc	No Calc	No Calc
Jckt	Jckt	1.6	No Calc	No Calc	No Calc
Jckt	40	14.68	No Calc	No Calc	No Calc
40	50	14.68	No Calc	No Calc	No Calc



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External Pressure Calculations: Step: 17 3:25pm Apr 10,2026

50	60	14.68	No Calc	No Calc	No Calc
60	70	14.68	No Calc	No Calc	No Calc
70	Jckt	14.68	No Calc	No Calc	No Calc
Jckt	Jckt	1.6	No Calc	No Calc	No Calc
Jckt	Tsht	0.19997	No Calc	No Calc	No Calc
Tsht	80	No Calc	No Calc	No Calc	No Calc
80	90	2.63303	No Calc	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

For vessels with Jackets, the External Design Pressure includes the core vacuum pressure. The External MAWP is the total allowable external pressure for that segment minus the core vacuum pressure. The External MAWP can be less than the External Design pressure and this is acceptable given that the actual thickness is greater than the required thickness.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume m ³	Corroded Metal Wgt. kgm	Corroded ID Volume m ³	Extra due Misc % kgm
10	20	7325.58	3.91322	7097.72	3.94193	732.558
20	30	18278.1	7.76433	17948.9	7.80579	1827.81
30	40	18278.1	5.85024	17948.9	5.89171	1827.81
40	50	6397.32	2.04759	6282.12	2.0621	639.732
50	60	11771.1	37.6756	115591	37.9426	11771.1
60	70	6397.32	2.04758	6282.12	2.0621	639.732
70	80	18278.1	5.85024	17948.9	5.89171	1827.81
80	90	18278.1	7.76433	17948.9	7.80579	1827.81
90	100	7325.58	3.91322	7097.72	3.94193	732.558
Total		218268	76.83	214146	77.35	21826



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. m	Y Offset, Dtl. Cent. m	Z Offset Dtl. Cent. m	Description
10	Insl	491.417	-0.4335	...	...	Ins: 10
10	Nozl	4341.45	-1.125	...	...	Manway-1
20	Liqd	314.374	1	0.325	...	Liquid: 20
20	Insl	900.765	1	...	...	Ins: 20
20	Nozl	1914.96	1	1.37029	...	N4- Tube outlet
30	Liqd	377.319	1	0.325	...	Liquid: 30
30	Insl	900.765	1	...	...	Ins: 30
30	Nozl	1421.13	1	1.37029	...	N1- Shell inlet
30	Jckt	28050.4	1	...	...	Jacket
40	Insl	315.268	0.35	...	...	Ins: 40
50	Sadl	1427.59	1.06	1.3795	...	Left Saddle
50	Sadl	1427.59	12.34	1.3795	...	Sdl 2 Fr60
50	Liqd	2429.93	6.44	0.325	...	Liquid: 50
50	Insl	5800.93	6.44	...	...	Ins: 50
60	Insl	315.268	0.35	...	...	Ins: 60
70	Liqd	377.319	1	0.175	...	Liquid: 70
70	Insl	900.765	1	...	...	Ins: 70
70	Nozl	1965.01	1	-1.37029	...	N2- Shell outle
70	Jckt	29426	1	...	...	Jacket
80	Liqd	314.374	1	0.325	...	Liquid: 80
80	Insl	900.765	1	...	...	Ins: 80
80	Nozl	1914.96	1	-1.37029	...	N3- tube inlet
90	Insl	491.417	0.6915	...	...	Ins: 90
90	Nozl	4341.45	1.383	...	...	Mayway-2
20	FTsh	12814	2.18	...	...	Tubes
20	Tube	62553.9	11.5	...	...	Tubesheet
20	RTsh	12814	20.82	...	...	Baffles
30	Baf1	73.8585	0.795	...	...	Baffles
30	Baf1	73.8585	1.36	...	...	Baffles
30	Baf1	73.8585	1.925	...	...	Baffles
30	Baf1	73.8585	2.49	...	...	Baffles
30	Baf1	73.8585	3.055	...	...	Baffles
30	Baf1	73.8585	3.62	...	...	Baffles
30	Baf1	73.8585	4.185	...	...	Baffles
30	Baf1	73.8585	4.75	...	...	Baffles



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30 Baf1	73.8585	5.315	...	...	Baffles
30 Baf1	73.8585	5.88	...	...	Baffles
30 Baf1	73.8585	6.445	...	...	Baffles
30 Baf1	73.8585	7.01	...	...	Baffles
30 Baf1	73.8585	7.575	...	...	Baffles
30 Baf1	73.8585	8.14	...	...	Baffles
30 Baf1	73.8585	8.705	...	...	Baffles
30 Baf1	73.8585	9.27	...	...	Baffles
30 Baf1	73.8585	9.835	...	...	Baffles
30 Baf1	73.8585	10.4	...	...	Baffles
30 Baf1	73.8585	10.965	...	...	Baffles
30 Baf1	73.8585	11.53	...	...	Baffles
30 Baf1	73.8585	12.095	...	...	Baffles
30 Baf1	73.8585	12.66	...	...	Baffles
30 Baf1	73.8585	13.225	...	...	Baffles
30 Baf1	73.8585	13.79	...	...	Baffles
30 Baf1	73.8585	14.355	...	...	Baffles
30 Baf1	73.8585	14.92	...	...	Baffles
30 Baf1	73.8585	15.485	...	...	Baffles
30 Baf1	73.8585	16.05	...	...	Baffles
30 Baf1	73.8585	16.615	...	...	Baffles
30 Baf1	73.8585	17.18	...	...	Baffles
30 Baf1	73.8585	17.745	...	...	Baffles
30 Baf1	73.8585	18.31	...	...	Baffles
30 Baf1	74.4954	18.875	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	2855.2
Liquid	3813.3
Insulation	11017.4
Nozzles	15898.9
Exchanger Components	90619.9
Liquid in Tubes	586.6
Jacket, Closure, Insul	57476.4

Sum of the Detail Weights 182267.7 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	240095.5	240095.5	240095.5	240095.5	240095.5	240095.5
Saddles	2855.2	2855.2	2855.2	2855.2	2855.2	2855.2
Nozzles	15898.9	15898.9	15898.9	15898.9	15898.9	15898.9
Exchanger	90619.9	90619.9	90619.9	90619.9	90619.9	90619.9
Jacket	57476.4	57476.4	57476.4	57476.4	57476.4	57476.4
Insulation	...	...	11017.4	11017.4	11017.4	11017.4
Ope. Liquid	...	...	...	...	...	3813.3
Tube Ope Lqd	...	...	...	...	...	586.6
Test Liquid	...	78650.9	...	...	...	...
Tube Tst Lqd	...	11281.6	...	...	...	...
Totals	406945.9	496878.4	417963.2	417963.2	417963.3	422363.2

Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:



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Fabricated Wt.	- Bare weight without removable internals	406945.9 kgm
Shop Test Wt.	- Fabricated weight + water (full)	496878.4 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	417963.2 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	417963.2 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	417963.3 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	422363.2 kgm
Oper. Wt. + CA	- Corr wt. + operating liquid	417828.5 kgm

Exchanger Tube Data

Volume of Exchanger tubes :	11.0 m ³
Weight of Ope Liq in tubes :	586.6 kgm
Weight of Water in tubes :	11281.6 kgm

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight includes test liquid in the jacket only for Type 1 jackets specified as vapor belts.

Surface Areas of Elements:

From	To	Outside Surface Area m ²	Inside Surface Area m ²
10	20	11.2422	9.77343
20	30	16.0935	14.1337
30	40	16.0935	14.1337
40	50	5.63274	4.94678
50	60	103.642	91.0208
60	70	5.63274	4.94678
70	80	16.0935	14.1337
80	90	16.0935	14.1337
90	100	11.2422	9.77343

Total	201.8	177.0 m ²
Total	2172.3	1905.6 ft ²

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Nozzle Flange MAWP: Step: 19 3:25pm Apr 10,2026

Nozzle Flange MAWP Results: (N/mm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
Manway-1	22.83	25.86	275	1500	GR 1.9	...	...	...	...
Mayway-2	22.83	25.86	275	1500	GR 1.9	...	...	...	...

Shellside Flange Rating**Channelside Flange Rating**Lowest Flange Pressure Rating was (Ope)[TubeSide]: 22.826 N/mm²Lowest Flange Pressure Rating was (Amb)[TubeSide]: 25.861 N/mm²

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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 Wind Load Calculation: Step: 20 3:25pm Apr 10,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	27 m/sec
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0.3 m
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 m
Distance Upwind of Crest Lh	0 m
Distance from Crest to the Vessel x	0 m
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 m

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.228 \cdot 0.973) / (1 + 1.7 \cdot 3.4 \cdot 0.228)))$
 $= \min(0.85, 0.911)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.616
Structure Height to Diameter ratio	9.942

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Zg = 274.32$ m

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1.8 + 0.3 = 2.1$ m
 $= 6.89$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (6.89 ft.) < 15 ft.
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15 / 274.32)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill



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 Wind Load Calculation: Step: 20 3:25pm Apr 10,2026

Wind Directionality Factor [Kd]:
 = 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:
 K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:
 $= (1 + K1 \cdot K2 \cdot K3)^2$
 $= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$
 = 1.0

Velocity Pressure evaluated at height z, Imperial Units [qz]:
 $= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$
 $= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$
 = 16.0 psf [0.766] kPa

Force on the first element [F]:
 $= qz \cdot G \cdot Cf \cdot \text{WindArea}$
 $= 16.0 \cdot 0.85 \cdot 0.616 \cdot 40.212$
 = 337.1 lbs. [1499.3] N

Element	Hgt (z) m	K1	K2	K3	Kz	Kzt	qz kPa
channel sphere-	2.1	0.000	0.000	0.000	0.849	1.000	0.766
Channel - rear	2.1	0.000	0.000	0.000	0.849	1.000	0.766
shell	2.1	0.000	0.000	0.000	0.849	1.000	0.766
shell cone -rea	2.1	0.000	0.000	0.000	0.849	1.000	0.766
Shell-2	2.1	0.000	0.000	0.000	0.849	1.000	0.766
shell cone- fro	2.1	0.000	0.000	0.000	0.849	1.000	0.766
shell- front	2.1	0.000	0.000	0.000	0.849	1.000	0.766
Channel front	2.1	0.000	0.000	0.000	0.849	1.000	0.766
Channel sphere-	2.1	0.000	0.000	0.000	0.849	1.000	0.766

Wind Load Calculation:

From	To	Wind Height m	Wind Diameter m	Wind Area m ²	Wind Pressure kPa	Element Wind Load N
10	20	2.1	3.384	3.73492	0.76606	1649.22
20	30	2.1	3.0744	6.14728	0.76606	2714.44
30	40	2.1	3.32208	6.64251	0.76606	2933.12
40	50	2.1	3.0744	2.15155	0.76606	950.052
50	60	2.1	3.0744	39.5885	0.76606	17481
60	70	2.1	3.0744	2.15155	0.76606	950.052
70	80	2.1	3.32208	6.64251	0.76606	2933.12
80	90	2.1	3.0744	6.14728	0.76606	2714.44
90	100	2.1	3.24	3.69778	0.76606	1632.82

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\ SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_n^x$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(9.59650.75)$
 = 0.109 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33.0)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.109$, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.157 \cdot 4069227 \cdot 1.25$
 = 239270.562 N

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.157 \cdot 4069227$
 = 127610.9 N

Earthquake Load Calculation:

From	To	Earthquake Height m	Earthquake Weight N	Element Ope Load N
10	20	1.125	369930	21751.9



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Earthquake Load Calculation: Step: 21 3:25pm Apr 10,2026

20	30	1.125	369930	21751.9
30	40	1.125	369930	21751.9
40	50	1.125	369930	21751.9
50	Sad1	1.125	369930	21751.9
Sad1	60	1.125	369930	21751.9
50	60	1.125	369930	21751.9
60	70	1.125	369930	21751.9
70	80	1.125	369930	21751.9
80	90	1.125	369930	21751.9
90	100	1.125	369930	21751.9

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 22 3:25pm Apr 10,2026

Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	12.018 m
Center of Gravity of the Liquid	11.758 m
Center of Gravity of the Insulation	11.758 m
Center of Gravity of the Nozzles	12.036 m
Center of Gravity of the Tubesheet(s)	11.758 m
Center of Gravity of the Tubes	11.758 m
Center of Gravity of the Jacket/Vapor Belt	11.953 m
Center of Gravity of the Baffles	12.455 m
Center of Gravity of Bare Shell New and Cold	11.758 m
Center of Gravity of Bare Shell Corroded	11.758 m
Vessel CG in the Operating Condition	11.820 m
Vessel CG in the Fabricated (Shop/Empty) Condition	11.801 m
Vessel CG in the Test Condition	11.794 m

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 Saddle Calcs: Operating Case: Step: 23 3:25pm Apr 10,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

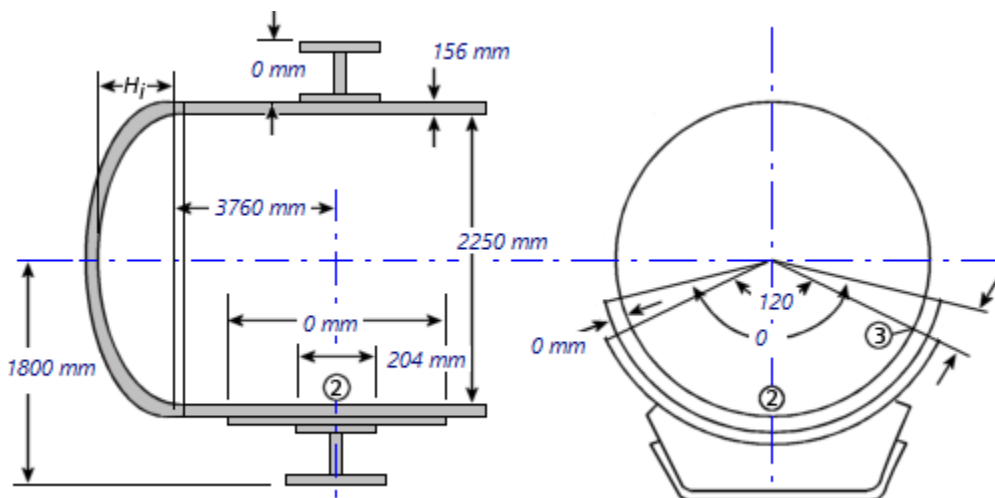
Horizontal Vessel Stress Calculations : Operating Case

Note:
 Wear Pad Width (254.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(204.0 + 1.56 \cdot \sqrt{1204.5 \cdot 153.0}, 2 \cdot 3760.0)$
 $= 873.6899 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	1204.50	mm
Shell Thickness used in this Case	t	156.000	mm
Stiffened Vessel Length per 4.15.6	L	18.28	m
Distance from Saddle to Vessel tangent	a1 or a	3760.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.16	m
Shell Allowable Stress used in Calculation		147988.73	kPa
Head Allowable Stress used in Calculation		147988.73	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1800.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	2751127.25	N
Pressure used in Analysis	P	18.001 N/mm ²

Horizontal Vessel Analysis Results:		Actual kPa	Allowable kPa
Long. Stress at Top of Midspan		69059.90	147988.73
Long. Stress at Bottom of Midspan		72648.31	147988.73



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Long. Stress at Top of Saddles	108033.32	147988.73
Long. Stress at Bottom of Saddles	50247.10	147988.73
Tangential Shear in Shell	9482.30	118390.98
Circ. Stress at Horn of Saddle	14462.63	184985.91
Circ. Compressive Stress in Shell	1564.71	147988.73

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3.0(33958.2/2 + 0) \cdot 1800.0/2236.0776$$

$$= 41003.6 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$$

$$= \max(7806.19, 0) \cdot 1800.0/11280.0$$

$$= 1245.7 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$$

$$= \max(239270.55, 0) \cdot 1800.0/11280.0$$

$$= 38181.5 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3.0(239271/2 + 0) \cdot 1800.0/2236.0776$$

$$= 288912.4 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 2462215 + \max(1246, 41004, 38181, 288912)$$

$$= 2751127.2 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08(0.0 + \pi/4(2.862 \cdot 1.2)^2)$$

$$= 7806.596 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	2765126.25	N
Transverse Shear Load	119635.27	N
Longitudinal Shear Load	239270.55	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(2250.0/2 + 156.0 + 10.0)\sin(120.0/2), 2 \cdot 1204.5)$$

$$= 2236.078 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(2a \cdot L))/(1 + (4\text{hm})/3L)]$$

$$= -2751127 \cdot 3.76 [1 - (1 - 3.76/18.28 + (1.205^2 - 1.164^2)/(2 \cdot 3.76 \cdot 18.28))/(1 + (4 \cdot 1.16)/(3 \cdot 18.28))]$$

$$= -2765170.5 \text{ N-m}$$



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Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L)$$

$$= 2751127 \cdot 18 / 4 \cdot ((1 + 2(1^2 - 1^2) / (18^2)) / (1 + (4 \cdot 1) / (3 \cdot 18)) - 4 \cdot 4 / 18)$$

$$= 1251672.0 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 18.001 \cdot 1204.5 / (2 \cdot 153.0) - 1251672 / (\pi \cdot 1204.5^2 \cdot 153.0)$$

$$= 69059.90 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 18.001 \cdot 1204.5 / (2 \cdot 153.0) + 1251672 / (\pi \cdot 1204.5^2 \cdot 153.0)$$

$$= 72648.31 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 18.001 \cdot 1204.5 / (2 \cdot 153.0) - 2765170.5 / (0.1066 \cdot \pi \cdot 1204.5^2 \cdot 153.0)$$

$$= 108033.32 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 18.001 \cdot 1204.5 / (2 \cdot 153.0) + 2765170.5 / (0.1923 \cdot \pi \cdot 1204.5^2 \cdot 153.0)$$

$$= 50247.10 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 2751127(18.28 - 2 \cdot 3.76) / (18.28 + (4 \cdot 1.16 / 3))$$

$$= 1492644.9 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 1492645 / (1204.5 \cdot 153.0)$$

$$= 9482.30 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1204.5 \cdot 153.0)}$$

$$= 334.845 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 2751127 \cdot 0.1 / (153.0(204.0 + 334.84 + 334.84))$$

$$= -1564.71 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -2751127 / (4 \cdot 153.0(204.0 + 334.845 + 334.845)) - 3 \cdot 0.0529 \cdot 2751127 / (2 \cdot 153.0^2)$$

$$= -14462.63 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(204.0 + 1.56 \cdot \sqrt{(1204.5 \cdot 153.0)}, 2 \cdot 3.76)$$

$$= 873.69 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 11280.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot Ls(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000013 \cdot 11280.0(302.0 - 21.1)$$

$$= 41.921 \text{ mm}$$



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Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	60.0000	mm
Baseplate Width	Bpwid	720.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	309976.4	kPa
Height of Web at Center	Hw,c	300.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1128.0 \cdot 153.0)} + 254.0)153.0^3)/12$$

$$= 269237152.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 0.26924\text{E}+09 + 0.138(218.815 - 76.5)^2$$

$$= 3064572160.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ² (2) mm ⁴
Shell	902.1	153.0	76.5	0.1	10558328.0	0.306E+10
Wearplate	254.0	10.0	158.0	0.0	401320.0	0.942E+07
Web	20.0	449.0	387.5	0.0	3479749.8	0.406E+09
BasePlate	720.0	60.0	642.0	0.0	27734400.0	0.775E+10
Totals	...	...	...	0.2	42173796.0	0.112E+11

Distance to Centroid [C1]:

$$= AY / A$$

$$= 42173796/0.193$$

$$= 218.815 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 2751127$$

$$= 559914.0000 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 10232.6240 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 185985.8438 \text{ kPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1800.0 - 1128.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 867.151 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 485727.1250 \text{ N-m}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 9460.8770 \text{ kPa}$$



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Allowed Stress, Sa = 2/3 * Yield Str = 206650.9219 kPa

Minimum Thickness of Baseplate per Moss:

= $\sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$
 = $\sqrt{\frac{3(2751127 + 13999) \cdot 720.0}{4 \cdot 2350.0 \cdot 206650.922}}$
 = 55.451 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= $2 \cdot \cos(90 - \text{Saddle Angle}/2) \cdot (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 = $2 \cdot \cos(90 - 120.0/2) \cdot (1125.0 + 156.0 + 10.0)$
 = 2236.077 mm

Distance between Ribs [e]:

= $\text{Web Length} / (\text{Nr ribs} - 1)$
 = $2236.0774 / (5 - 1)$
 = 559.019 mm

Baseplate Pressure Area [Ap]:

= $e \cdot \text{Bpwid} / 2$
 = $559.0193 \cdot 720.0 / 2$
 = 0.201 m²

Bearing Pressure [Bp]:

= $(Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 = $2765126 / (2350.0 \cdot 720.0)$
 = 1634640.625 N/m²

Axial Load [P]:

= $\text{Ap} \cdot \text{Bp}$
 = $0.2 \cdot 1634641$
 = 328884.938 N

Area of the Rib and Web [Ar]:

= Rib Area + Web Area
 = $0.004 + 0.006$
 = 0.009 m²

Compressive Stress [Sc]:

= P / Ar
 = $328884.9 / 0.0093$
 = 35478.637 kPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	204.0	...	...	...	0.141E+08

Rib dimension [D]:

= Saddle Width - Web Thickness
 = $204.0 - 20.0$
 = 184.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = $0.0 / 0.009$
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = $204.0 / 2$



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= 102.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{14149426 / 0.009}$
 = 39.068 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1800.0 - cos(120.0/2)(1125.0 + 156.0 + 10.0) - 60.0
 = 1094.500 mm

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 199955 / 309976.4}$
 = 112.838

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 1094.5/39.068
 = 28.015

Bending Moment [Rm]:
 $= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$
 $= 239270.5 / (2 \cdot 2350.0) \cdot 559.019 \cdot 1094.5 / 2$
 = 15580.452 N-m

Compressive Allowable, $KL/r < Cc$ (28.015 < 112.8376) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (28.01)^2 / (2 \cdot 112.84^2)) 309976 /$
 $(5/3 + 3 \cdot (28.01) / (8 \cdot 112.84) - (28.01^3) / (8 \cdot 112.84^3))$
 = 170903 kPa

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= Sc / Sca + (Rm \cdot C1 / I) / Sba$
 $= 35478.64 / 170902.75 + (15580.45 \cdot 102.0 / 14149440) / 206650.92$
 = 0.751

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	184.0	0.0	0.0	0.0	0.141E+08
Web	559.0	20.0	0.0	0.0	0.0	0.373E+06
Totals	...	...	...	...	...	0.145E+08

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 = 0.0/0.015
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 204.0/2
 = 102.000 mm

Length of Inner Rib [L]:
 = Saddle Height - Outside Radius - Bpthk
 = 1800.0 - cos(1291.0/2)(60.0 + 0.0 + 0.0) - 0.0
 = 449.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{14508773 / 0.015}$



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= 31.246 mm

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 449.0/31.246
 = 14.370

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)
 = 239270.547/(2 • 2350.0)
 = 50.909 N/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
 = 50.909 • 559.019 • 449.0
 = 12783.230 N-m

Bending Stress due to Transverse Force and Weight Load [σB,base,c]:

= Bending Moment / Section Modulus
 = 12783.229/142243.0
 = 89834.953 kPa

Compressive Allowable, KL/r < Cc (14.3696 < 112.8376) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2•Cc²))Fy/(5/3+3•(KL/r)/(8•Cc)-(KL/r)³/(8•Cc³))
 = (1-(14.37)²/(2 • 112.84²))309976/
 (5/3+3•(14.37)/(8 • 112.84)-(14.37³)/(8•112.84³))
 = 179366 kPa

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba
 = 26660.88/179366.09 + (12783.23 • 102.0/14508773)/206650.92
 = 0.583

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	137991.98 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0016 m ²
BasePlate Allowable Stress	S	147988.45 kPa
Area Available in a Single Bolt	BltArea	0.0014 m ²
Saddle Load Q0 (Weight)	Q0	2476213.8 N
Saddle Load QL (Wind/Seismic contribution)	QL	38181.5 N
Maximum Transverse Force	Ft	119635.3 N
Maximum Longitudinal Force	F1	239270.5 N
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

= 924894.1 N

Shear Stress in a Single Bolt, Longitudinal Direction [τb,l]:

= max(Ff, F1) / (Bolt Area • Number of Bolts)
 = 924894/(0.001 • 8)
 = 82134.3 kPa. Must be less than 82795.2 kPa.

Shear Stress in a Single Bolt, Transverse Direction [τb,t]:

= Ft / (Bolt Area • Number of Bolts)
 = 119635.3/(0.001 • 8)
 = 10624.1 kPa. Must be less than 82795.2 kPa.

Bolt Area Calculation per Dennis R. Moss



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Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (BoltShearAllowable \cdot Nbolts)$
 = $239270.55 / (82795.2 \cdot 8.0)$
 = 0.0004 m²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $1800.0 \cdot 119635.27 + 0.0$
 = 215430.80 N-m

Eccentricity (e):
 = $Rmom / QO$
 = $215430.8 / 2476214$
 = 86.96 mm < $B_{plen} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0.0, 0.0004, 0.0]$
 = 0.0004 m²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: A

Govern. thk, $t_g = 10.0$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = -1.0$, Temp. Reduction = 0 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (254.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = $\min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$
 = $\min(204.0 + 1.56 \cdot \sqrt{1204.5 \cdot 153.0}, 2 \cdot 3760.0)$
 = 873.6899 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1204.50	mm
Shell Thickness used in this Case	t	156.000	mm
Stiffened Vessel Length per 4.15.6	L	18.28	m
Distance from Saddle to Vessel tangent	a1 or a	3760.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.16	m
Shell Allowable Stress used in Calculation		147988.73	kPa
Head Allowable Stress used in Calculation		147988.73	kPa
Circumferential Efficiency in Plane of Saddle		1.00	



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Circumferential Efficiency at Mid-Span 1.00
 Distance from Saddle Base to Centerline B 1800.00 mm
 Coefficient of Friction mu 0.40

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 2587148.50 N
 Pressure used in Analysis P 18.001 N/mm²

Horizontal Vessel Analysis Results:	Actual kPa	Allowable kPa
Long. Stress at Top of Midspan	69166.85	147988.73
Long. Stress at Bottom of Midspan	72541.37	147988.73
Long. Stress at Top of Saddles	105817.28	147988.73
Long. Stress at Bottom of Saddles	51475.36	147988.73
Tangential Shear in Shell	8917.11	118390.98
Circ. Stress at Horn of Saddle	13600.60	184985.91
Circ. Compressive Stress in Shell	1471.45	147988.73

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (33958.2/2 + 0) \cdot 1800.0 / 2236.0776$
 $= 41003.6 \text{ N}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(7806.19, 0) \cdot 1800.0 / 11280.0$
 $= 1245.7 \text{ N}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(239270.55, 0) \cdot 1800.0 / 11280.0$
 $= 38181.5 \text{ N}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (239271/2 + 0) \cdot 1800.0 / 2236.0776$
 $= 288912.4 \text{ N}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 2298236 + \max(1246, 41004, 38181, 288912)$
 $= 2587148.5 \text{ N}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766.08 (0.0 + \pi/4 (2.862 \cdot 1.2)^2)$
 $= 7806.596 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 2601147.50 N
 Transverse Shear Load 119635.27 N
 Longitudinal Shear Load 239270.55 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2 (\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2 (2250.0/2 + 156.0 + 10.0) \sin(120.0/2), 2 \cdot 1204.5)$



= 2236.078 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
 K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405
 K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(Rm^2 - hm^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm)}{3L} \right) \right]$$

$$= -2587149 \cdot 3.76 \left[1 - \left(1 - \frac{3.76}{18.28} + \frac{(1.205^2 - 1.164^2)}{(2 \cdot 3.76 \cdot 18.28)} \right) / \left(1 + \frac{(4 \cdot 1.16)}{(3 \cdot 18.28)} \right) \right]$$

$$= -2600354.8 \text{ N-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{1 + 2(Rm^2 - hm^2)/(L^2)}{(1 + (4hm^2)/(3L))} - 4a/L \right)$$

$$= 2587149 \cdot 18 / 4 \left(\frac{1 + 2(1^2 - 1^2)/(18^2)}{(1 + (4 \cdot 1)/(3 \cdot 18))} - 4 \cdot 4 / 18 \right)$$

$$= 1177067.1 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 18.001 \cdot 1204.5 / (2 \cdot 153.0) - 1177067 / (\pi \cdot 1204.5^2 \cdot 153.0)$$

$$= 69166.85 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 18.001 \cdot 1204.5 / (2 \cdot 153.0) + 1177067 / (\pi \cdot 1204.5^2 \cdot 153.0)$$

$$= 72541.37 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 18.001 \cdot 1204.5 / (2 \cdot 153.0) - 2600354.8 / (0.1066 \cdot \pi \cdot 1204.5^2 \cdot 153.0)$$

$$= 105817.28 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot Rm / (2t) + M1 / (K1* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 18.001 \cdot 1204.5 / (2 \cdot 153.0) + 2600354.8 / (0.1923 \cdot \pi \cdot 1204.5^2 \cdot 153.0)$$

$$= 51475.36 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 2587149(18.28 - 2 \cdot 3.76) / (18.28 + (4 \cdot 1.16 / 3))$$

$$= 1403677.0 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 1403677 / (1204.5 \cdot 153.0)$$

$$= 8917.11 \text{ kPa}$$

Decay Length (4.15.20) [x1, x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1204.5 \cdot 153.0)}$$

$$= 334.845 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 2587149 \cdot 0.1 / (153.0(204.0 + 334.84 + 334.84))$$

$$= -1471.45 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -2587149 / (4 \cdot 153.0(204.0 + 334.845 + 334.845)) - 3 \cdot 0.0529 \cdot 2587149 / (2 \cdot 153.0^2)$$



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= -13600.60 kPa

Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(204.0 + 1.56 \cdot \sqrt{1204.5 \cdot 153.0}, 2 \cdot 3.76)$
 $= 873.69 \text{ mm}$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	60.0000	mm
Baseplate Width	Bpwid	720.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	261980.0	kPa
Height of Web at Center	Hw,c	300.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})t^3)/12$
 $= ((1.56 \cdot \sqrt{1128.0 \cdot 153.0}) + 254.0)153.0^3/12$
 $= 269237152.000 \text{ mm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 0.26924\text{E}+09 + 0.138(218.815 - 76.5)^2$
 $= 3064572160.000 \text{ mm}^4$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	902.1	153.0	76.5	0.1	10558328.0	0.306E+10
Wearplate	254.0	10.0	158.0	0.0	401320.0	0.942E+07
Web	20.0	449.0	387.5	0.0	3479749.8	0.406E+09
BasePlate	720.0	60.0	642.0	0.0	27734400.0	0.775E+10
Totals	...	...	...	0.2	42173796.0	0.112E+11

Distance to Centroid [C1]:
 $= AY / A$
 $= 42173796/0.193$
 $= 218.815 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 120.0/2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$
 $= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.204 \cdot 2587149$
 $= 526540.8125 \text{ N}$

Tension Stress, St = (Fh/As) = 9622.7168 kPa
 Allowed Stress, Sa = 0.6 * Yield Str = 157188.0312 kPa



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Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 1800.0 - 1128.0 \cdot \sin(120.0/2) / 1.0472$
 $= 867.151 \text{ mm}$

Bending Moment, $M = F_h \cdot d = 456775.7812 \text{ N-m}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 8896.9707 \text{ kPa}$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 174653.3594 \text{ kPa}$

Minimum Thickness of Baseplate per Moss:

$= \sqrt[3]{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt[3]{(3(2587149 + 13999) \cdot 720.0) / (4 \cdot 2350.0 \cdot 174653.359)}$
 $= 58.501 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120.0/2) (1125.0 + 156.0 + 10.0)$
 $= 2236.077 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 2236.0774 / (5 - 1)$
 $= 559.019 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 559.0193 \cdot 720.0 / 2$
 $= 0.201 \text{ m}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 2601148 / (2350.0 \cdot 720.0)$
 $= 1537702.500 \text{ N/m}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 0.2 \cdot 1537703$
 $= 309381.250 \text{ N}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 0.004 + 0.006$
 $= 0.009 \text{ m}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 309381.2 / 0.0093$
 $= 33374.668 \text{ kPa}$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	204.0	...	...	...	0.141E+08

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 204.0 - 20.0$
 $= 184.000 \text{ mm}$



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Distance to Centroid from Datum [ytot]:

= AY / A
 = 0.0/0.009
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 204.0/2
 = 102.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{14149426 / 0.009}$
 = 39.068 mm

Length of Outer Rib [L]:

= Saddle Height - $\cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 = $1800.0 - \cos(120.0/2) (1125.0 + 156.0 + 10.0) - 60.0$
 = 1094.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 199955 / 261980.0}$
 = 122.739

Slenderness ratio [KL/r]:

= KL/r
 = $1 \cdot 1094.5 / 39.068$
 = 28.015

Bending Moment [Rm]:

= $FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$
 = $239270.5 / (2 \cdot 2350.0) \cdot 559.019 \cdot 1094.5 / 2$
 = 15580.452 N-m

Compressive Allowable, $KL/r < Cc$ ($28.015 < 122.7394$) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (28.01)^2 / (2 \cdot 122.74^2)) 261980 /$
 = $(5/3 + 3 \cdot (28.01) / (8 \cdot 122.74) - (28.01^3) / (8 \cdot 122.74^3))$
 = 145739 kPa

AISC Unity Check of Outside Ribs (must be ≤ 1)

= $Sc/Sca + (Rm \cdot C1 / I) / Sba$
 = $33374.67 / 145738.94 + (15580.45 \cdot 102.0 / 14149440) / 174653.36$
 = 0.872

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	184.0	0.0	0.0	0.0	0.141E+08
Web	559.0	20.0	0.0	0.0	0.0	0.373E+06
Totals	...	...	...	...	...	0.145E+08

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0.0/0.015
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 204.0/2
 = 102.000 mm



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Length of Inner Rib [L]:
 = Saddle Height - Outside Radius - Bpthk
 = 1800.0 - cos(1291.0/2)(60.0 + 0.0 + 0.0) - 0.0
 = 449.000 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{14508773/0.015}$
 = 31.246 mm

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 449.0/31.246
 = 14.370

Unit Force [Force,u]:
 = FI / (2 • Baseplate Length)
 = 239270.547/(2 • 2350.0)
 = 50.909 N/mm

Moment at base of inner Rib [Mbase,c]:
 = Unit Force • e • L
 = 50.909 • 559.019 • 449.0
 = 12783.230 N-m

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 = Bending Moment / Section Modulus
 = 12783.229/142243.0
 = 89834.953 kPa

Compressive Allowable, $KL/r < C_c$ (14.3696 < 122.7394) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 = $(1 - (14.37)^2 / (2 \cdot 122.74^2)) 261980 / (5/3 + 3 \cdot (14.37) / (8 \cdot 122.74) - (14.37)^3 / (8 \cdot 122.74^3))$
 = 152122 kPa

AISC Unity Check of Inside Ribs (must be ≤ 1)
 = $S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$
 = 25079.83/152121.94 + (12783.23 • 102.0/14508773)/174653.36
 = 0.679

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	137991.98 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0016 m²
BasePlate Allowable Stress	S	137989.20 kPa
Area Available in a Single Bolt	BltArea	0.0014 m²
Saddle Load Q0 (Weight)	Q0	2312235.0 N
Saddle Load QL (Wind/Seismic contribution)	QL	38181.5 N
Maximum Transverse Force	Ft	119635.3 N
Maximum Longitudinal Force	F1	239270.5 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
 = FI / (Bolt Area • Number of Bolts)
 = 239270.5/(0.001 • 8)
 = 21248.2 kPa. Must be less than 82795.2 kPa.

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:



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= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $119635.3 / (0.001 \cdot 8)$
 = 10624.1 kPa. Must be less than 82795.2 kPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 = $239270.55 / (82795.2 \cdot 8.0)$
 = 0.0004 m²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $1800.0 \cdot 119635.27 + 0.0$
 = 215430.80 N-m

Eccentricity (e):
 = R_{mom} / QO
 = $215430.8 / 2312235$
 = 93.13 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0.0, 0.0004, 0.0]$
 = 0.0004 m²

MDMT Calculations for: Sdl 2 Fr60, Saddle to Wearplate Junction, Curve: A

Govrn. thk, tg = 10.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = -1.0$, Temp. Reduction = 0 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

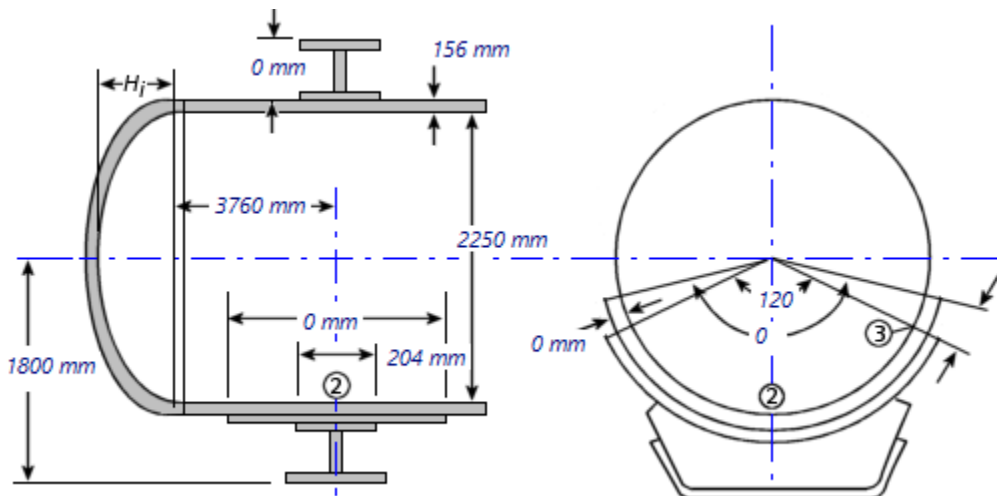
Horizontal Vessel Stress Calculations : Test Case

Note:
 Wear Pad Width (254.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(204.0 + 1.56 \cdot \sqrt{1203.0 \cdot 156.0}, 2 \cdot 3760.0)$
 $= 879.8023 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	1203.00	mm
Shell Thickness used in this Case	t	156.000	mm
Stiffened Vessel Length per 4.15.6	L	18.28	m
Distance from Saddle to Vessel tangent	a1 or a	3760.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.16	m
Shell Allowable Stress used in Calculation		278978.75	kPa
Head Allowable Stress used in Calculation		278978.75	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1800.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	2630011.25	N
Pressure used in Analysis	P	23.413 N/mm ²

Horizontal Vessel Analysis Results:	Actual kPa	Allowable kPa
Long. Stress at Top of Midspan	88582.48	278978.75
Long. Stress at Bottom of Midspan	91958.46	278978.75



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Long. Stress at Top of Saddles	125206.82	278978.75
Long. Stress at Bottom of Saddles	70906.59	278978.75
Tangential Shear in Shell	8902.51	165217.41
Circ. Stress at Horn of Saddle	13358.53	418468.09
Circ. Compressive Stress in Shell	1456.87	278978.75

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$$

$$= 3.0(16979.1/2 + 0) \cdot 1800.0/2236.0776$$

$$= 20501.8\ N$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(3903.1, 0) \cdot 1800.0/11280.0$$

$$= 622.8\ N$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 2609510 + \max(623, 20502, 0, 0)$$

$$= 2630011.2\ N$$

Longitudinal Wind Force [Fl]:

$$= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$$

$$= 1.1 \cdot 766.08(0.0 + \pi/4(2.862 \cdot 1.2)^2)$$

$$= 7806.596\ N$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	2644010.25	N
Transverse Shear Load	8489.55	N
Longitudinal Shear Load	3903.10	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(2250.0/2 + 156.0 + 10.0) \sin(120.0/2), 2 \cdot 1203.0)$$

$$= 2236.078\ mm$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -2630011 \cdot 3.76 [1 - (1 - 3.76/18.28 + (1.203^2 - 1.163^2) / (2 \cdot 3.76 \cdot 18.28)) / (1 + (4 \cdot 1.16)/(3 \cdot 18.28))]$$

$$= -2642713.5\ N \cdot m$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 \{ (1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L \}$$

$$= 2630011 \cdot 18/4 \{ (1 + 2(1^2 - 1^2)/(18^2)) / (1 + (4 \cdot 1)/(3 \cdot 18)) - 4 \cdot 4/18 \}$$

$$= 1197679.1\ N \cdot m$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 23.413 \cdot 1203.0 / (2 \cdot 156.0) - 1197679 / (\pi \cdot 1203.0^2 \cdot 156.0)$$

$$= 88582.48\ kPa$$



Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$
 $= 23.413 \cdot 1203.0 / (2 \cdot 156.0) + 1197679 / (\pi \cdot 1203.0^2 \cdot 156.0)$
 $= 91958.46 \text{ kPa}$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:
 $= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$
 $= 23.413 \cdot 1203.0 / (2 \cdot 156.0) - 2642713.5 / (0.1066 \cdot \pi \cdot 1203.0^2 \cdot 156.0)$
 $= 125206.82 \text{ kPa}$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:
 $= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$
 $= 23.413 \cdot 1203.0 / (2 \cdot 156.0) + 2642713.5 / (0.1923 \cdot \pi \cdot 1203.0^2 \cdot 156.0)$
 $= 70906.59 \text{ kPa}$

Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L-2a) / (L + (4 \cdot h_m / 3))$
 $= 2630011(18.28 - 2 \cdot 3.76) / (18.28 + (4 \cdot 1.16 / 3))$
 $= 1427076.2 \text{ N}$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:
 $= K_2 \cdot T / (R_m \cdot t)$
 $= 1.1707 \cdot 1427076 / (1203.0 \cdot 156.0)$
 $= 8902.51 \text{ kPa}$

Decay Length (4.15.20) [x_1, x_2]:
 $= 0.78 \cdot \sqrt{R_m \cdot t}$
 $= 0.78 \cdot \sqrt{1203.0 \cdot 156.0}$
 $= 337.901 \text{ mm}$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:
 $= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$
 $= -0.7603 \cdot 2630011 \cdot 0.1 / (156.0(204.0 + 337.9 + 337.9))$
 $= -1456.87 \text{ kPa}$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:
 $= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$
 $= -2630011 / (4 \cdot 156.0(204.0 + 337.901 + 337.901)) -$
 $3 \cdot 0.0529 \cdot 2630011 / (2 \cdot 156.0^2)$
 $= -13358.53 \text{ kPa}$

Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(204.0 + 1.56 \cdot \sqrt{1203.0 \cdot 156.0}, 2 \cdot 3.76)$
 $= 879.80 \text{ mm}$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	60.0000	mm
Baseplate Width	Bpwid	720.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	309976.4	kPa
Height of Web at Center	Hw,c	300.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{R \cdot t}) + \text{WearPlateWidth})^3 / 12$
 $= ((1.56 \cdot \sqrt{1125.0 \cdot 156.0}) + 254.0)^3 / 12$
 $= 287112000.000 \text{ mm}^4$



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Resolved Inertia for the Shell [I,shell]:
 = ShellInertia + AShell(C1 - Y)²
 = 0.28711E+09 + 0.142(218.154 - 78.0)²
 = 3068083712.000 mm⁴

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	907.5	156.0	78.0	0.1	11042780.0	0.307E+10
Wearplate	254.0	10.0	161.0	0.0	408940.0	0.832E+07
Web	20.0	449.0	390.5	0.0	3506689.8	0.418E+09
BasePlate	720.0	60.0	645.0	0.0	27864000.0	0.788E+10
Totals	...	...	...	0.2	42822412.0	0.114E+11

Distance to Centroid [C1]:
 = AY / A
 = 42822412/0.196
 = 218.154 mm

Angle [β]:
 = 180 - Saddle Angle/2
 = 180 - 120.0/2
 = 120.0

Saddle Splitting Coefficient [K1]:
 = (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))
 = (1 + cos(120.0) - 0.5•sin(120.0)²)/(π - 2.094 + sin(120.0)cos(120.0))
 = 0.2035

Saddle Splitting Force [Fh]:
 = K1 • Q
 = 0.204 • 2630011
 = 535264.3125 N

Tension Stress, St = (Fh/As) = 9782.1406 kPa
 Allowed Stress, Sa = 0.6 * Yield Str = 185985.8438 kPa

Saddle Splitting Dimension [d]:
 = B - R • sin(theta/2)/(theta/2 in radians)
 = 1800.0 - 1125.0 • sin(120.0/2)/1.0472
 = 869.632 mm

Bending Moment, M = Fh * d = 465671.9688 N-m

Bending Stress, Sb = (M * C1 / I) = 8925.2061 kPa
 Allowed Stress, Sa = 2/3 * Yield Str = 206650.9219 kPa

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{\frac{3(Q + \text{SaddleWt})\text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt[3]{\frac{3(2630011 + 13999)720.0}{4 \cdot 2350.0 \cdot 206650.922}}$$
 = 54.223 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 = 2 • cos(90 - Saddle Angle/2)(Inside Radius + Shell Thk + Wear Plate Thk)
 = 2 • cos(90 - 120.0/2)(1125.0 + 156.0 + 10.0)
 = 2236.077 mm

Distance between Ribs [e]:
 = Web Length / (Nribs - 1)
 = 2236.0774/(5 - 1)
 = 559.019 mm



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Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 559.0193 \cdot 720.0 / 2$$

$$= 0.201 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 2644010 / (2350.0 \cdot 720.0)$$

$$= 1563041.375 \text{ N/m}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 0.2 \cdot 1563041$$

$$= 314479.344 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 0.004 + 0.006$$

$$= 0.009 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 314479.3 / 0.0093$$

$$= 33924.625 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	204.0	...	...	...	0.141E+08

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 204.0 - 20.0$$

$$= 184.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 0.009$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 204.0 / 2$$

$$= 102.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (14149426 / 0.009) }$$

$$= 39.068 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1800.0 - \cos(120.0 / 2) (1125.0 + 156.0 + 10.0) - 60.0$$

$$= 1094.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199955 / 309976.4) }$$

$$= 112.838$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 1094.5/39.068$$

$$= 28.015$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 3903.1 / (2 \cdot 2350.0) \cdot 559.019 \cdot 1094.5/2$$

$$= 254.156 \text{ N-m}$$

Compressive Allowable, $KL/r < Cc$ (28.015 < 112.8376) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (28.01)^2 / (2 \cdot 112.84^2)) 309976 /$$

$$(5/3 + 3 \cdot (28.01) / (8 \cdot 112.84) - (28.01^3) / (8 \cdot 112.84^3))$$

$$= 170903 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 33924.62/170902.75 + (254.16 \cdot 102.0/14149440) / 206650.92$$

$$= 0.207$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	184.0	0.0	0.0	0.0	0.141E+08
Web	559.0	20.0	0.0	0.0	0.0	0.373E+06
Totals	...	...	...	...	...	0.145E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/0.015$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204.0/2$$

$$= 102.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1800.0 - \cos(1291.0/2) (60.0 + 0.0 + 0.0) - 0.0$$

$$= 449.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (14508773/0.015) }$$

$$= 31.246 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 449.0/31.246$$

$$= 14.370$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 3903.096 / (2 \cdot 2350.0)$$

$$= 0.830 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.83 \cdot 559.019 \cdot 449.0$$

$$= 208.526 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B, \text{base}, c$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



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= 208.526/142243.0
 = 1465.431 kPa

Compressive Allowable, $KL/r < C_c$ (14.3696 < 112.8376) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 = $(1 - (14.37)^2 / (2 \cdot 112.84^2)) 309976 / (5/3 + 3 \cdot (14.37) / (8 \cdot 112.84) - (14.37)^3 / (8 \cdot 112.84^3))$
 = 179366 kPa

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$
 = 25493.1/179366.09 + (208.53 • 102.0/14508773)/206650.92
 = 0.149

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48296.32 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0016 m ²
BasePlate Allowable Stress	S	147988.73 kPa
Area Available in a Single Bolt	BltArea	0.0014 m ²
Saddle Load QO (Weight)	QO	2623508.5 N
Saddle Load QL (Wind/Seismic contribution)	QL	622.8 N
Maximum Transverse Force	Ft	8489.6 N
Maximum Longitudinal Force	F1	3903.1 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 = $F1 / (Bolt\ Area \cdot Number\ of\ Bolts)$
 = 3903.1/(0.001 • 8)
 = 346.6 kPa. Must be less than 28977.8 kPa.

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 = $Ft / (Bolt\ Area \cdot Number\ of\ Bolts)$
 = 8489.6/(0.001 • 8)
 = 753.9 kPa. Must be less than 28977.8 kPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F1 / (BoltShearAllowable \cdot Nbolts)$
 = 3903.1/(28977.79 • 8.0)
 = 0.0000 m²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot Ft + Sum\ of\ X\ Moments$
 = 1800.0 • 8489.55 + 0.0
 = 15287.39 N-m

Eccentricity (e):
 = $Rmom / QO$
 = 15287.39/2623509
 = 5.82 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)



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Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 0., 0.0]
 = 0.0000 m²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (254.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = $\min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$
 = $\min(204.0 + 1.56 \cdot \sqrt{1203.0 \cdot 156.0}, 2 \cdot 3760.0)$
 = 879.8023 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1203.00	mm
Shell Thickness used in this Case	t	156.000	mm
Stiffened Vessel Length per 4.15.6	L	18.28	m
Distance from Saddle to Vessel tangent	a1 or a	3760.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	1.16	m
Shell Allowable Stress used in Calculation		278978.75	kPa
Head Allowable Stress used in Calculation		278978.75	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1800.00	mm
Coefficient of Friction	mu	0.40	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		2397624.00	N
Pressure used in Analysis	P	23.413	N/mm²

Horizontal Vessel Analysis Results:	Actual kPa	Allowable kPa
Long. Stress at Top of Midspan	88731.63	278978.75
Long. Stress at Bottom of Midspan	91809.31	278978.75
Long. Stress at Top of Saddles	122119.84	278978.75
Long. Stress at Bottom of Saddles	72617.59	278978.75
Tangential Shear in Shell	8115.88	165217.41
Circ. Stress at Horn of Saddle	12178.18	418468.09
Circ. Compressive Stress in Shell	1328.14	278978.75

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 = $Ftr (Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 = $3.0 (16979.1/2 + 0) \cdot 1800.0/2236.0776$
 = 20501.8 N

Longitudinal Saddle Reaction Force [Fwl]:
 = $\max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / Ls$



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$$= \max(3903.1, 0) \cdot 1800.0/11280.0$$

$$= 622.8 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 2377123 + \max(623, 20502, 0, 0)$$

$$= 2397624.0 \text{ N}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08(0.0 + \pi/4(2.862 \cdot 1.2)^2)$$

$$= 7806.596 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	2411623.00	N
Transverse Shear Load	8489.55	N
Longitudinal Shear Load	3903.10	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(2250.0/2 + 156.0 + 10.0)\sin(120.0/2), 2 \cdot 1203.0)$$

$$= 2236.078 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -2397624 \cdot 3.76 [1 - (1 - 3.76/18.28 + (1.203^2 - 1.163^2)/(2 \cdot 3.76 \cdot 18.28))/(1 + (4 \cdot 1.16)/(3 \cdot 18.28))]$$

$$= -2409203.8 \text{ N-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 2397624 \cdot 18/4((1 + 2(1^2 - 1^2)/(18^2))/(1 + (4 \cdot 1)/(3 \cdot 18)) - 4 \cdot 4/18)$$

$$= 1091852.5 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 23.413 \cdot 1203.0/(2 \cdot 156.0) - 1091853/(\pi \cdot 1203.0^2 \cdot 156.0)$$

$$= 88731.63 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 23.413 \cdot 1203.0/(2 \cdot 156.0) + 1091853/(\pi \cdot 1203.0^2 \cdot 156.0)$$

$$= 91809.31 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 23.413 \cdot 1203.0/(2 \cdot 156.0) - 2409203.8/(0.1066 \cdot \pi \cdot 1203.0^2 \cdot 156.0)$$

$$= 122119.84 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 23.413 \cdot 1203.0/(2 \cdot 156.0) + 2409203.8/(0.1923 \cdot \pi \cdot 1203.0^2 \cdot 156.0)$$

$$= 72617.59 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:



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$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 2397624(18.28 - 2 \cdot 3.76)/(18.28 + (4 \cdot 1.16/3))$$

$$= 1300980.2 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 1300980 / (1203.0 \cdot 156.0)$$

$$= 8115.88 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1203.0 \cdot 156.0)}$$

$$= 337.901 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 2397624 \cdot 0.1 / (156.0(204.0 + 337.9 + 337.9))$$

$$= -1328.14 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q/(4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -2397624 / (4 \cdot 156.0(204.0 + 337.901 + 337.901)) - 3 \cdot 0.0529 \cdot 2397624 / (2 \cdot 156.0^2)$$

$$= -12178.18 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(204.0 + 1.56 \cdot \sqrt{(1203.0 \cdot 156.0)}, 2 \cdot 3.76)$$

$$= 879.80 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	60.0000	mm
Baseplate Width	Bpwid	720.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	259980.2	kPa
Height of Web at Center	Hw,c	300.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1125.0 \cdot 156.0)} + 254.0)156.0^3)/12$$

$$= 287112000.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 0.28711\text{E}+09 + 0.142(218.154 - 78.0)^2$$

$$= 3068083712.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	907.5	156.0	78.0	0.1	11042780.0	0.307E+10
Wearplate	254.0	10.0	161.0	0.0	408940.0	0.832E+07
Web	20.0	449.0	390.5	0.0	3506689.8	0.418E+09
BasePlate	720.0	60.0	645.0	0.0	27864000.0	0.788E+10
Totals	...	...	...	0.2	42822412.0	0.114E+11

Distance to Centroid [C1]:



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$$= AY / A$$

$$= 42822412 / 0.196$$

$$= 218.154 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 2397624$$

$$= 487968.5000 \text{ N}$$

Tension Stress, $St = (Fh / As) = 8917.7930 \text{ kPa}$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 155988.1250 \text{ kPa}$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1800.0 - 1125.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 869.632 \text{ mm}$$

Bending Moment, $M = Fh \cdot d = 424525.2812 \text{ N-m}$

Bending Stress, $Sb = (M \cdot C1 / I) = 8136.5771 \text{ kPa}$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 173320.1406 \text{ kPa}$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(2397624 + 13999)720.0 / (4 \cdot 2350.0 \cdot 173320.141)}$$

$$= 56.546 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1125.0 + 156.0 + 10.0)$$

$$= 2236.077 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 2236.0774 / (5 - 1)$$

$$= 559.019 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{\text{pwid}} / 2$$

$$= 559.0193 \cdot 720.0 / 2$$

$$= 0.201 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 2411623 / (2350.0 \cdot 720.0)$$

$$= 1425662.625 \text{ N/m}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 0.2 \cdot 1425663$$

$$= 286839.156 \text{ N}$$

Area of the Rib and Web [Ar]:



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= Rib Area + Web Area
 = 0.004 + 0.006
 = 0.009 m²

Compressive Stress [Sc]:
 = P/Ar
 = 286839.2/0.0093
 = 30942.928 kPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	204.0	...	...	...	0.141E+08

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 204.0 - 20.0
 = 184.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0.0/0.009
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 204.0/2
 = 102.000 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{14149426/0.009}$
 = 39.068 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1800.0 - cos(120.0/2)(1125.0 + 156.0 + 10.0) - 60.0
 = 1094.500 mm

Intermediate Term [Cc]:
 = $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 199955/259980.2}$
 = 123.211

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 1094.5/39.068
 = 28.015

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 3903.1/(2 • 2350.0) • 559.019 • 1094.5/2
 = 254.156 N-m

Compressive Allowable, KL/r < Cc (28.015 < 123.2105) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (28.01)^2 / (2 \cdot 123.21^2)) 259980 / (5/3 + 3 \cdot (28.01) / (8 \cdot 123.21) - (28.01)^3 / (8 \cdot 123.21^3))$
 = 144682 kPa

AISC Unity Check of Outside Ribs (must be ≤ 1)
 = Sc/Sca + (Rm • C1 / I) / Sba



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$$= 30942.93/144681.61 + (254.16 \cdot 102.0/14149440)/173320.14$$

$$= 0.224$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	184.0	0.0	0.0	0.0	0.141E+08
Web	559.0	20.0	0.0	0.0	0.0	0.373E+06
Totals	...	...	...	...	...	0.145E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/0.015$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 204.0/2$$

$$= 102.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1800.0 - \cos(1291.0/2)(60.0 + 0.0 + 0.0) - 0.0$$

$$= 449.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{14508773/0.015}$$

$$= 31.246 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 449.0/31.246$$

$$= 14.370$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 3903.096/(2 \cdot 2350.0)$$

$$= 0.830 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.83 \cdot 559.019 \cdot 449.0$$

$$= 208.526 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 208.526/142243.0$$

$$= 1465.431 \text{ kPa}$$

Compressive Allowable, $KL/r < C_c$ (14.3696 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2/(2 \cdot C_c^2))F_y/(5/3 + 3 \cdot (KL/r)/(8 \cdot C_c) - (KL/r)^3/(8 \cdot C_c^3))$$

$$= (1 - (14.37)^2/(2 \cdot 123.21^2))259980/$$

$$(5/3 + 3 \cdot (14.37)/(8 \cdot 123.21) - (14.37^3)/(8 \cdot 123.21^3))$$

$$= 150983 \text{ kPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I)/Sba$$

$$= 23252.46/150983.28 + (208.53 \cdot 102.0/14508773)/173320.14$$

$$= 0.162$$

Input Data for Base Plate Bolting Calculations:



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Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48296.32 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0016 m²
BasePlate Allowable Stress	S	137989.50 kPa
Area Available in a Single Bolt	BltArea	0.0014 m²
Saddle Load Q0 (Weight)	Q0	2391121.2 N
Saddle Load QL (Wind/Seismic contribution)	QL	622.8 N
Maximum Transverse Force	Ft	8489.6 N
Maximum Longitudinal Force	F1	3903.1 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3903.1 / (0.001 \cdot 8)$$

$$= 346.6 \text{ kPa. Must be less than } 28977.8 \text{ kPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [τb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 8489.6 / (0.001 \cdot 8)$$

$$= 753.9 \text{ kPa. Must be less than } 28977.8 \text{ kPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 3903.1 / (28977.79 \cdot 8.0)$$

$$= 0.0000 \text{ m}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1800.0 \cdot 8489.55 + 0.0$$

$$= 15287.39 \text{ N-m}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 15287.39 / 2391121$$

$$= 6.39 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0., 0.0]$$

$$= 0.0000 \text{ m}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: Manway-1

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Input, Nozzle Desc: Manway-1

From: 10

Pressure for Reinforcement Calculations	P	18.0010	N/mm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147988.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa
Inside Diameter of Hemispherical Head	D	2250.00	mm
Head Finished (Minimum) Thickness	t	75.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

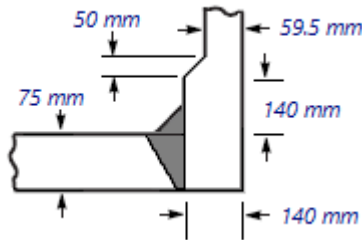
Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147988.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	140.0000	mm
Height of Beveled Transition	L'	50.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		140.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	450.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		54.4000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		1500	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



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 Nozzle Calcs.: Manway-1 Nozl: 13 3:25pm Apr 10,2026



Reinforcement CALCULATION, Description: Manway-1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 19.314 in.
 Actual Thickness Used in Calculation 2.343 in.

Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 18.0 \cdot 1128.0 / (2 \cdot 147989 \cdot 1.0 - 0.2 \cdot 18.0)$
 $= 69.4445 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 18.0 \cdot 248.29 / (147989 \cdot 1.0 - 0.6 \cdot 18.0)$
 $= 32.5769 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 80.488 / 0.5773$
 $= 139.4090 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	993.1512	mm
Parallel to Vessel Wall, opening length	d	496.5756	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	180.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 147988.7 / 147988.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 147988.7 / 147988.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



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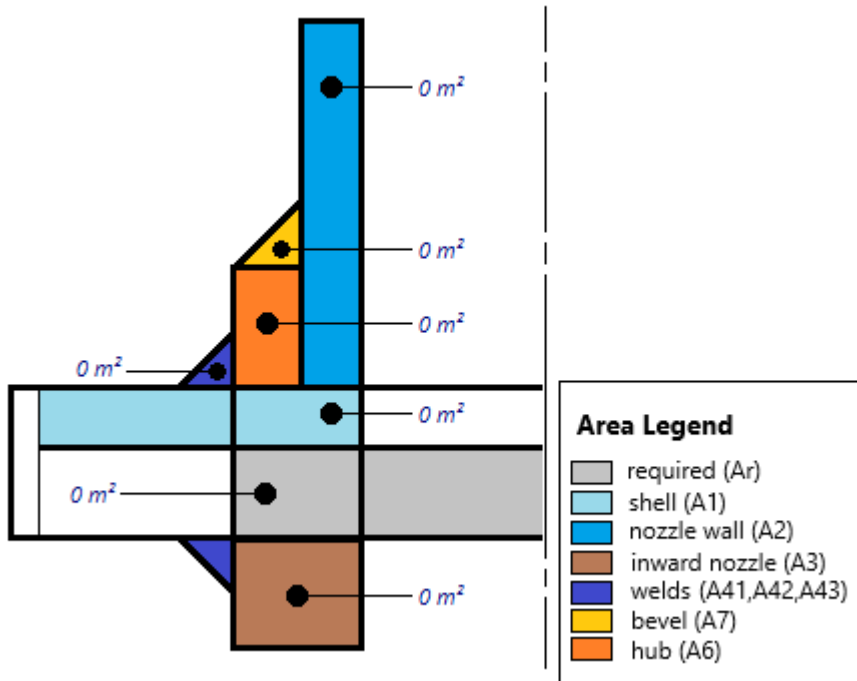
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Nozzle Calcs.: Manway-1

Nozl: 13 3:25pm Apr 10, 2026



Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	0.034	NA	NA
Area in Shell	A1	0.001	NA	NA
Area in Nozzle Wall	A2	0.009	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	0.026	NA	NA
TOTAL AREA AVAILABLE	Atot	0.036	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (496.5756 \cdot 69.4445 \cdot 1.0 + 2 \cdot 137.0 \cdot 69.4445 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 0.034 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 496.576(1.0 \cdot 72.0 - 1.0 \cdot 69.444) - 2 \cdot 137.0$$

$$(1.0 \cdot 72.0 - 1.0 \cdot 69.4445) \cdot (1 - 1.0)$$

$$= 0.001 \text{ m}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 180.0)(56.51 - 32.58)1.0$$

$$= 0.009 \text{ m}^2$$



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 Nozzle Calcs.: Manway-1 Nozl: 13 3:25pm Apr 10, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(180.0, 450.0, 140.0)) \cdot (140.0 - 59.5) \cdot 1.0$$

$$= 0.026 \text{ m}^2 \text{ Includes Bevel Area of } 0.004 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 35.5769 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 35.5769 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 59.512 = 52.073 \text{ mm} \rightarrow \text{OK}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, $t_g = 52.073$, $t_r = 32.577$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.664$, Temp. Reduction = $19 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-1 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:A)

Govrn. thk, $t_g = 75.0$, $t_r = 69.444$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.965$, Temp. Reduction = $2 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	44 °C
Min Metal Temp. at Required thickness (UCS 66.1)	42 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 42 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-35 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $18.00 / 25.86 = 0.696$

Weld Size Calculations, Description: Manway-1

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - t_r)) S_v)$$

$$= \max(0, (0.0345 - 0.0013 + 2 \cdot 137.0 \cdot 1.0 \cdot (1.0 \cdot 72.0 - 69.4445)) 147989)$$

$$= 5018998.00 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.



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 Nozzle Calcs.: Manway-1 Nozl: 13 3:25pm Apr 10,2026

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot S_v$$

$$= (0.035 + 0.0001 - 0.0 \cdot 1.0) \cdot 147989$$

$$= 5196738.00 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S_v$$

$$= (0.035 + 0.0 + 0.0001 + (0.0197)) \cdot 147989$$

$$= 8116181.50 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S$$

$$= (0.035 + 0.0 + 0.0001 + 0.0 + (0.0197)) \cdot 147989$$

$$= 8116181.50 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 770.5756 \cdot 10.0 \cdot 0.49 \cdot 147989$$

$$= 877704. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 316.7878) \cdot (140.0 - 3.0) \cdot 0.7 \cdot 147989$$

$$= 14123874. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 770.5756 \cdot (54.4 - 0.0) \cdot 0.74 \cdot 147989$$

$$= 7210788. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (877704 + 14123874) = 15001578 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (877704 + 0 + 7210788 + 0) = 8088492 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (877704 + 7210788 + 0) = 8088492 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 15001578N	, must exceed W = 5018998 N	or W1 = 5196738 N
Path 2-2 = 8088491 N	, must exceed W = 5018998 N	or W2 = 8116181 N
Path 3-3 = 8088491 N	, must exceed W = 5018998 N	or W3 = 8116181 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 18 N/mm²

The Drop for this Nozzle is : 68.0334 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 593.0334 mm

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 Nozzle Calcs.: N4- Tube outlet Nozl: 14 3:25pm Apr 10,2026

Input, Nozzle Desc: N4- Tube outlet

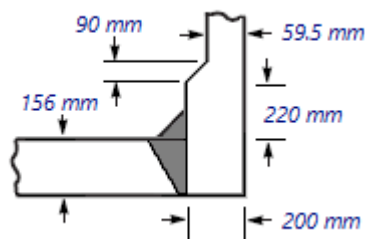
From: 20

Pressure for Reinforcement Calculations	P	18.0010	N/mm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147988.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa
Inside Diameter of Cylindrical Shell	D	2250.00	mm
Shell Finished (Minimum) Thickness	t	156.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1.26	m
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147988.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	220.0000	mm
Height of Beveled Transition	L'	90.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		200.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	50.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N4- Tube outlet



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 Nozzle Calcs.: N4- Tube outlet Nozl: 14 3:25pm Apr 10,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 19.314 in.
 Actual Thickness Used in Calculation 2.343 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 18.0 \cdot 1128.0 / (147989 \cdot 1.0 - 0.6 \cdot 18.0)$
 $= 148.0005 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 18.0 \cdot 248.29 / (147989 \cdot 1.0 - 0.6 \cdot 18.0)$
 $= 32.5769 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 140.488 / 0.5773$
 $= 243.3321 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	993.1512	mm
Parallel to Vessel Wall, opening length	d	496.5756	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	382.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 147988.7 / 147988.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 147988.7 / 147988.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



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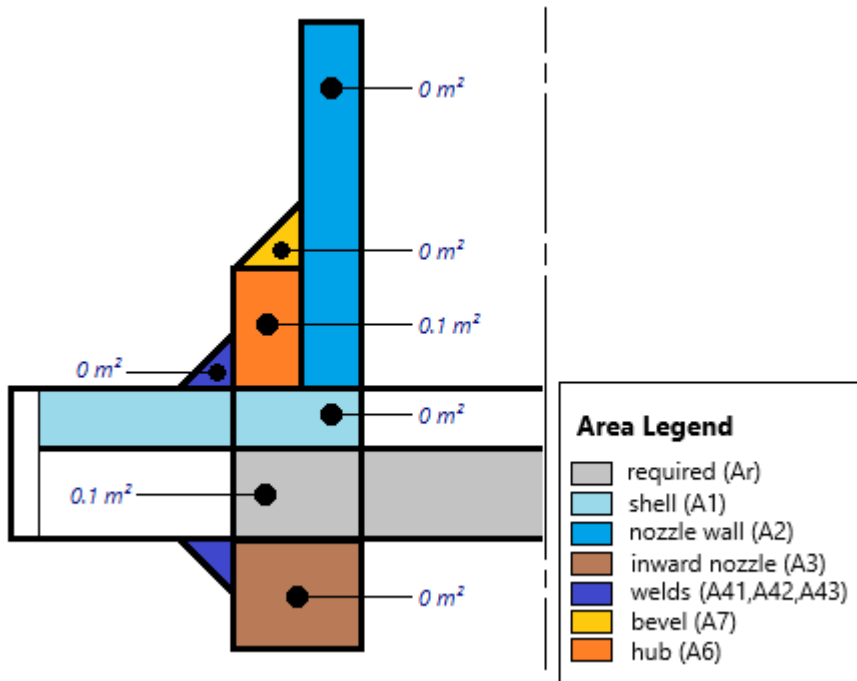
FileName : 121-C_NVS_MANWAY+JACKET_Final

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Nozzle Calcs.: N4- Tube outlet

Nozl: 14

3:25pm Apr 10, 2026



Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	0.073	NA	NA
Area in Shell	A1	0.002	NA	NA
Area in Nozzle Wall	A2	0.018	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	0.074	NA	NA
TOTAL AREA AVAILABLE	Atot	0.096	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (496.5756 \cdot 148.0005 \cdot 1.0 + 2 \cdot 197.0 \cdot 148.0005 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 0.073 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 496.576(1.0 \cdot 153.0 - 1.0 \cdot 148.001) - 2 \cdot 197.0$$

$$(1.0 \cdot 153.0 - 1.0 \cdot 148.0005) \cdot (1 - 1.0)$$

$$= 0.002 \text{ m}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 382.5)(56.51 - 32.58)1.0$$

$$= 0.018 \text{ m}^2$$



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 Nozzle Calcs.: N4- Tube outlet Nozl: 14 3:25pm Apr 10,2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$
 $= 20.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 0.000 \text{ m}^2$

Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$
 $= (2 \cdot \min(382.5, 500.0, 220.0)) \cdot (200.0 - 59.5) \cdot 1.0$
 $= 0.074 \text{ m}^2$ Includes Bevel Area of 0.013 m²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 35.5769 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 151.0005 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 151.0005 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[11.331, \max(151.0005, 4.5)]$
 $= 11.3312 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(35.5769, 11.3312)$
 $= 35.5769 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 59.512 = 52.073 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 47003.3,	Allowable	: 147988.7 kPa	Passed
Expansion	: 0.0,	Allowable	: 322968.5 kPa	Passed
Occasional	: 35501.5,	Allowable	: 196825.0 kPa	Passed
Shear	: 7678.0,	Allowable	: 103592.1 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Weld Size Calculations, Description: N4- Tube outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$14.1400 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) S_v)$
 $= \max(0, (0.0735 - 0.0025 + 2 \cdot 197.0 \cdot 1.0 \cdot (1.0 \cdot 153.0 - 148.0005)) / 147989)$



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 Nozzle Calcs.: N4- Tube outlet Nozl: 14 3:25pm Apr 10,2026

= 10800017.00 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - C_{an}/.707)^2 \cdot fr2) \cdot S_v$$

$$= (0.0927 + 0.0004 - 0.0 \cdot 1.0) \cdot 147989$$

$$= 13787597.00 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S_v$$

$$= (0.0927 + 0.0 + 0.0004 + (0.0603)) \cdot 147989$$

$$= 22708412.00 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (0.0927 + 0.0 + 0.0004 + 0.0 + (0.0603)) \cdot 147989$$

$$= 22708412.00 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 890.5756 \cdot 20.0 \cdot 0.49 \cdot 147989$$

$$= 2028774. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - C_{an}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 346.7878) \cdot (200.0 - 3.0) \cdot 0.7 \cdot 147989$$

$$= 22232838. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 890.5756 \cdot (50.0 - 0.0) \cdot 0.74 \cdot 147989$$

$$= 7659657. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (2028774 + 22232838) = 24261614 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (2028774 + 0 + 7659657 + 0) = 9688431 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (2028774 + 7659657 + 0) = 9688431 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 24261614N , must exceed W = 10800017 N or W1 = 13787597 N
 Path 2-2 = 9688431 N , must exceed W = 10800017 N or W2 = 22708412 N
 Path 3-3 = 9688431 N , must exceed W = 10800017 N or W3 = 22708412 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 17 N/mm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 91.8767 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 747.8766 mm



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 Nozzle Calcs.: N4- Tube outlet Nozl: 14 3:25pm Apr 10,2026

Input Echo, WRC297 Item 1, Description: N4- Tube outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2250.000	mm
Shell Thickness	156.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 147988.73	kPa
Vessel Hot S.I. Allowable	Smh 147988.73	kPa
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	490.576	mm
Nozzle Thickness	200.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 147988.73	kPa
Nozzle Hot S.I. Allowable	SNmh 147988.73	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	18.00	N/mm ²
Radial Load	P	-74967.03	N
Circumferential Shear	Vc	59973.27	N
Longitudinal Shear	Vl	59973.27	N
Circumferential Moment	Mc	107995.79	N-m
Longitudinal Moment	Ml	107995.79	N-m
Torsional Moment	Mt	125996.74	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_v/T_v (16.423) is < 20
 check the limitations of WRC bulletin 297.

Warning - The ratio, D_n/T_n (4.453) is < 10
 check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 890.576 + 2 \cdot 1.65 \cdot \sqrt{(1204.5(156.0 - 3.0))}$
 $= 2307.227 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2409.000 mm
Nozzle Outside Diameter	(d) =	890.576 mm
Vessel Thickness used	(T) =	153.000 mm
Nozzle Thickness used	(t) =	197.000 mm
T / t	=	0.777
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.467$$

$$Nr/P = 0.095$$



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Nozzle Calcs.: N4- Tube outlet Nozl: 14 3:25pm Apr 10,2026

Mr/P = 0.159
MØ/P = 0.050
NØ/P = 0.111
MrD/Mc = 0.279
NrDL/Mc = 0.124
MØd/Mc = 0.096
NØDL/Mc = 0.053
MrD/Ml = 0.149
NrDL/Ml = 0.085
MØD/Ml = 0.052
NØDL/Ml = 0.134

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	351	351	351	351
Outplane Bending (P)	958	-958	958	-958
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-689	-689	689	689
Outplane Bending (ML)	-1606	1606	1606	-1606
Normal Pressure Stress	124275	142276	124275	142276

Outplane Stress Summary 123289 142587 127881 140753

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	296	296	296	296
Inplane Bending (P)	3047	-3047	3047	-3047
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-441	-441	441	441
Inplane Bending (ML)	-4633	4633	4633	-4633
Inplane Pressure Stress	62134	62134	62134	62134

Inplane Stress Summary 60403 63575 70552 55191

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	275	275	-275	-275
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	654	654	654	654

Shear Stress Summary 930 930 379 379

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 123296 142594 127881 140753

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Nozzle Calcs.: N4- Tube outlet Nozl: 14 3:25pm Apr 10,2026

Outplane Membrane (P)	351	351	351	351
Outplane Bending (P)	958	-958	958	-958
Outplane Membrane (Mc)	-268	-268	268	268
Outplane Bending (Mc)	-2964	2964	2964	-2964
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	62134	62134	62134	62134

Outplane Stress Summary	60210	64223	66677	58831
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	296	296	296	296
Inplane Bending (P)	3047	-3047	3047	-3047
Inplane Membrane (Mc)	-634	-634	634	634
Inplane Bending (Mc)	-8666	8666	8666	-8666
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	124275	142276	124275	142276

Inplane Stress Summary	118318	147558	136919	131493
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-275	-275	275	275
Torsional Shear (Mt)	654	654	654	654

Shear Stress Summary	379	379	930	930
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	118318	147558	136926	131500
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	351	351	351	351
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-689	-689	689	689
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	19484	41084	19484	41084

Hoop Stress Summary	19146	40747	20525	42126
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	172	172	172	172
Axial Bending (P)	1289	-1289	1289	-1289



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Nozzle Calcs.: N4- Tube outlet Nozl: 14 3:25pm Apr 10,2026

Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1716	-1716	1716	1716
Axial Bending (ML)	-1992	1992	1992	-1992
Axial Pressure Stress	9742	9742	9742	9742

Axial Stress Summary	7494	8900	14913	8349
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	213	213	-213	-213
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	510	510	510	510
Shear Stress Summary	723	723	296	296

Nozzle Stresses (kPa):

LONGITUDINAL PLANE				
(Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	19187	40760	20539	42126

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE				
(Stresses in the	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
hoop direction)				
Hoop Membrane (P)	351	351	351	351
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-268	-268	268	268
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	19484	41084	19484	41084
Hoop Stress Summary	19566	41167	20104	41705

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
pipe cross-section)				
Axial Membrane (P)	172	172	172	172
Axial Bending (P)	1289	-1289	1289	-1289
Axial Membrane (Mc)	-1716	-1716	1716	1716
Axial Bending (Mc)	-4067	4067	4067	-4067
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	9742	9742	9742	9742
Axial Stress Summary	5419	10976	16988	6274

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-213	-213	213	213
Shear due to Torsion	510	510	510	510
Shear Stress Summary	296	296	723	723

Nozzle Stresses (kPa):



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 Nozzle Calcs.: N4- Tube outlet Nozl: 14 3:25pm Apr 10,2026

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	19566	41167	20263	41719

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	124275	142276	124275	142276	124275	142276	124275	142276
Circ. Pl (SUS)	-338	-338	1040	1040	-338	-338	930	930
Circ. Q (SUS)	-648	648	2564	-2564	-5619	5619	11713	-11713
Long. Pm (SUS)	62134	62134	62134	62134	62134	62134	62134	62134
Long. Pl (SUS)	-145	-145	737	737	83	83	619	619
Long. Q (SUS)	-1586	1586	7680	-7680	-2006	2006	3922	-3922
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	275	275	-275	-275	-275	-275	275	275
Shear Q (SUS)	654	654	654	654	654	654	654	654
Pm (SUS)	124275	142276	124275	142276	124275	142276	124275	142276
Pm+Pl (SUS)	123938	141938	125316	143316	123938	141938	125206	143206
Pm+Pl+Q (Total)	123302	142596	127881	140753	118320	147558	136930	131504

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	142276	147988	Passed
Pm+Pl (SUS)	143316	221983	Passed
Pm+Pl+Q (TOTAL)	147558	443966	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	19484	41084	19484	41084	19484	41084	19484	41084
Circ. Pl (SUS)	-338	-338	1040	1040	83	83	619	619
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	9742	9742	9742	9742	9742	9742	9742	9742
Long. Pl (SUS)	-1544	-1544	1888	1888	-1544	-1544	1888	1888
Long. Q (SUS)	-703	703	3281	-3281	-2778	2778	5356	-5356
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	213	213	-213	-213	-213	-213	213	213
Shear Q (SUS)	510	510	510	510	510	510	510	510
Pm (SUS)	19484	41084	19484	41084	19484	41084	19484	41084
Pm+Pl (SUS)	19150	40747	20529	42125	19570	41168	20108	41704



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Nozzle Calcs.: N4- Tube outlet Nozl: 14 3:25pm Apr 10,2026

Pm+Pl+Q (Total)	19190	40762	20539	42126	19573	41169	20262	41717
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Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	41084	147988	Passed
Pm+Pl (SUS)	42125	221983	Passed
Pm+Pl+Q (TOTAL)	42126	443966	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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 Nozzle Calcs.: N1- Shell inlet Nozl: 15 3:25pm Apr 10,2026

Input, Nozzle Desc: N1- Shell inlet

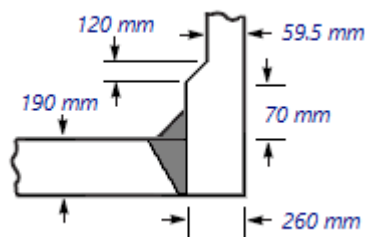
From: 30

Pressure for Reinforcement Calculations	P	18.0010	N/mm ²
Temperature for Internal Pressure	Temp	302	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147988.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa
Inside Diameter of Cylindrical Shell	D	2820.00	mm
Shell Finished (Minimum) Thickness	t	190.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3.26	m
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147988.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	120.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		260.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	50.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N1- Shell inlet



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 Nozzle Calcs.: N1- Shell inlet Nozl: 15 3:25pm Apr 10,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	19.314 in.
Actual Thickness Used in Calculation	2.343 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 18.0 \cdot 1413.0 / (147989 \cdot 1.0 - 0.6 \cdot 18.0)$
 $= 185.3943 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 18.0 \cdot 248.29 / (147989 \cdot 1.0 - 0.6 \cdot 18.0)$
 $= 32.5769 \text{ mm}$

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 191.776 / 0.5773$
 $= 332.1652 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	993.1512	mm
Parallel to Vessel Wall, opening length	d	496.5756	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	331.2805	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 147988.7 / 147988.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 147988.7 / 147988.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



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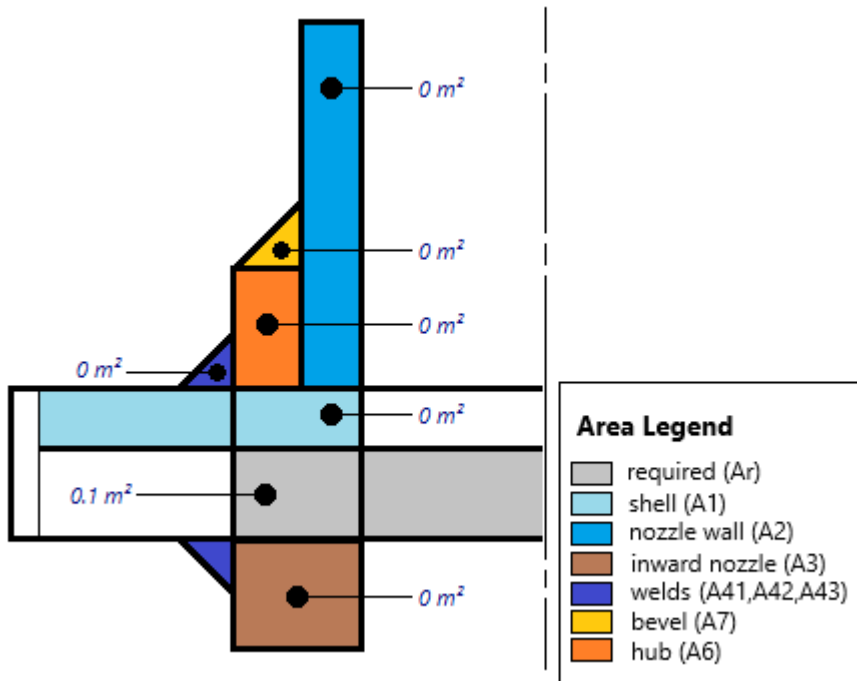
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Nozzle Calcs.: N1- Shell inlet

Nozl: 15

3:25pm

Apr 10, 2026



Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	0.092	NA	NA
Area in Shell	A1	0.001	NA	NA
Area in Nozzle Wall	A2	0.012	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	0.050	NA	NA
TOTAL AREA AVAILABLE	Atot	0.063	NA	NA

Additional Area Needed, Ar - Atot: 0.029 m²

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Insufficient.

RECOMMENDATION: Add a Reinforcing Pad.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on the Estimated Diameter Limit: 991.5637 77.0638 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (496.5756 \cdot 185.3943 \cdot 1.0 + 2 \cdot 248.2878 \cdot 185.3943 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 0.092 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 496.576(1.0 \cdot 187.0 - 1.0 \cdot 185.394) - 2 \cdot 248.288$$

$$(1.0 \cdot 187.0 - 1.0 \cdot 185.3943) \cdot (1 - 1.0)$$

$$= 0.001 \text{ m}^2$$



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 Nozzle Calcs.: N1- Shell inlet Nozl: 15 3:25pm Apr 10,2026

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot h_o)(t_n - t_{rn})f_r2$$

$$= (2 \cdot 250.0)(56.51 - 32.58)1.0$$

$$= 0.012 \text{ m}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (W_o^2 - \text{Area Lost}) \cdot f_r2 + (W_i\text{-can}/0.707)^2 - \text{Area Lost} \cdot f_r2$$

$$= (20.0^2 - 0.0004) \cdot 1.0 + (0.0^2 - 0.0) \cdot 1.0$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_r2$$

$$= (2 \cdot \min(250.0, 250.0, 70.0)) \cdot (251.3 - 59.5) \cdot 1.0$$

$$= 0.050 \text{ m}^2 \text{ Includes Bevel Area of } 0.023 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 35.5769 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 188.3943 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 188.3943 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(188.3943, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(35.5769, 11.3312)$$

$$= 35.5769 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 59.512 = 52.073 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 47003.3,	Allowable	: 147988.7 kPa	Passed
Expansion	: 0.0,	Allowable	: 322968.5 kPa	Passed
Occasional	: 35501.5,	Allowable	: 196825.0 kPa	Passed
Shear	: 7678.0,	Allowable	: 103592.1 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Weld Size Calculations, Description: N1- Shell inlet

Intermediate Calc. for nozzle/shell welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = $0.7 \cdot W_o$ mm



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 Nozzle Calcs.: N1- Shell inlet Nozl: 15 3:25pm Apr 10,2026

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (0.092 - 0.0008 + 2 \cdot 248.2878 \cdot 1.0 \cdot (1.0 \cdot 187.0 - 185.3943)) \cdot 147989)$$

$$= 13623810.00 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (0.0618 + 0.0 - 0.0 \cdot 1.0) \cdot 147989$$

$$= 9149796.00 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (0.0618 + 0.0 + 0.0 + (0.0928)) \cdot 147989$$

$$= 22891606.00 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (0.0618 + 0.0 + 0.0 + 0.0 + (0.0928)) \cdot 147989$$

$$= 22891606.00 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 993.1512 \cdot 20.0 \cdot 0.49 \cdot 147989$$

$$= 2262446. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 372.4317) \cdot (251.2878 - 3.0) \cdot 0.7 \cdot 147989$$

$$= 30093094. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 993.1512 \cdot (50.0 - 0.0) \cdot 0.74 \cdot 147989$$

$$= 8541889. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (2262446 + 30093094) = 32355542 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (2262446 + 0 + 8541889 + 0) = 10804335 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (2262446 + 8541889 + 0) = 10804335 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 32355542N , must exceed W = 13623810 N or W1 = 9149796 N
 Path 2-2 = 10804335N , must exceed W = 13623810 N or W2 = 22891606 N
 Path 3-3 = 10804335N , must exceed W = 13623810 N or W3 = 22891606 N

Maximum Allowable Pressure for this Nozzle at this Location:



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 Nozzle Calcs.: N1- Shell inlet Nozl: 15 3:25pm Apr 10,2026

Converged Maximum Allowable Pressure in the Operating case: 14 N/mm²

Warning: The converged MAWP of nozzle junction was less than the design pressure.

The Drop for this Nozzle is : 90.3362 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 530.3362 mm

Input Echo, WRC297 Item 1, Description: N1- Shell inle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2820.000	mm
Shell Thickness	190.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 147988.73	kPa
Vessel Hot S.I. Allowable	Smh 147988.73	kPa

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	490.576	mm
Nozzle Thickness	260.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 147988.73	kPa
Nozzle Hot S.I. Allowable	SNmh 147988.73	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	18.00	N/mm ²
Radial Load	P	-74967.03	N
Circumferential Shear	Vc	59973.27	N
Longitudinal Shear	Vl	59973.27	N
Circumferential Moment	Mc	107995.79	N-m
Longitudinal Moment	Ml	107995.79	N-m
Torsional Moment	Mt	125996.74	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (16.842) is < 20
 check the limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (3.887) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 1010.576 + 2 \cdot 1.65 \cdot \sqrt{(1506.5(190.0 - 3.0))}$$

$$= 2762.113 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3013.000 mm
Nozzle Outside Diameter	(d) =	1010.576 mm
Vessel Thickness used	(T) =	187.000 mm
Nozzle Thickness used	(t) =	257.000 mm



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Nozzle Calcs.: N1- Shell inlet Nozl: 15 3:25pm Apr 10,2026

T / t = 0.728
d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.346$$

Nr/P = 0.097
Mr/P = 0.171
M0/P = 0.053
N0/P = 0.115
MrD/Mc = 0.286
NrDL/Mc = 0.118
M0d/Mc = 0.097
N0DL/Mc = 0.051
MrD/Ml = 0.160
NrDL/Ml = 0.088
M0D/Ml = 0.055
N0DL/Ml = 0.133

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	241	241	241	241
Outplane Bending (P)	682	-682	682	-682
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-399	-399	399	399
Outplane Bending (ML)	-999	999	999	-999
Normal Pressure Stress	127563	145565	127563	145565
Outplane Stress Summary	127088	145724	129887	144524

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	206	206	206	206
Inplane Bending (P)	2199	-2199	2199	-2199
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-268	-268	268	268
Inplane Bending (ML)	-2923	2923	2923	-2923
Inplane Pressure Stress	63781	63781	63781	63781
Inplane Stress Summary	62995	64443	69380	59134

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	199	199	-199	-199
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	413	413	413	413
Shear Stress Summary	613	613	213	213

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	127088	145724	129887	144524



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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	241	241	241	241
Outplane Bending (P)	682	-682	682	-682
Outplane Membrane (Mc)	-151	-151	151	151
Outplane Bending (Mc)	-1771	1771	1771	-1771
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	63781	63781	63781	63781
Outplane Stress Summary	62782	64960	66629	61720

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	206	206	206	206
Inplane Bending (P)	2199	-2199	2199	-2199
Inplane Membrane (Mc)	-358	-358	358	358
Inplane Bending (Mc)	-5246	5246	5246	-5246
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	127563	145565	127563	145565
Inplane Stress Summary	124364	148461	135575	138684

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-199	-199	199	199
Torsional Shear (Mt)	413	413	413	413
Shear Stress Summary	213	213	613	613

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	124364	148461	135575	138684

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	241	241	241	241
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-399	-399	399	399
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13747	35348	13747	35348
Hoop Stress Summary	13589	35190	14389	35989

Nozzle Stresses (kPa):



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Nozzle Calcs.: N1- Shell inlet Nozl: 15 3:25pm Apr 10,2026

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	117	117	117	117
Axial Bending (P)	827	-827	827	-827
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1130	-1130	1130	1130
Axial Bending (ML)	-1116	1116	1116	-1116
Axial Pressure Stress	6873	6873	6873	6873
Axial Stress Summary	5570	6149	10066	6177

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	144	144	-144	-144
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	303	303	303	303
Shear Stress Summary	448	448	158	158

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	13609	35196	14389	35989

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	241	241	241	241
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-151	-151	151	151
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13747	35348	13747	35348
Hoop Stress Summary	13837	35438	14140	35741

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	117	117	117	117
Axial Bending (P)	827	-827	827	-827
Axial Membrane (Mc)	-1130	-1130	1130	1130
Axial Bending (Mc)	-2199	2199	2199	-2199
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6873	6873	6873	6873
Axial Stress Summary	4488	7232	11148	5095

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0



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Shear due to (V1)	-144	-144	144	144
Shear due to Torsion	303	303	303	303
Shear Stress Summary	158	158	448	448

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	13837	35438	14202	35741

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	127563	145565	127563	145565	127563	145565	127563	145565
Circ. Pl (SUS)	-158	-158	640	640	-152	-152	564	564
Circ. Q (SUS)	-317	317	1681	-1681	-3047	3047	7445	-7445
Long. Pm (SUS)	63781	63781	63781	63781	63781	63781	63781	63781
Long. Pl (SUS)	-62	-62	474	474	90	90	392	392
Long. Q (SUS)	-724	724	5122	-5122	-1089	1089	2453	-2453
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	199	199	-199	-199	-199	-199	199	199
Shear Q (SUS)	413	413	413	413	413	413	413	413
Pm (SUS)	127563	145565	127563	145565	127563	145565	127563	145565
Pm+Pl (SUS)	127405	145407	128203	146205	127411	145413	128127	146129
Pm+Pl+Q (Total)	127093	145728	129884	144524	124364	148460	135577	138688

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	145565	147988	Passed
Pm+Pl (SUS)	146205	221983	Passed
Pm+Pl+Q (TOTAL)	148460	443966	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13747	35348	13747	35348	13747	35348	13747	35348
Circ. Pl (SUS)	-158	-158	640	640	90	90	392	392
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6873	6873	6873	6873	6873	6873	6873	6873
Long. Pl (SUS)	-1013	-1013	1247	1247	-1013	-1013	1247	1247
Long. Q (SUS)	-289	289	1943	-1943	-1372	1372	3026	-3026



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Nozzle Calcs.: N1- Shell inlet Nozl: 15 3:25pm Apr 10,2026

Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	144	144	-144	-144	-144	-144	144	144
Shear Q (SUS)	303	303	303	303	303	303	303	303
Pm (SUS)	13747	35348	13747	35348	13747	35348	13747	35348
Pm+Pl (SUS)	13591	35190	14390	35988	13839	35438	14142	35740
Pm+Pl+Q (Total)	13613	35196	14392	35988	13839	35438	14204	35746

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	35348	147988	Passed
Pm+Pl (SUS)	35988	221983	Passed
Pm+Pl+Q (TOTAL)	35988	443966	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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 Nozzle Calcs.: N2- Shell outle Nozl: 16 3:25pm Apr 10,2026

Input, Nozzle Desc: N2- Shell outle

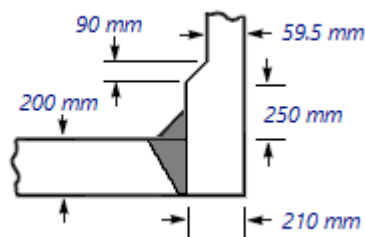
From: 70

Pressure for Reinforcement Calculations	P	18.0022	N/mm ²
Temperature for Internal Pressure	Temp	302	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147988.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa
Inside Diameter of Cylindrical Shell	D	2820.00	mm
Shell Finished (Minimum) Thickness	t	200.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		19.90	m
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147988.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	250.0000	mm
Height of Beveled Transition	L'	90.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		210.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	530.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	50.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	50.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N2- Shell outle



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 Nozzle Calcs.: N2- Shell outle Nozl: 16 3:25pm Apr 10,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 19.314 in.
 Actual Thickness Used in Calculation 2.343 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 18.0 \cdot 1413.0 / (147989 \cdot 1.0 - 0.6 \cdot 18.0)$
 $= 185.4073 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 18.0 \cdot 248.29 / (147989 \cdot 1.0 - 0.6 \cdot 18.0)$
 $= 32.5792 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 150.488 / 0.5773$
 $= 260.6526 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1003.6000	mm
Parallel to Vessel Wall	Rn+tn+t	501.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	401.9331	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 147988.7 / 147988.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 147988.7 / 147988.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



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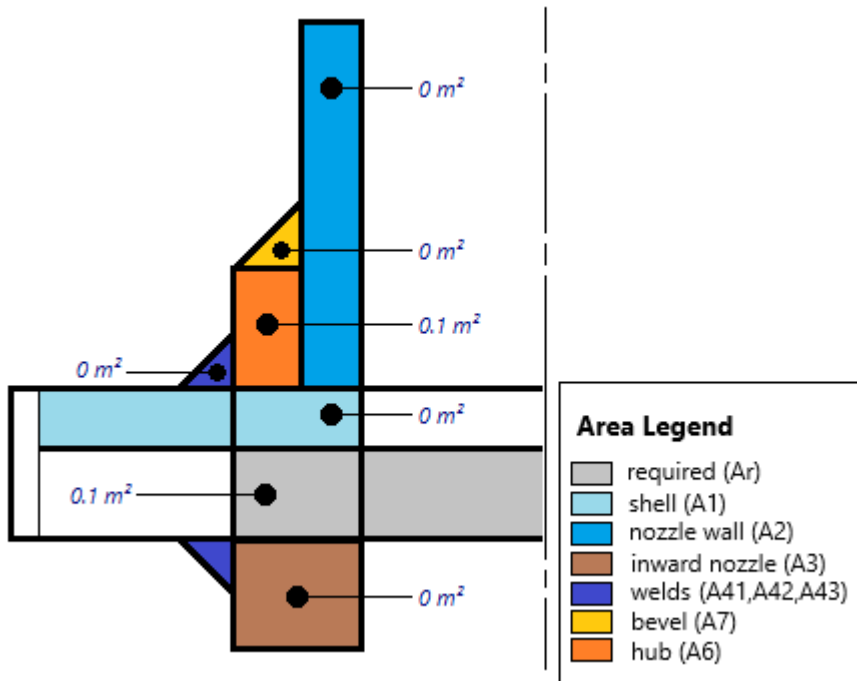
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Nozzle Calcs.: N2- Shell outle

Nozl:

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Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	0.092	NA	NA
Area in Shell	A1	0.006	NA	NA
Area in Nozzle Wall	A2	0.019	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.002	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	0.089	NA	NA
TOTAL AREA AVAILABLE	Atot	0.116	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (496.5756 \cdot 185.4073 \cdot 1.0 + 2 \cdot 207.0 \cdot 185.4073 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 0.092 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 507.024(1.0 \cdot 197.0 - 1.0 \cdot 185.407) - 2 \cdot 207.0$$

$$(1.0 \cdot 197.0 - 1.0 \cdot 185.4073) \cdot (1 - 1.0)$$

$$= 0.006 \text{ m}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 401.93)(56.51 - 32.58)1.0$$

$$= 0.019 \text{ m}^2$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (Wo^2 - \text{Area Lost}) \cdot fr2 + (Wi\text{-}can/0.707)^2 - \text{Area Lost}) \cdot fr2$$

$$= (50.0^2 - 0.) \cdot 1.0 + (0.0^2 - 0.0) \cdot 1.0$$

$$= 0.002 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(401.9, 530.0, 250.0)) \cdot (210.0 - 59.5) \cdot 1.0$$

$$= 0.089 \text{ m}^2 \text{ Includes Bevel Area of } 0.014 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 35.5792 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 188.4073 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 188.4073 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(188.4073, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(35.5792, 11.3312)$$

$$= 35.5792 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 59.512 = 52.073 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 47005.6,	Allowable	: 147988.7 kPa	Passed
Expansion	: 0.0,	Allowable	: 322966.2 kPa	Passed
Occasional	: 35503.8,	Allowable	: 196825.0 kPa	Passed
Shear	: 7678.0,	Allowable	: 103592.1 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Weld Size Calculations, Description: N2- Shell outle

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$35.3500 = 0.7 \cdot Wo \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr))Sv)$$

$$= \max(0, (0.092 - 0.0059 + 2 \cdot 207.0 \cdot 1.0 \cdot (1.0 \cdot 197.0 - 185.4073))147989)$$



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= 13465179.00 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - C_{an}/.707)^2 \cdot fr2) \cdot S_v$$

$$= (0.108 + 0.0025 - 0.0 \cdot 1.0) \cdot 147989$$

$$= 16354473.00 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S_v$$

$$= (0.108 + 0.0 + 0.0025 + (0.0815)) \cdot 147989$$

$$= 28423812.00 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (0.108 + 0.0 + 0.0025 + 0.0 + (0.0815)) \cdot 147989$$

$$= 28423812.00 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 910.5756 \cdot 50.0 \cdot 0.49 \cdot 147989$$

$$= 5185838. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - C_{an}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 351.7878) \cdot (210.0 - 3.0) \cdot 0.7 \cdot 147989$$

$$= 23698232. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 910.5756 \cdot (50.0 - 0.0) \cdot 0.74 \cdot 147989$$

$$= 7831674. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (5185838 + 23698232) = 28884070 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (5185838 + 0 + 7831674 + 0) = 13017512 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (5185838 + 7831674 + 0) = 13017512 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 28884070N , must exceed W = 13465179 N or W1 = 16354473 N
 Path 2-2 = 13017512N , must exceed W = 13465179 N or W2 = 28423812 N
 Path 3-3 = 13017512N , must exceed W = 13465179 N or W3 = 28423812 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 18 N/mm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 75.5289 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 805.5289 mm



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Input Echo, WRC297 Item 1, Description: N2- Shell outl:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2820.000	mm
Shell Thickness	200.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 147988.73	kPa
Vessel Hot S.I. Allowable	Smh 147988.73	kPa

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	490.576	mm
Nozzle Thickness	210.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 147988.73	kPa
Nozzle Hot S.I. Allowable	SNmh 147988.73	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	18.00	N/mm ²
Radial Load	P	-74967.03	N
Circumferential Shear	Vc	59973.27	N
Longitudinal Shear	Vl	59973.27	N
Circumferential Moment	Mc	107995.79	N-m
Longitudinal Moment	Ml	107995.79	N-m
Torsional Moment	Mt	125996.74	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_v/T_v (16.100) is < 20
 check the limitations of WRC bulletin 297.

Warning - The ratio, D_n/T_v (4.553) is < 5
 check the limitations of WRC bulletin 297.

Warning - The ratio, D_n/T_n (4.336) is < 10
 check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 910.576 + 2 \cdot 1.65 \cdot \sqrt{1511.5(200.0 - 3.0)}$
 $= 2711.316 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3023.000 mm
Nozzle Outside Diameter	(d) =	910.576 mm
Vessel Thickness used	(T) =	197.000 mm
Nozzle Thickness used	(t) =	207.000 mm
T / t	=	0.952
d / t	=	10.000

$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.180$



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Nr/P = 0.115
 Mr/P = 0.172
 MØ/P = 0.058
 NØ/P = 0.116
 MrD/Mc = 0.277
 NrDL/Mc = 0.141
 MØd/Mc = 0.103
 NØDL/Mc = 0.042
 MrD/Ml = 0.178
 NrDL/Ml = 0.102
 MØD/Ml = 0.065
 NØDL/Ml = 0.127

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	220	220	220	220
Outplane Bending (P)	661	-661	661	-661
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-386	-386	386	386
Outplane Bending (ML)	-1192	1192	1192	-1192
Normal Pressure Stress	120696	138698	120696	138698
Outplane Stress Summary	120000	139064	123158	137450

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	220	220	220	220
Inplane Bending (P)	1992	-1992	1992	-1992
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-310	-310	310	310
Inplane Bending (ML)	-3268	3268	3268	-3268
Inplane Pressure Stress	60348	60348	60348	60348
Inplane Stress Summary	58983	61534	66139	55618

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	206	206	-206	-206
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	489	489	489	489
Shear Stress Summary	696	696	282	282

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	120007	139064	123158	137450

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to	Cu Left	Cl Left	Du Right	Dl Right
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Nozzle Calcs.: N2- Shell outle Nozl: 16 3:25pm Apr 10,2026

circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	220	220	220	220
Outplane Bending (P)	661	-661	661	-661
Outplane Membrane (Mc)	-124	-124	124	124
Outplane Bending (Mc)	-1882	1882	1882	-1882
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	60348	60348	60348	60348
Outplane Stress Summary	59224	61665	63237	58149

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	220	220	220	220
Inplane Bending (P)	1992	-1992	1992	-1992
Inplane Membrane (Mc)	-427	-427	427	427
Inplane Bending (Mc)	-5081	5081	5081	-5081
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	120696	138698	120696	138698
Inplane Stress Summary	117401	141580	128418	132272

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-206	-206	206	206
Torsional Shear (Mt)	489	489	489	489
Shear Stress Summary	282	282	696	696

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	117401	141580	128425	132272

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	220	220	220	220
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-386	-386	386	386
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	18284	39885	18284	39885
Hoop Stress Summary	18119	39719	18891	40491

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	158	158	158	158
Axial Bending (P)	1199	-1199	1199	-1199
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1592	-1592	1592	1592
Axial Bending (ML)	-2109	2109	2109	-2109
Axial Pressure Stress	9142	9142	9142	9142
Axial Stress Summary	6798	8618	14202	7584

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	199	199	-199	-199
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	461	461	461	461
Shear Stress Summary	661	661	261	261

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	18153	39733	18904	40491

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	220	220	220	220
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-124	-124	124	124
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	18284	39885	18284	39885
Hoop Stress Summary	18381	39981	18629	40229

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	158	158	158	158
Axial Bending (P)	1199	-1199	1199	-1199
Axial Membrane (Mc)	-1592	-1592	1592	1592
Axial Bending (Mc)	-3433	3433	3433	-3433
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	9142	9142	9142	9142
Axial Stress Summary	5474	9942	15526	6260

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-199	-199	199	199
Shear due to Torsion	461	461	461	461
Shear Stress Summary	261	261	661	661



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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	18381	39981	18760	40236

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	120696	138698	120696	138698	120696	138698	120696	138698
Circ. Pl (SUS)	-166	-166	606	606	-207	-207	647	647
Circ. Q (SUS)	-531	531	1853	-1853	-3089	3089	7073	-7073
Long. Pm (SUS)	60348	60348	60348	60348	60348	60348	60348	60348
Long. Pl (SUS)	-90	-90	530	530	96	96	344	344
Long. Q (SUS)	-1276	1276	5260	-5260	-1221	1221	2543	-2543
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	206	206	-206	-206	-206	-206	206	206
Shear Q (SUS)	489	489	489	489	489	489	489	489
Pm (SUS)	120696	138698	120696	138698	120696	138698	120696	138698
Pm+Pl (SUS)	120530	138532	121302	139304	120489	138491	121343	139345
Pm+Pl+Q (Total)	120006	139069	123156	137451	117401	141581	128423	132278

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	138698	147988	Passed
Pm+Pl (SUS)	139345	221983	Passed
Pm+Pl+Q (TOTAL)	141581	443966	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	18284	39885	18284	39885	18284	39885	18284	39885
Circ. Pl (SUS)	-166	-166	606	606	96	96	344	344
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	9142	9142	9142	9142	9142	9142	9142	9142
Long. Pl (SUS)	-1434	-1434	1750	1750	-1434	-1434	1750	1750
Long. Q (SUS)	-910	910	3308	-3308	-2234	2234	4632	-4632
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	199	199	-199	-199	-199	-199	199	199
Shear Q (SUS)	461	461	461	461	461	461	461	461
Pm (SUS)	18284	39885	18284	39885	18284	39885	18284	39885



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Pm+Pl (SUS)	18121	39720	18894	40492	18383	39982	18633	40230
Pm+Pl+Q (Total)	18156	39733	18904	40493	18385	39983	18762	40241

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	39885	147988	Passed
Pm+Pl (SUS)	40492	221983	Passed
Pm+Pl+Q (TOTAL)	40493	443966	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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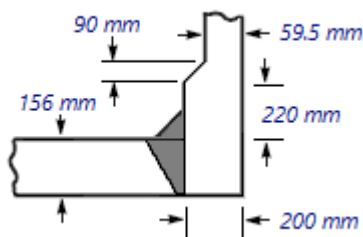
Input, Nozzle Desc: N3- tube inlet From: 80

Pressure for Reinforcement Calculations	P	18.0019	N/mm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147988.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa
Inside Diameter of Cylindrical Shell	D	2250.00	mm
Shell Finished (Minimum) Thickness	t	156.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		22.26	m
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147988.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	220.0000	mm
Height of Beveled Transition	L'	90.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		200.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	50.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N3- tube inlet



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ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 19.314 in.
 Actual Thickness Used in Calculation 2.343 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 18.0 \cdot 1128.0 / (147989 \cdot 1.0 - 0.6 \cdot 18.0)$
 $= 148.0077 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 18.0 \cdot 248.29 / (147989 \cdot 1.0 - 0.6 \cdot 18.0)$
 $= 32.5785 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 140.488 / 0.5773$
 $= 243.3321 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	993.1512	mm
Parallel to Vessel Wall, opening length	d	496.5756	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	382.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 147988.7 / 147988.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 147988.7 / 147988.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



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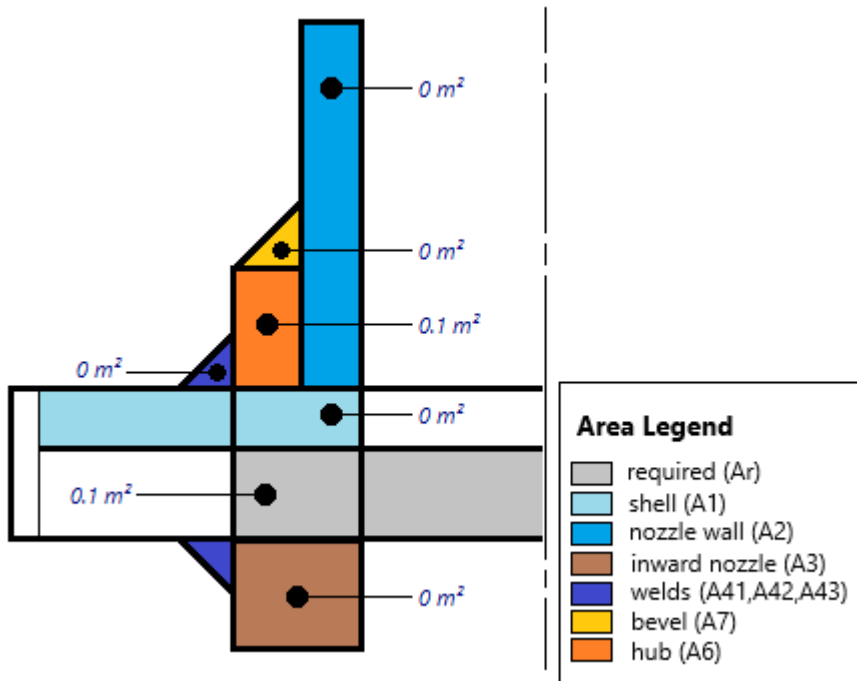
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Nozzle Calcs.: N3- tube inlet

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Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	0.073	NA	NA
Area in Shell	A1	0.002	NA	NA
Area in Nozzle Wall	A2	0.018	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	0.074	NA	NA
TOTAL AREA AVAILABLE	Atot	0.096	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (496.5756 \cdot 148.0077 \cdot 1.0 + 2 \cdot 197.0 \cdot 148.0077 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 0.073 \text{ m}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 496.576(1.0 \cdot 153.0 - 1.0 \cdot 148.008) - 2 \cdot 197.0 \\
 &\quad (1.0 \cdot 153.0 - 1.0 \cdot 148.0077) \cdot (1 - 1.0) \\
 &= 0.002 \text{ m}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tlnp)(tn - trn)fr2 \\
 &= (2 \cdot 382.5)(56.51 - 32.58)1.0 \\
 &= 0.018 \text{ m}^2
 \end{aligned}$$



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 Nozzle Calcs.: N3- tube inlet Nozl: 17 3:25pm Apr 10,2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$
 $= 20.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 0.000 \text{ m}^2$

Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$
 $= (2 \cdot \min(382.5, 500.0, 220.0)) \cdot (200.0 - 59.5) \cdot 1.0$
 $= 0.074 \text{ m}^2$ Includes Bevel Area of 0.013 m²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 35.5785 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 151.0077 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 151.0077 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[11.331, \max(151.0077, 4.5)]$
 $= 11.3312 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(35.5785, 11.3312)$
 $= 35.5785 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 59.512 = 52.073 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 47004.9,	Allowable	: 147988.7 kPa	Passed
Expansion	: 0.0,	Allowable	: 322966.9 kPa	Passed
Occasional	: 35503.1,	Allowable	: 196825.0 kPa	Passed
Shear	: 7678.0,	Allowable	: 103592.1 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Weld Size Calculations, Description: N3- tube inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$14.1400 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) S_v)$
 $= \max(0, (0.0735 - 0.0025 + 2 \cdot 197.0 \cdot 1.0 \cdot (1.0 \cdot 153.0 - 148.0077)) / 147989)$



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= 10800659.00 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - C_{an}/1.707)^2 \cdot fr2) \cdot S_v$$

$$= (0.0927 + 0.0004 - 0.0 \cdot 1.0) \cdot 147989$$

$$= 13787418.00 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S_v$$

$$= (0.0927 + 0.0 + 0.0004 + (0.0603)) \cdot 147989$$

$$= 22708234.00 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (0.0927 + 0.0 + 0.0004 + 0.0 + (0.0603)) \cdot 147989$$

$$= 22708234.00 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 890.5756 \cdot 20.0 \cdot 0.49 \cdot 147989$$

$$= 2028774. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - C_{an}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 346.7878) \cdot (200.0 - 3.0) \cdot 0.7 \cdot 147989$$

$$= 22232838. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 890.5756 \cdot (50.0 - 0.0) \cdot 0.74 \cdot 147989$$

$$= 7659657. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (2028774 + 22232838) = 24261614 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (2028774 + 0 + 7659657 + 0) = 9688431 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (2028774 + 7659657 + 0) = 9688431 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 24261614N , must exceed W = 10800659 N or W1 = 13787418 N
 Path 2-2 = 9688431 N , must exceed W = 10800659 N or W2 = 22708234 N
 Path 3-3 = 9688431 N , must exceed W = 10800659 N or W3 = 22708234 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 17 N/mm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 91.8767 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 747.8766 mm



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 Nozzle Calcs.: N3- tube inlet Nozl: 17 3:25pm Apr 10,2026

Input Echo, WRC297 Item 1, Description: N3- tube inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2250.000	mm
Shell Thickness	156.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 147988.73	kPa
Vessel Hot S.I. Allowable	Smh 147988.73	kPa
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	490.576	mm
Nozzle Thickness	200.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 147988.73	kPa
Nozzle Hot S.I. Allowable	SNmh 147988.73	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	18.00	N/mm ²
Radial Load	P	-74967.03	N
Circumferential Shear	Vc	59973.27	N
Longitudinal Shear	Vl	59973.27	N
Circumferential Moment	Mc	107995.79	N-m
Longitudinal Moment	Ml	107995.79	N-m
Torsional Moment	Mt	125996.74	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_v/T_v (16.423) is < 20
 check the limitations of WRC bulletin 297.

Warning - The ratio, D_n/T_n (4.453) is < 10
 check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 890.576 + 2 \cdot 1.65 \cdot \sqrt{(1204.5(156.0 - 3.0))}$
 $= 2307.227 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2409.000 mm
Nozzle Outside Diameter	(d) =	890.576 mm
Vessel Thickness used	(T) =	153.000 mm
Nozzle Thickness used	(t) =	197.000 mm
T / t	=	0.777
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.467$$

$$Nr/P = 0.095$$



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Mr/P = 0.159
MØ/P = 0.050
NØ/P = 0.111
MrD/Mc = 0.279
NrDL/Mc = 0.124
MØd/Mc = 0.096
NØDL/Mc = 0.053
MrD/Ml = 0.149
NrDL/Ml = 0.085
MØD/Ml = 0.052
NØDL/Ml = 0.134

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	351	351	351	351
Outplane Bending (P)	958	-958	958	-958
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-689	-689	689	689
Outplane Bending (ML)	-1606	1606	1606	-1606
Normal Pressure Stress	124282	142283	124282	142283
Outplane Stress Summary	123296	142594	127887	140760

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	296	296	296	296
Inplane Bending (P)	3047	-3047	3047	-3047
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-441	-441	441	441
Inplane Bending (ML)	-4633	4633	4633	-4633
Inplane Pressure Stress	62141	62141	62141	62141
Inplane Stress Summary	60410	63582	70559	55198

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	275	275	-275	-275
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	654	654	654	654
Shear Stress Summary	930	930	379	379

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	123303	142601	127887	140760

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Nozzle Calcs.: N3- tube inlet Nozl: 17 3:25pm Apr 10,2026

Outplane Membrane (P)	351	351	351	351
Outplane Bending (P)	958	-958	958	-958
Outplane Membrane (Mc)	-268	-268	268	268
Outplane Bending (Mc)	-2964	2964	2964	-2964
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	62141	62141	62141	62141

Outplane Stress Summary	60217	64230	66684	58838
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	296	296	296	296
Inplane Bending (P)	3047	-3047	3047	-3047
Inplane Membrane (Mc)	-634	-634	634	634
Inplane Bending (Mc)	-8666	8666	8666	-8666
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	124282	142283	124282	142283

Inplane Stress Summary	118325	147565	136926	131500
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-275	-275	275	275
Torsional Shear (Mt)	654	654	654	654

Shear Stress Summary	379	379	930	930
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	118325	147565	136933	131507
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Hoop Membrane (P)	351	351	351	351
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-689	-689	689	689
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	19491	41091	19491	41091

Hoop Stress Summary	19153	40753	20532	42132
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Axial Membrane (P)	172	172	172	172
Axial Bending (P)	1289	-1289	1289	-1289



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Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1716	-1716	1716	1716
Axial Bending (ML)	-1992	1992	1992	-1992
Axial Pressure Stress	9742	9742	9742	9742

Axial Stress Summary	7494	8900	14913	8349
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	213	213	-213	-213
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	510	510	510	510
Shear Stress Summary	723	723	296	296

Nozzle Stresses (kPa):

LONGITUDINAL PLANE				
(Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	19194	40767	20545	42132

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE				
(Stresses in the	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
hoop direction)				
Hoop Membrane (P)	351	351	351	351
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-268	-268	268	268
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	19491	41091	19491	41091
Hoop Stress Summary	19573	41174	20111	41712

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
pipe cross-section)				
Axial Membrane (P)	172	172	172	172
Axial Bending (P)	1289	-1289	1289	-1289
Axial Membrane (Mc)	-1716	-1716	1716	1716
Axial Bending (Mc)	-4067	4067	4067	-4067
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	9742	9742	9742	9742
Axial Stress Summary	5419	10976	16988	6274

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-213	-213	213	213
Shear due to Torsion	510	510	510	510
Shear Stress Summary	296	296	723	723

Nozzle Stresses (kPa):



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CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	19573	41174	20270	41726

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	124282	142283	124282	142283	124282	142283	124282	142283
Circ. Pl (SUS)	-338	-338	1040	1040	-338	-338	930	930
Circ. Q (SUS)	-648	648	2564	-2564	-5619	5619	11713	-11713
Long. Pm (SUS)	62141	62141	62141	62141	62141	62141	62141	62141
Long. Pl (SUS)	-145	-145	737	737	83	83	619	619
Long. Q (SUS)	-1586	1586	7680	-7680	-2006	2006	3922	-3922
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	275	275	-275	-275	-275	-275	275	275
Shear Q (SUS)	654	654	654	654	654	654	654	654
Pm (SUS)	124282	142283	124282	142283	124282	142283	124282	142283
Pm+Pl (SUS)	123945	141945	125323	143323	123945	141945	125213	143213
Pm+Pl+Q (Total)	123309	142603	127888	140760	118327	147565	136937	131511

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	142283	147988	Passed
Pm+Pl (SUS)	143323	221983	Passed
Pm+Pl+Q (TOTAL)	147565	443966	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	19491	41091	19491	41091	19491	41091	19491	41091
Circ. Pl (SUS)	-338	-338	1040	1040	83	83	619	619
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	9742	9742	9742	9742	9742	9742	9742	9742
Long. Pl (SUS)	-1544	-1544	1888	1888	-1544	-1544	1888	1888
Long. Q (SUS)	-703	703	3281	-3281	-2778	2778	5356	-5356
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	213	213	-213	-213	-213	-213	213	213
Shear Q (SUS)	510	510	510	510	510	510	510	510
Pm (SUS)	19491	41091	19491	41091	19491	41091	19491	41091
Pm+Pl (SUS)	19157	40754	20536	42132	19577	41175	20115	41711



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Nozzle Calcs.: N3- tube inlet Nozl: 17 3:25pm Apr 10,2026

Pm+Pl+Q (Total)	19197	40769	20546	42133	19580	41176	20269	41724
-----------------	-------	-------	-------	-------	-------	-------	-------	-------

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	41091	147988	Passed
Pm+Pl (SUS)	42132	221983	Passed
Pm+Pl+Q (TOTAL)	42133	443966	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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 Nozzle Calcs.: Mayway-2 Nozl: 18 3:25pm Apr 10,2026

Input, Nozzle Desc: Mayway-2

From: 90

Pressure for Reinforcement Calculations	P	18.0010	N/mm ²
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147988.73	kPa
Parent Allowable Stress At Ambient	Sva	147988.73	kPa
Inside Diameter of Hemispherical Head	D	2250.00	mm
Head Finished (Minimum) Thickness	t	75.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

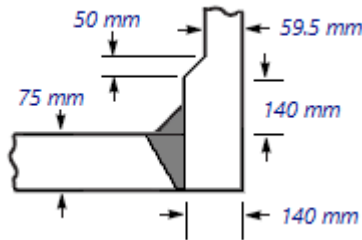
Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147988.73	kPa
Allowable Stress At Ambient	Sna	147988.73	kPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	140.0000	mm
Height of Beveled Transition	L'	50.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		140.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	450.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		54.4000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		1500	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



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 Nozzle Calcs.: Mayway-2 Nozl: 18 3:25pm Apr 10,2026



Reinforcement CALCULATION, Description: Mayway-2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 19.314 in.
 Actual Thickness Used in Calculation 2.343 in.

Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 18.0 \cdot 1128.0 / (2 \cdot 147989 \cdot 1.0 - 0.2 \cdot 18.0)$
 $= 69.4445 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 18.0 \cdot 248.29 / (147989 \cdot 1.0 - 0.6 \cdot 18.0)$
 $= 32.5769 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 80.488 / 0.5773$
 $= 139.4090 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	993.1512	mm
Parallel to Vessel Wall, opening length	d	496.5756	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	180.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 147988.7 / 147988.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 147988.7 / 147988.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



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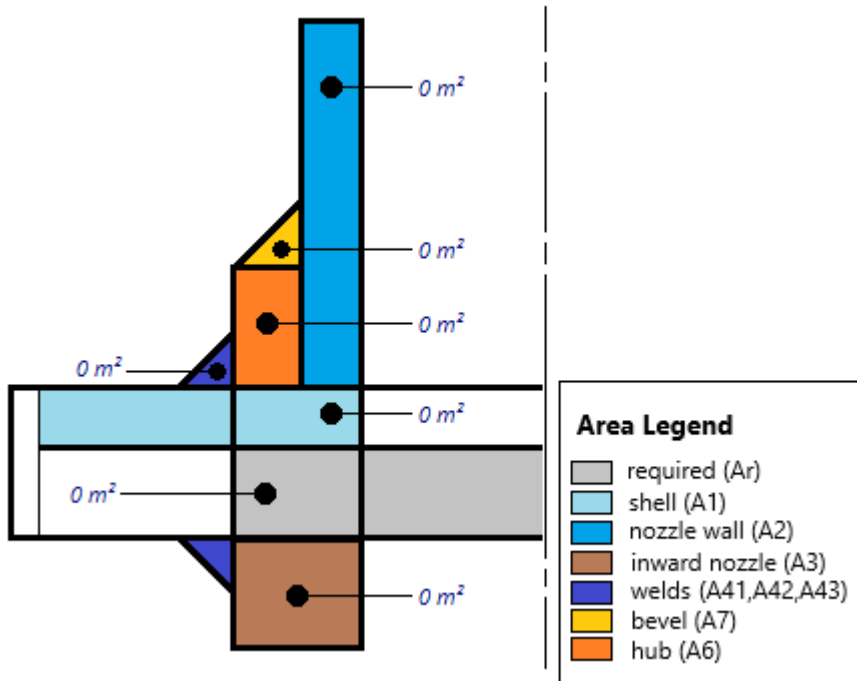
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Nozzle Calcs.: Mayway-2

Nozl: 18 3:25pm Apr 10, 2026



Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	0.034	NA	NA
Area in Shell	A1	0.001	NA	NA
Area in Nozzle Wall	A2	0.009	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	0.026	NA	NA
TOTAL AREA AVAILABLE	Atot	0.036	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (496.5756 \cdot 69.4445 \cdot 1.0 + 2 \cdot 137.0 \cdot 69.4445 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 0.034 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 496.576(1.0 \cdot 72.0 - 1.0 \cdot 69.444) - 2 \cdot 137.0$$

$$(1.0 \cdot 72.0 - 1.0 \cdot 69.4445) \cdot (1 - 1.0)$$

$$= 0.001 \text{ m}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 180.0)(56.51 - 32.58)1.0$$

$$= 0.009 \text{ m}^2$$



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 Nozzle Calcs.: Mayway-2 Nozl: 18 3:25pm Apr 10,2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 0.000 \text{ m}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr_2$$

$$= (2 \cdot \min(180.0, 450.0, 140.0)) \cdot (140.0 - 59.5) \cdot 1.0$$

$$= 0.026 \text{ m}^2 \text{ Includes Bevel Area of } 0.004 \text{ m}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 35.5769 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 35.5769 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 59.512 = 52.073 \text{ mm} \rightarrow \text{OK}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, $t_g = 52.073$, $t_r = 32.577$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.664$, Temp. Reduction = $19 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	18 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-1 $^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, $t_g = 75.0$, $t_r = 69.444$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.965$, Temp. Reduction = $2 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	44 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	42 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : 42 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-35 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $18.00 / 25.86 = 0.696$

Weld Size Calculations, Description: Mayway-2

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - t_r)) S_v)$$

$$= \max(0, (0.0345 - 0.0013 + 2 \cdot 137.0 \cdot 1.0 \cdot (1.0 \cdot 72.0 - 69.4445)) 147989)$$

$$= 5018998.00 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.



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 Nozzle Calcs.: Mayway-2 Nozl: 18 3:25pm Apr 10,2026

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot S_v$$

$$= (0.035 + 0.0001 - 0.0 \cdot 1.0) \cdot 147989$$

$$= 5196738.00 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S_v$$

$$= (0.035 + 0.0 + 0.0001 + (0.0197)) \cdot 147989$$

$$= 8116181.50 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S$$

$$= (0.035 + 0.0 + 0.0001 + 0.0 + (0.0197)) \cdot 147989$$

$$= 8116181.50 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 770.5756 \cdot 10.0 \cdot 0.49 \cdot 147989$$

$$= 877704. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 316.7878) \cdot (140.0 - 3.0) \cdot 0.7 \cdot 147989$$

$$= 14123874. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 770.5756 \cdot (54.4 - 0.0) \cdot 0.74 \cdot 147989$$

$$= 7210788. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (877704 + 14123874) = 15001578 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (877704 + 0 + 7210788 + 0) = 8088492 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (877704 + 7210788 + 0) = 8088492 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 15001578N	, must exceed W = 5018998 N	or W1 = 5196738 N
Path 2-2 = 8088491 N	, must exceed W = 5018998 N	or W2 = 8116181 N
Path 3-3 = 8088491 N	, must exceed W = 5018998 N	or W3 = 8116181 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 18 N/mm²

The Drop for this Nozzle is : 68.0334 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 593.0334 mm

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: shell

Shell Maximum Design Pressure	Psd,max	18.00 N/mm ²
Shell Maximum Operating Pressure	Psox,max	18.00 N/mm ²
Shell Minimum Operating Pressure	Psox,min	15.87 N/mm ²
Shell Thickness	ts	156.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2250.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	302.00 °C
Shell Material		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K11789
Shell Allowable Stress at Temperature	Ss	147988.73 kPa
Shell Allowable Stress at Ambient		147988.73 kPa

Channel Data:

Channel Description: Channel - rear

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	18.00 N/mm ²
Channel Maximum Operating Pressure	Ptox,max	18.00 N/mm ²
Channel Minimum Operating Pressure	Ptox,min	16.37 N/mm ²
Channel Thickness	tc	156.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2250.000 mm
Channel Design Temperature	TEMPC	275.00 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	147988.73 kPa
Channel Allowable Stress at Ambient		147988.73 kPa

Tube Data:

Number of Tube Holes	Nt	3439
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	19000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		18280.00 mm
Design Temperature of the Tubes		275.00 °C
Tube Material		SA-213 T11
Tube Material UNS Number		K11597
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		110493.01 kPa
Tube Allowable Stress At Ambient		117991.02 kPa
Tube Yield Stress At design Temperature	Syt	165990.17 kPa
Tube Pitch (Center to Center Spacing)	P	35.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	3.0000 mm
Groove Weld Leg	ag	3.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1078.200 mm



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Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	100.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	3.6 m ²
Perimeter of Tube Layout	Cp	6982.6 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	275.00 °C
Tubesheet Material		SA-336 F11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K11572
Tubesheet Allowable Stress at Temperature	S	147988.73 kPa
Tubesheet Allowable Stress at Ambient	Tt	147988.73 kPa
Thickness of Tubesheet	h	360.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	2562.000 mm

Area of the Untubed Lanes	AL	0.000 m ²
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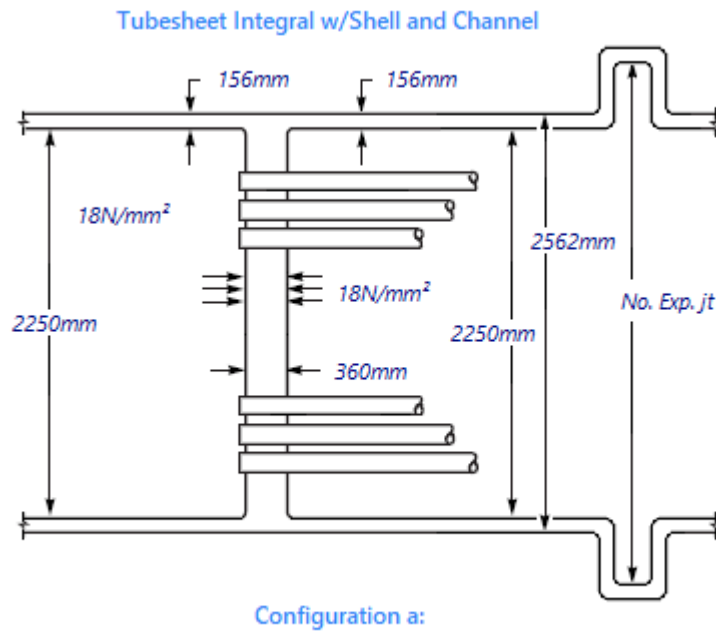
Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	795.000 mm
Tube End condition corresponding to Span (l)	k	0.80
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		Yes
Differential Design Pressure	Pdif	2.4501 N/mm ²
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	155.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	135.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



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ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. along Len	155.0 °C	0.0000124585 /°C
Elastic Mod. at Design Temperature	302.0 °C	0.18632E+06 N/mm²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	155.0 °C	0.19613E+06 N/mm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20409E+06 N/mm²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	275.0 °C	0.18799E+06 N/mm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20409E+06 N/mm²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. along Len	135.0 °C	0.0000123289 /°C
Elastic Mod. at Design Temperature	275.0 °C	0.18799E+06 N/mm²
Elastic Mod. at Metal Temp. along Len	135.0 °C	0.19737E+06 N/mm²
Elastic Mod. at Tubsht. Design Temp.	275.0 °C	0.18799E+06 N/mm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20409E+06 N/mm²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	275.0 °C	0.18799E+06 N/mm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20409E+06 N/mm²

Note:



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The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Differential pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (2.45 \cdot (19.05/2 - 0)) / (110493.01 \cdot 1.0 + 0.4 \cdot 2.45)$
 $= 0.2093 + 0.0000 = 0.2093 \text{ mm}$

Tube Required Thickness under External Pressure (Differential pressure) :

Determine Parameter [Mx]:
 $= L / \sqrt{(Ro \cdot t)}$
 $= 18280.0 / \sqrt{(9.525 \cdot 2.11)}$
 $= 4077.580$

Compute the Factor [Ch]:
 Since $Mx > 2 \cdot (Do/t)^{0.94}$, $Ch = 0.55 / (Do/t)$
 $= 0.55 / (19.05/2.11)$
 $= 0.0609$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:
 $= 1.6 \cdot Ch \cdot Ey (t / Do)$
 $= 1.6 \cdot 0.061 \cdot 187993 (2.11/19.05)$
 $= 2029437.375 \text{ kPa}$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:
 $= Fhe / Fy$
 $= 2029437.4 / 165990.2$
 $= 12.226$

Predicted Buckling Stress [Fic]:
 Since $Fr \geq 2.439$, $Fic = Fy$
 $= 165990.172 \text{ kPa}$

Factor of Safety [FS]:
 Since $Fic [165990.17] \geq Fy [165990.17]$, $FS = 1.667$

Allowable Circumferential Compressive Stress [Fha]:
 Since $Fr > 2.439$, $Fha = Fy/FS$
 $= 165990.172 / 1.667$
 $= 99574.188 \text{ kPa}$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot Fha (t / Do)$
 $= 2 \cdot 99.58 (2.11/19.05)$
 $= 22.059 \text{ N/mm}^2$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.4782 mm
Allowable Internal Pressure at Corroded thickness	26.86 N/mm ²
Required Internal Design Pressure	2.45 N/mm ²
Allowable External Pressure at Corroded thickness	22.06 N/mm ²
Required External Design Pressure	2.45 N/mm ²
Required Thickness due to Shell Side pressure	0.4782 mm

Note: The tubes were evaluated using the specified differential pressure.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Note: Shell/Channel Design Pressures modified by Input differential pressure:
 Pdif = 2.450 Ps = 18.001 Pt = 15.551 N/mm²



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Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 100.0/360.0$$

$$= 0.2778 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole diameter $[d^*]$:

$$= \max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(19.05 - 2 \cdot 2.11 (187993/187993) \cdot (0.278); 19.05 - 2 \cdot 2.11)$$

$$= 18.1748 \text{ mm}$$

Equivalent outer tube limit circle diameter $[Do]$:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 1078.2 + 19.05$$

$$= 2175.450 \text{ mm}$$

Determine the basic ligament efficiency for shear $[\mu]$:

$$= (p - dt) / p$$

$$= (35.0 - 19.05)/35.0$$

$$= 0.456$$

Equivalent outer tube limit radius $[ao]$:

$$= Do / 2$$

$$= 2175.45/2$$

$$= 1087.7250 \text{ mm}$$

Effective tube pitch $[p^*]$:

$$= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 35.0 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 1000248 ; 4 \cdot 2175.45 \cdot 35.0) / (\pi \cdot 2175.45^2)) }$$

$$= 35.0000 \text{ mm}$$

Effective ligament efficiency for bending $[\mu^*]$:

$$= (p^* - d^*) / p^*$$

$$= (35.0 - 18.1748)/35.0$$

$$= 0.4807$$

Ratio $[ps]$:

$$= as / ao$$

$$= 1125.0/1087.725$$

$$= 1.0343$$

Ratio $[pc]$:

$$= ac / ao$$

$$= 1125.0/1087.725$$

$$= 1.0343$$

Compute parameter $[xt]$:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 3439 ((19.05 - 2 \cdot 2.11) / (2 \cdot 1087.725))^2$$

$$= 0.8402$$

Determine parameter $[xs]$:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 3439 (19.05 / (2 \cdot 1087.725))^2$$

$$= 0.7363$$



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Determine the value [h'g]:
 $= \max((h_g - CATC); 0)$ (For pressure only cases)
 $= \max((0 - 0); 0)$
 $= 0.000 \text{ mm}$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:
 $= \pi \cdot ts(D_s + ts)E_s/L$
 $= \pi \cdot 156.0 (2250.0 + 156.0)186318/18280.0$
 $= 12017452.0000 \text{ N/mm}$

Axial Tube Stiffness [Kt]:
 $= \pi \cdot tt(D_t - tt)E_t/L$
 $= \pi \cdot 2.11 (19.05 - 2.11)187993/18280.0$
 $= 1154.7175 \text{ N/mm}$

Stiffness Factor [Ks,t]:
 $= K_s / (N_t \cdot K_t)$
 $= 12017452 / (3439 \cdot 1154.718)$
 $= 3.02625$

Rigidity Ratio [J]:
 $= 1 / (1 + K_s/K_t)$
 $= 1 / (1 + 12017452/0)$
 $= 1.0000 (= 1 \text{ if No Exp. Jt.})$

Shell coefficient [β_s]:
 $= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + ts)ts)$
 $= ((12(1 - 0.3^2))^{1/4}) / ((2250.0 + 156.0)156.0)$
 $= 0.0030 \text{ 1/mm}$

Shell coefficient [ks]:
 $= \beta_s \cdot E_s \cdot ts^3 / (6(1 - \nu_s^2))$
 $= 0.003 \cdot 186318 \cdot 156.0^3 / (6(1 - 0.3^2))$
 $= 384398912.0000 \text{ N/mm}^2 \cdot \text{mm}^2$

Shell coefficient [λ_s]:
 $= 6(D_s + ts) \cdot ks/h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2/2)$
 $= 6(2250.0 + 156.0) \cdot 0.38440\text{E}+09/360.0^3 \cdot (1 + 360.0 \cdot 0.003 + (360.0 \cdot 0.003)^2/2)$
 $= 313842.6562 \text{ N/mm}^2$

Shell coefficient [δ_s]:
 $= D_s^2 / (4 \cdot E_s \cdot ts) \cdot (1 - D_s \cdot \nu_s / (2(D_s + ts)))$
 $= 2250.0^2 / (4 \cdot 186318 \cdot 156.0) \cdot (1 - 2250.0 \cdot 0.3 / (2(2250.0 + 156.0)))$
 $= 0.37438\text{E}-04 \text{ mm/kPa}$

Channel coefficient [β_c]:
 $= ((12(1 - \nu_c^2))^{1/4}) / ((D_c + tc)tc)$
 $= ((12(1 - 0.3^2))^{1/4}) / ((2250.0 + 156.0)156.0)$
 $= 0.0030 \text{ 1/mm}$

Channel coefficient [kc]:
 $= \beta_c \cdot E_c \cdot tc^3 / (6(1 - \nu_c^2))$
 $= 0.003 \cdot 187993 \cdot 156.0^3 / (6(1 - 0.3^2))$
 $= 387855392.0000 \text{ N/mm}^2 \cdot \text{mm}^2$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:
 $CqDelta = CqTheta = CmTheta = CmDelta = 1$

Channel coefficient [λ_c]:
 $= 6(D_c + tc/\cos(\theta)) \cdot kc/h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta)/2 + (h \cdot \beta_c)^2/2 \cdot CqDelta)$
 $= 6(2250.0 + 156.0/\cos(0)) \cdot 0.38786\text{E}+09/360.0^3 \cdot (1.0 + 360.0 \cdot 0.003 \cdot (1.0 + 1.0)/2 + (360.0 \cdot 0.003)^2/2 \cdot 1.0)$



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$$= 316664.7 \text{ N/mm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot T_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 2250.0^2 / (4 \cdot 187993 \cdot 156.0) \cdot (1 - 2250.0 \cdot 0.3 / (2(2250.0 + 156.0)))$$

$$= 0.37104\text{E-}04 \text{ mm/kPa}$$

Channel coefficient [ζ_{etac}]:

$$\zeta_{etac} = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 10.285714 ; \mu^* = 0.480721 ; E^* = 97558 \text{ N/mm}^2$$

$$E^*/E = 0.518946 ; v^* = 0.300894$$

Note: As h/p (10.286) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$= ((24(1 - v^*)N_t \cdot E_t \cdot t_t(dt - t_t)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.301^2)3439 \cdot 187993 \cdot 2.11$$

$$(19.05 - 2.11)1087.725^2) / (97558 \cdot 18280.0 \cdot 360.0^3))^{1/4}$$

$$= 1.6365$$

Values from Table 4.18.3

$$Z_d = 0.332842 ; Z_v = 0.178731 ; Z_m = 0.728097$$

$$Z_a = 0.582977\text{E}+00 ; Z_w = 0.178731$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2562.0 / 2175.45$$

$$= 1.1777$$

Compute coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.3) / 97558 \cdot (313842.66 + 316664.72 +$$

$$187993 \cdot \ln(1.18))$$

$$= 4.7386$$

Compute parameter [ϕ]:

$$= (1 + v^*)F$$

$$= (1 + 0.3009)4.7386$$

$$= 6.1644$$

Compute parameter [Q_1]:

$$= (\psi_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.0343 - 1 - 6.1644 \cdot 0.1787) / (1 + 6.1644 \cdot 0.7281)$$

$$= -0.194505$$

Compute Parameter [Q_{z1}]:

$$= (Z_d + Q_1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.33284 + -0.19451 \cdot 0.17873) / 2 \cdot 1.63649^4$$

$$= 1.0689$$

Compute parameter [Q_{z2}]:

$$= (Z_v + Q_1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.17873 + -0.19451 \cdot 0.7281) / 2 \cdot 1.63649^4$$

$$= 0.1331$$

Compute parameter [U]:

$$= (Z_w + (\psi_s - 1)Z_m)X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.1787 + (1.0343 - 1)0.7281)1.63649^4 / (1 + 6.1644 \cdot 0.7281)$$

$$= 0.2662$$



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ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:
 = 0 (config a)

Compute Parameter [γ]:
 = 0.000 mm (For Pressure only cases)

Calculate parameter [ωs]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.0343 \cdot 0.38440E+09 \cdot 0.003 \cdot 0.374E-04 (1 + 360.0 \cdot 0.003)$$

$$= 91334.7891 \text{ mm}^2$$

Calculate parameter [ω*s]:

$$= a_o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s$$

$$= 1087.725^2 (1.034^2 - 1) \cdot (1.034 - 1) / 4 - 91334.789$$

$$= -90628.1719 \text{ mm}^2$$

Parameter [ωc]:

$$= \rho_c \cdot k_c [\delta_c (C_m \theta + \beta_c + C_q \Delta \cdot h \cdot \beta_c^2) - \zeta_c (C_m \theta + h \cdot C_q \theta + h \cdot \beta_c / 2)]$$

$$= 1.034 \cdot 0.38786E+09 [0.371E-04 (1.0 + 0.003 \cdot 1.0 \cdot 360.0) - 0 (1.0/360.0 + 1.0 \cdot 0.003/2)]$$

$$= 91334.7969 \text{ mm}^2$$

Parameter [ω*c]:

$$= a_o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega_c$$

$$= 1087.72498^2 [(1.03427^2 + 1) \cdot (1.03427 - 1) / 4 - (1.03427 - 1) / 2] - 91334.79688$$

$$= -90628.1797 \text{ mm}^2$$

Pressure [P*S]:
 = 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:
 = 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/K_{st} \cdot (D_s/D_o)^2] v_s - [(\rho_s^2 - 1) / (J \cdot K_{st})] - [(1 - J) / (2J \cdot K_{st})] [(D_j)^2 - (D_s)^2 / D_o^2] \}$$

$$= 18.001 \{ 0.736 + 2(1 - 0.736) 0.3 + [2/3.026 \cdot (2250.0/2175.45)^2] 0.3 - [(1.034^2 - 1) / (1.0 \cdot 3.026)] - [(1 - 1.0) / (2 \cdot 1.0 \cdot 3.026)] [(0^2 - (2250.0)^2 / 2175.45^2)] \}$$

$$= 19.5054 \text{ N/mm}^2$$

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot K_{st})] P_t$$

$$= [0.84 + 2(1 - 0.84) 0.3 + 1/(1 \cdot 3.026)] 15.551$$

$$= 19.6955 \text{ N/mm}^2$$

Pressure [P_γ]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2)$$

$$= 3439 \cdot 1154.718 \cdot 0 / (3.142 \cdot 1087.725^2)$$

$$= 0.0000 \text{ N/mm}^2$$

Pressure [P_w]:

$$= -\gamma_b \cdot U \cdot W^* / (2 \cdot \pi \cdot a_o^2)$$

$$= -0 \cdot 0.266 \cdot 0 / (2 \cdot \pi \cdot 1087.725^2)$$

$$= -0.0000 \text{ N/mm}^2$$

Calculate the pressure [Prim]:

$$= -(U/a_o^2) \cdot (\omega^*s \cdot P_s - \omega^*c \cdot P_t)$$

$$= -(0.266/1087.725^2) \cdot (-90628.172 \cdot 18.001 - -90628.18 \cdot 15.551)$$

$$= 0.0500 \text{ N/mm}^2$$



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Calculate the pressure [Pw]:

$$= (U/ao^2) \cdot (\omega_s \cdot P^*s - \omega_c \cdot P^*c)$$

$$= (0.266/1087.725^2) \cdot (91334.7891 \cdot 0 - 91334.7969 \cdot 0)$$

$$= 0.0000 \text{ N/mm}^2$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (ps - 1) Qz2)) \cdot (P^*s - P^*t + P_y + P_\omega + P_w + P_{rim})$$

$$= 0.1000E+01 \cdot 3.026 / (1 + 1.0 \cdot 3.026 (1.069 + (1.034 - 1) 0.133)) \cdot (19.505 - 19.695 + 0 + 0 + 0 + 0.05)$$

$$= -0.0998 \text{ N/mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega^*s \cdot P^*s - \omega^*c \cdot P^*t) - (\omega_s \cdot P^*s - \omega_c \cdot P^*c)) + W^* \cdot y_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-90628.172 \cdot 18.001 - -90628.18 \cdot 15.551) - (91334.789 \cdot 0 - 91334.797 \cdot 0)) \cdot 1 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 6.16436 \cdot 0.7281)$$

$$= -40456.0508 \text{ N}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.195 + 2 \cdot -40456.051 / (-0.1 \cdot 1087.725^2)$$

$$= 0.49054$$

Fm Value from Table 4.18.3 = 0.401322

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu^*) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.4013 / 0.4807) \cdot (2 \cdot 1087.725 / (360.0 - 0))^2 \cdot -0.1$$

$$= -4565.1118 \text{ kPa}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 147988.73 = 221983.11 \text{ kPa}$$

Note: Tubesheet Bending Stress is probably low, use the following req. thk:

$$\text{Tubesheet thickness (Incl. Corr.)} = 3.8100 \text{ mm}$$

$$\text{Tubesheet Bending Stress} = 74589.4531 \text{ kPa}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / ao) \cdot \min(0.8S; 0.533Sy)$$

$$= (2 \cdot 0.456 \cdot 360.0 / 1087.725) \cdot 118390.984$$

$$= 35712.8672 \text{ kPa}$$

Shear Stress is not required to be computed; [Pe] <= τ limit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\text{Tubesheet thickness (Incl. Corr.)} = 3.8100 \text{ mm}$$

$$\text{Tubesheet Shear Stress} = 2995.0361 \text{ kPa}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(3.81; 3.81)$$

$$= 3.8100 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:



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$$F_{tmin} = 0.4678, F_{tmax} = 1.508$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s) \\
= ((18.0 \cdot 0.7363 - 15.55 \cdot 0.8402) - (-0.1) \cdot 0.468) / \\
(0.8402 - 0.7363) \\
= 2262.5342 \text{ kPa}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s) \\
= ((18.0 \cdot 0.7363 - 15.55 \cdot 0.8402) - (-0.1) \cdot 1.508) / \\
(0.8402 - 0.7363) \\
= 3262.1230 \text{ kPa}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 3262.123 \text{ kPa}$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 110493.0078 \text{ kPa}$$

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area} \\
= 3262.12 \cdot 0.0001 \\
= 366.30 \text{ N}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t \\
= \pi (19.05 - 2.11) 2.11 \\
= 0.000 \text{ m}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S \\
= \pi \cdot 2.11 (19.05 - 2.11) 110493.008 \\
= 12407.058 \text{ N}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\
= 0.85 \cdot \pi \cdot 3.0 (19.05 + 0.67 \cdot 3.0) 110493.01 \\
= 12407.0576 \text{ N}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\
= 110493.008 / 110493.008 \\
= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t) \\
= 0.55 \cdot \pi \cdot 3.0 (19.05 + 0.67 \cdot 3.0) 110493.01 \\
= 12061.8867 \text{ N}$$

Minimum Length of Weld [ar]:

$$= 2 \sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) \cdot f_w \cdot f_d) - 0.75 \cdot d_o} \\
= 2 \sqrt{((0.75 \cdot 19.05)^2 + 1.07 \cdot 2.11 \cdot (19.05 - 2.11) \cdot 1.0 \cdot 1.0) - 0.75 \cdot 19.05} \\
= 2.562 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm

Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 12407.06 is $\geq$ tube strength 12407.06 N



Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= S_s \cdot E_{sw} = 147988.73 \cdot 1.0 = 147988.73 \text{ kPa}$

Axial Membrane Stress in Shell $[\sigma_s, m]$:
 $= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 1087.725^2 / ((2250.0 + 156.0) 156.0) \cdot [-0.1 + (1.034^2 - 1)$
 $(18.0 - 15.55)] + 1125.0^2 \cdot 15.55 / ((2250.0 + 156.0) 156.0)$
 $= 52657.9648 \text{ kPa}$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 156.0 mm, the minimum Shell length adjacent to the tubesheet should be 1066.415 mm

Shell Membrane Stress due to Joint Interaction $[\sigma_s, m]$:
 $= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 1087.725^2 / ((2250.0 + 156.0) 156.0) [-0.1 + (1.034^2 - 1)$
 $(18.0 - 15.55)] + 1125.0^2 \cdot 15.55 / ((2250.0 + 156.0) 156.0)$
 $= 52657.9648 \text{ kPa}$

Shell Bending Stress due to Joint Interaction $[\sigma_s, b]$:
 $= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P_s / (E_s \cdot t_s)] +$
 $6(1 - \nu^2) / (E_s) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$
 $= 6 \cdot 0.38440E+09 / 156.0^2 \{ 0.003 [0.374E-04 \cdot 18.001 + 1125.0^2$
 $\cdot 0 / (29065652)] +$
 $6(1 - 0.3^2) / (97558.6) (1087.72 / 360.0)^3 (1 + 360.0 \cdot$
 $0. / 2) [-0.1 (0.179 + 0.728 \cdot -0.195) +$
 $2 / 1087.72^2 \cdot 0.728 \cdot -40456.051] \}$
 $= 177489.8438 \text{ kPa}$

Shell Stress Summation vs. Allowable
 $:: |\sigma_s, m| + |\sigma_s, b| \leq 1.5 \cdot S_s$
 $:: |52657.96| + |177489.84| \leq 221983.11 \text{ kPa}$
 $:: 230147.81, \text{ must be } \leq 221983.11 \text{ kPa (See Elastic-Plastic iteration below)}$

Note:
 For a given Channel thickness of 156.0 mm, the minimum Channel length adjacent to the tubesheet should be 1066.415 mm

Channel Membrane Stress due to Joint Interaction $[\sigma_c, m]$:
 $= a_c^2 \cdot P_t / ((D_c + t_c) t_c \cdot \cos(\theta))$
 $= 1125.0^2 \cdot 15.55 / ((2250.0 + 156.0) 156.0 \cdot \cos(0))$
 $= 52434.270 \text{ kPa}$

Channel Bending Stress due to Joint Interaction $[\sigma_c, b]$:
 $= 6 \cdot k_c / t_c^2 \cdot \{ (\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t$
 $- 6(1 - \nu^2) / (E_s) \cdot (a_o / h)^3 \cdot (C_m \theta +$
 $h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$
 $= 6 \cdot 0.38786E+09 / 156.0^2 \cdot \{ (0.371E-04 \cdot 1.0 \cdot 0.003 - 0 \cdot 1.0) \cdot 15.551$
 $- 6(1 - 0.301^2) / 97558 \} \cdot$
 $(1087.725 / 360.0)^3 \cdot (1.0 + 360.0 \cdot 0.003 / 2 \cdot 1.0) \cdot [-0.1 (0.17873 + 0.7281$
 $\cdot -0.195) + 2 / 1087.725^2 \cdot 0.7281 \cdot -40456.051] \}$
 $= 175807.0938 \text{ kPa}$

Channel Stress Summation vs. Allowable
 $:: |\sigma_c, m| + |\sigma_c, b| \leq 1.5 \cdot S_c$
 $:: |52434.27| + |175807.09| \leq 221983.11 \text{ kPa}$
 $:: 228241.36 \text{ must be } \leq 221983.11 \text{ kPa (See Elast-Plast iteration below)}$



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Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	4565.1	<= 221983.1 kPa	Pass
Tubesheet Shear Stress	312.6	<= 118391.0 kPa	Pass
Maximum Tube Stress	3262.1	<= 110493.0 kPa	Pass
Maximum Force on any one Tube	366.3	<= 12407.1 N	Pass
Axial Membrane Stress in Shell	52658.0	<= 147988.7 kPa	Pass
Shell Stress (jt. inter.)	230147.8	<= 221983.1 kPa	See below
Channel Stress (jt. inter.)	228241.4	<= 221983.1 kPa	See below

Additional Results for the 2nd Elastic-Plastic Iteration.



These calculations are not intended to be used in the creep range. Check Part 4.18 in Section VIII Division 2.

Compute Allowable [Sstars]:
 = min(Sys, SPS,s/2)
 = min(245676, 491353/2) = 245676.95 kPa

Compute Allowable [S*c]:
 = min(Syc, SPS,c/2)
 = min(248996, 497993/2) = 248996.52 kPa

Compute Factor [facts]:
 = min([1.4 - 0.4• abs(σs,b) / Sstars], 1)
 = min([1.4 - 0.4• abs(177489.84)/245676.95], 1) = 1.0

Compute Factor [factc]:
 = min([1.4 - 0.4• abs(σc,b) / Sstarc], 1)
 = min([1.4 - 0.4• abs(175807.09)/248996.52], 1) = 1.0

Note: Since facts & factc are both >= 1, the design Passes.

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Shell Stress (jt. inter.)	230147.8	<= 491353.9 kPa	Pass
Channel Stress (jt. inter.)	228241.4	<= 497993.0 kPa	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	3.810	360.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	3.000	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses		Case		Pass/ Fail
		Bend	Allwd	Shear	Allwd	Type	
D3uc	3.810 ...	4565	221983	313	118391	Ps+Pt-Th	D3 Ok
O3uc	49.179 ...	46514	497970	11808	118391	Ps+Pt+Th	O3 Ok



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04uc	47.651	...	41948	497970	11496	118391	Ps+Pt+Th	04	Ok
D3c	9.810	...	5007	221983	413	118391	Ps+Pt-Th-c	D3	Ok
03c	55.487	...	48066	497970	12016	118391	Ps+Pt+Th-c	03	Ok
04c	53.437	...	43060	497970	11603	118391	Ps+Pt+Th-c	04	Ok

Max: 55.4870 ... mm 0.097 0.101 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:		Fail
D3uc	52658	147988	...	...		...	...	...	...			Ok
03uc	41085	491353	...	...		...	...	...	...			Ok
04uc	11573	491353	-11573	-101197		...	...	...	...			Ok
D3c	53857	147988	...	...		...	...	...	...			Ok
03c	42160	491353	...	...		...	...	...	...			Ok
04c	11697	491353	-11697	-100940		...	...	...	...			Ok
Max	RATIO	0.364		0.116		...			...			

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D3uc	3262	110493	...	...	366	12407	230148	491353	228241	497993	Ok
03uc	41037	220986	...	...	4608	24814	188010	491353	259081	497993	Ok
04uc	37775	220986	...	...	4242	24814	42138	491353	30839	497993	Ok
D3c	3590	110493	...	...	403	12407	235260	491353	233895	497993	Ok
03c	41217	220986	...	...	4628	24814	191703	491353	266041	497993	Ok
04c	37628	220986	...	...	4225	24814	43557	491353	32146	497993	Ok
Max	RATIO	0.187		...		0.187		0.479		0.534	

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	55.487	360.000	Ok
Tube thickness :	0.478	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	3.000	Ok



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Tube-tubesheet groove weld leg :	1.281	3.000	Ok
----------------------------------	-------	-------	----

Min Shell length of thk, (156.000) adj. to tubesheet: 1066.415 mm

Min Channel length of thk, (156.000) adj. to tubesheet: 1066.415 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -4.826 mm
 Due to Pressure Shell Compresses by : -2.073 mm
 Due to Pressure + Thermal Shell Compresses by : -6.394 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	18.431	1.000	21.067	1.000
Tubesheet Shear Stress	40.651	1.000	47.489	1.000
Tube Tensile Stress	93.643	1.000	132.945	1.000
Tube Compressive Stress	735.395	1.000	64.669	1.000
Tube-Tubesheet Joint load	93.644	1.000	132.945	1.000
Shell Stress (Axial, Junction)	18.431	1.000	21.067	1.000
Tubesheet-Channel Junction	18.431	1.000	21.067	1.000
Tube Pressure Stress	26.858	1.000	22.059	1.000
Minimum MAWP	18.431		21.067	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	18.962	1.000	21.606	1.000
Tubesheet Shear Stress	41.254	1.000	48.054	1.000
Tube Tensile Stress	100.876	1.000	141.230	1.000
Tube Compressive Stress	6045.78	1.000	76.570	1.000
Tube-Tubesheet Joint load	100.876	1.000	141.230	1.000
Shell Stress (Axial, Junction)	18.962	0.286	21.606	0.531
Tubesheet-Channel Junction	18.962	0.520	21.606	0.195
Tube Pressure Stress	28.680	1.000	27.241	1.000
Minimum MAPnc	18.962		21.606	

(*) All load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.5846	0.5848	...	...	...	...
2	-2.0724	-2.0731	...	...	...	...
3	-1.5674	-1.5679	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	-1.5674	-1.5679	-4.8246	-4.8262	...	...
8	...	...	-4.8246	-4.8262	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:



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Governing thickness on the shell side per figure UCS-66.3 (e):

= max(tubesheet thk/4, min(tubesheet thk, shell thickness))
= max(360.0/4, min(360.0, 156.0))
= 156.000 mm

MDMT reduction ratio per UCS 66 (b)(1)(b):

= min(ps/Shellside MAPnc, 1)
= min(18.0/21.61, 1)
= 0.833

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

= max(tubesheet thk/4, min(tubesheet thk, channel thickness))
= max(360.0/4, min(360.0, 156.0))
= 156.000 mm

MDMT reduction ratio per UCS 66 (b)(1)(b):

= min(pt/Tubeside MAPnc, 1)
= min(18.0/18.96, 1)
= 0.949

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

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Input Echo, Tubesheet Number 2, Description:

Shell Data:

Main Shell Description: shell

Shell Maximum Design Pressure	Psd,max	18.00 N/mm ²
Shell Maximum Operating Pressure	Psox,max	18.00 N/mm ²
Shell Minimum Operating Pressure	Psox,min	15.87 N/mm ²
Shell Thickness	ts	156.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2250.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	302.00 °C
Shell Material		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K11789
Shell Allowable Stress at Temperature	Ss	147988.73 kPa
Shell Allowable Stress at Ambient		147988.73 kPa

Channel Data:

Channel Description: Channel - rear

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	18.00 N/mm ²
Channel Maximum Operating Pressure	Ptox,max	18.00 N/mm ²
Channel Minimum Operating Pressure	Ptox,min	16.37 N/mm ²
Channel Thickness	tc	156.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2250.000 mm
Channel Design Temperature	TEMPC	275.00 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	147988.73 kPa
Channel Allowable Stress at Ambient		147988.73 kPa

Tube Data:

Number of Tube Holes	Nt	3439
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	19000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		18280.00 mm
Design Temperature of the Tubes		275.00 °C
Tube Material		SA-213 T11
Tube Material UNS Number		K11597
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		110493.01 kPa
Tube Allowable Stress At Ambient		117991.02 kPa
Tube Yield Stress At design Temperature	Syt	165990.17 kPa
Tube Pitch (Center to Center Spacing)	P	35.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	3.0000 mm
Groove Weld Leg	ag	3.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1078.200 mm



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Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	100.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	3.6 m ²
Perimeter of Tube Layout	Cp	6982.6 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	275.00 °C
Tubesheet Material		SA-336 F11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K11572
Tubesheet Allowable Stress at Temperature	S	147988.73 kPa
Tubesheet Allowable Stress at Ambient	Tt	147988.73 kPa
Thickness of Tubesheet	h	360.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	2562.000 mm
Area of the Untubed Lanes	AL	0.000 m ²

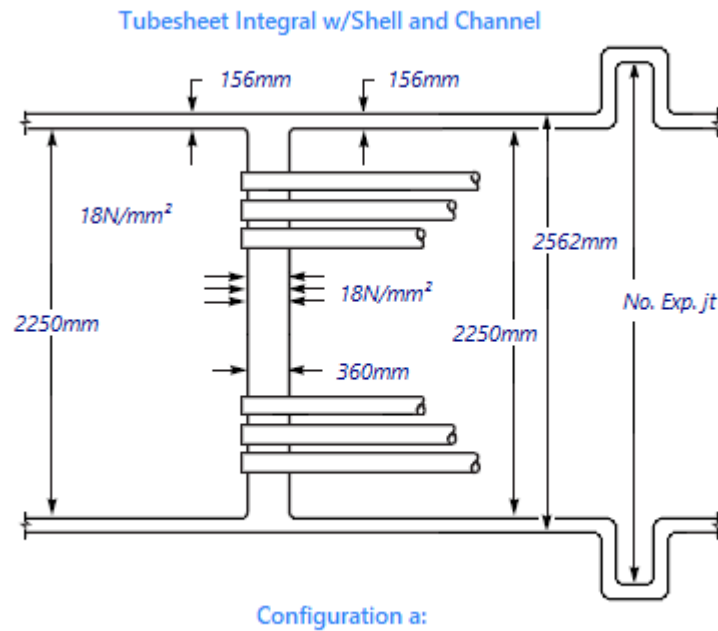
Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	795.000 mm
Tube End condition corresponding to Span (l)	k	0.80
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	155.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	135.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



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ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. along Len	155.0 °C	0.0000124585 /°C
Elastic Mod. at Design Temperature	302.0 °C	0.18632E+06 N/mm ²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	155.0 °C	0.19613E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20409E+06 N/mm ²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	275.0 °C	0.18799E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20409E+06 N/mm ²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. along Len	135.0 °C	0.0000123289 /°C
Elastic Mod. at Design Temperature	275.0 °C	0.18799E+06 N/mm ²
Elastic Mod. at Metal Temp. along Len	135.0 °C	0.19737E+06 N/mm ²
Elastic Mod. at Tubsht. Design Temp.	275.0 °C	0.18799E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20409E+06 N/mm ²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	275.0 °C	0.18799E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20409E+06 N/mm ²

Note:



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The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (18.0 \cdot (19.05/2 - 0)) / (110493.01 \cdot 1.0 + 0.4 \cdot 18.0)$
 $= 1.4568 + 0.0000 = 1.4568 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:
 $= L / \sqrt{(Ro \cdot t)}$
 $= 18280.0 / \sqrt{(9.525 \cdot 2.11)}$
 $= 4077.580$

Compute the Factor [Ch]:
 Since $Mx > 2 \cdot (Do/t)^{0.94}$, $Ch = 0.55 / (Do/t)$
 $= 0.55 / (19.05/2.11)$
 $= 0.0609$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:
 $= 1.6 \cdot Ch \cdot Ey(t / Do)$
 $= 1.6 \cdot 0.061 \cdot 187993 (2.11/19.05)$
 $= 2029437.375 \text{ kPa}$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:
 $= Fhe / Fy$
 $= 2029437.4 / 165990.2$
 $= 12.226$

Predicted Buckling Stress [Fic]:
 Since $Fr \geq 2.439$, $Fic = Fy$
 $= 165990.172 \text{ kPa}$

Factor of Safety [FS]:
 Since $Fic [165990.17] \geq Fy [165990.17]$, $FS = 1.667$

Allowable Circumferential Compressive Stress [Fha]:
 Since $Fr > 2.439$, $Fha = Fy/FS$
 $= 165990.172 / 1.667$
 $= 99574.188 \text{ kPa}$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot Fha(t / Do)$
 $= 2 \cdot 99.58 (2.11/19.05)$
 $= 22.059 \text{ N/mm}^2$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.7218 mm
Allowable Internal Pressure at Corroded thickness	26.86 N/mm ²
Required Internal Design Pressure	18.00 N/mm ²
Allowable External Pressure at Corroded thickness	22.06 N/mm ²
Required External Design Pressure	18.00 N/mm ²
Required Thickness due to Shell Side pressure	1.7218 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	351.110 ...	211461	221983	51662 118391	Ps+Pt-Th	D1 Ok



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D2uc	327.350	...	185469	221983	44317	118391	Ps+Pt-Th	D2	Ok
D3uc	76.355	...	26799	221983	7344	118391	Ps+Pt-Th	D3	Ok
O1uc	251.498	...	252146	497970	63158	118391	Ps+Pt+Th	O1	Ok
O2uc	167.892	...	144784	497970	32821	118391	Ps+Pt+Th	O2	Ok
O3uc	90.897	...	68748	497970	18840	118391	Ps+Pt+Th	O3	Ok
O4uc	47.651	...	41948	497970	11496	118391	Ps+Pt+Th	O4	Ok
D1c	355.869	...	216894	221983	52426	118391	Ps+Pt-Th-c	D1	Ok
D2c	332.004	...	189940	221983	44877	118391	Ps+Pt-Th-c	D2	Ok
D3c	83.707	...	28015	221983	7549	118391	Ps+Pt-Th-c	D3	Ok
O1c	256.630	...	258531	497970	64029	118391	Ps+Pt+Th-c	O1	Ok
O2c	173.504	...	148384	497970	33274	118391	Ps+Pt+Th-c	O2	Ok
O3c	97.058	...	71075	497970	19152	118391	Ps+Pt+Th-c	O3	Ok
O4c	53.437	...	43060	497970	11603	118391	Ps+Pt+Th-c	O4	Ok

Max: 355.8694 ... mm 0.977 0.541 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass Fail
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	4730	147988	...	...		...	...	...	...	Ok
D2uc	48572	147988	...	...		...	...	...	...	Ok
D3uc	53302	147988	...	...		...	...	...	...	Ok
O1uc	6843	491353	-6843	-101197		...	...	...	...	Ok
O2uc	36998	491353	...	...		...	...	...	...	Ok
O3uc	41728	491353	...	...		...	...	...	...	Ok
O4uc	11573	491353	-11573	-101197		...	...	...	...	Ok
D1c	4929	147988	...	...		...	...	...	...	Ok
D2c	49599	147988	...	...		...	...	...	...	Ok
D3c	54528	147988	...	...		...	...	...	...	Ok
O1c	6768	491353	-6768	-100940		...	...	...	...	Ok
O2c	37902	491353	...	...		...	...	...	...	Ok
O3c	42831	491353	...	...		...	...	...	...	Ok
O4c	11697	491353	-11697	-100940		...	...	...	...	Ok
Max	RATIO	0.368		0.116		...			...	

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress	Channel Stress		Pass Fail
	Ten	Allwd	Cmp	Allwd	Ld	Allwd		Stress	Allwd	



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D1uc	21094	110493	-238	-53747	2369	12407	60023	221983	305976	497993	Ok
D2uc	14961	110493	-14961	-53747	1680	12407	273829	491353	36088	221983	Ok
D3uc	6133	110493	...	...	689	12407	223266	491353	269888	497993	Ok
O1uc	58870	220986	...	...	6610	24814	92701	491353	336816	497993	Ok
O2uc	34134	220986	...	...	3833	24814	231691	491353	5249	497993	Ok
O3uc	43909	220986	...	...	4930	24814	181128	491353	300727	497993	Ok
O4uc	37775	220986	...	...	4242	24814	42138	491353	30839	497993	Ok
D1c	21240	110493	-1316	-53747	2385	12407	65022	491353	316723	497993	Ok
D2c	14759	110493	-14759	-53747	1657	12407	282915	491353	39719	221983	Ok
D3c	6481	110493	...	...	728	12407	227751	491353	277004	497993	Ok
O1c	58867	220986	...	...	6610	24814	98720	491353	348869	497993	Ok
O2c	34872	220986	...	...	3916	24814	239358	491353	7573	497993	Ok
O3c	44108	220986	...	...	4953	24814	184194	491353	309150	497993	Ok
O4c	37628	220986	...	...	4225	24814	43557	491353	32146	497993	Ok
Max RATIO	0.266		0.278		0.266		0.576		0.701		

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	355.869	360.000	Ok
Tube thickness :	1.722	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	3.000	Ok
Tube-tubesheet groove weld leg :	1.281	3.000	Ok

Min Shell length of thk, (156.000) adj. to tubesheet: 1066.415 mm

Min Channel length of thk, (156.000) adj. to tubesheet: 1066.415 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -4.826 mm
 Due to Pressure Shell Compresses by : -2.073 mm
 Due to Pressure + Thermal Shell Compresses by : -6.899 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	18.431	1.000	21.067	1.000
Tubesheet Shear Stress	36.667	1.000	47.489	1.000
Tube Tensile Stress	93.643	1.000	132.945	1.000
Tube Compressive Stress	735.395	1.000	64.669	1.000
Tube-Tubesheet Joint load	93.644	1.000	132.945	1.000
Shell Stress (Axial, Junction)	18.431	1.000	21.067	1.000
Tubesheet-Channel Junction	18.431	1.000	21.067	1.000
Tube Pressure Stress	26.858	1.000	22.059	1.000
Minimum MAWP	18.431		21.067	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	18.962	1.000	21.606	1.000
Tubesheet Shear Stress	41.254	1.000	48.054	1.000
Tube Tensile Stress	100.876	1.000	141.230	1.000
Tube Compressive Stress	6045.78	1.000	76.570	1.000
Tube-Tubesheet Joint load	100.876	1.000	141.230	1.000



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Shell Stress (Axial, Junction)	18.962	0.286	21.606	0.531
Tubesheet-Channel Junction	18.962	0.520	21.606	0.195
Tube Pressure Stress	28.680	1.000	27.241	1.000
Minimum MAPnc	18.962		21.606	

(*) All load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.5846	0.5848	...	...	...	...
2	-2.0724	-2.0731	...	...	...	...
3	-1.4878	-1.4883	...	...	...	...
4	...	...	...	...	...	...
5	0.5846	0.5848	-4.8246	-4.8262	...	...
6	-2.0724	-2.0731	-4.8246	-4.8262	...	...
7	-1.4878	-1.4883	-4.8246	-4.8262	...	...
8	...	...	-4.8246	-4.8262	...	...

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: One or more Nozzles failed Code Requirements!**

Please review all reports carefully!

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